CHEMISTRY GRADE 10: ACIDS AND BASES

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.1 (10 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Chemistry
Strand	Strand 3.0: Acids, Bases and Salts
Sub-Strand	Sub-Strand 3.1: Acids and Bases
Total Duration	10 lessons × 40 minutes = 400 minutes total
Sub-Strand Content	– Simple classification of common substances as acids or bases (e.g., lemon juice, vinegar, milk, baking soda, soap) – Definition of acids and bases using the Arrhenius concept (acids produce H^+ ions; bases produce OH^- ions in aqueous solution) – Physical properties of acids and bases (taste, feel, colour change with indicators) – Chemical properties of acids: reactions with metals, bases, metal carbonates, and metal hydrogen carbonates – Chemical properties of bases: reactions with acids and indicators – Commercial indicators and their colour changes (litmus, phenolphthalein, methyl orange) – The pH scale: concept, range (0–14), and relationship to acidity/alkalinity – Measuring pH using pH indicator paper and universal indicator/pH chart – Strong vs. weak acids and bases (degree of ionisation; examples relevant to Kenyan industry and agriculture) – Uses of acids and bases in everyday life, Kenyan industry, and agriculture
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) classify common substances found in the Kenyan environment as acids or bases based on their observable properties; b) define acids and bases using the Arrhenius definition and write ionic equations showing H^+ and OH^- ion production; c) describe and write balanced chemical equations for the reactions of acids with metals, bases, metal carbonates, and metal hydrogen carbonates; d) use commercial indicators and pH indicator paper to determine and compare the pH values of given solutions; e) distinguish between strong and weak acids and bases in terms of degree of ionisation and relate this to their pH values; f) explain the uses of acids and bases in everyday life, Kenyan agriculture (e.g., soil pH management in Kericho tea farms), and industry.
Core Competencies	– Communication and Collaboration: Learners discuss predictions, share experimental observations in groups, and present findings on the Driving Question Board — developing scientific vocabulary around acids and bases. – Critical Thinking and Problem Solving: Learners analyse anomalous results from indicator and pH experiments, evaluate evidence to distinguish strong from weak acids, and construct explanations grounded in the Arrhenius definition. – Creativity and Imagination: Learners design their own natural indicators from locally available materials (e.g., red cabbage, hibiscus/rosella) and propose models to explain acid-base behaviour at the particle level. – Digital Literacy: Learners engage with video resources on pH and uses of acids, interpret digital pH simulations, and record

	data in structured digital or paper formats on the ARES platform. – Learning to Learn: Learners revise their mental models progressively across all 10 lessons, reflecting on how each new piece of evidence refines their understanding of acids and bases. – Self-Efficacy: Learners gain confidence in laboratory practice through structured experiments — handling indicators, conducting acid-metal reactions, and measuring pH safely.
Core Values	– Responsibility: Learners handle laboratory acids (e.g., dilute H_2SO_4, HCl) and bases (NaOH) safely, following MSDS guidelines and proper disposal procedures — building a culture of personal and environmental responsibility. – Respect: Learners respect diverse group contributions during sensemaking discussions and the Driving Question Board sessions, valuing peers' predictions even when they differ. – Unity: Collaborative experiments (e.g., testing pH of multiple solutions) require coordinated group work, reinforcing the value of working together toward shared scientific understanding. – Social Justice: Learners connect soil acidity and liming practices in Kenyan smallholder farming (Rift Valley, Kisii) to food security, understanding how chemistry knowledge can address inequalities in agricultural productivity. – Integrity: Learners record experimental results honestly, even when observations do not match their initial predictions, fostering scientific honesty and intellectual integrity.
Science & Engineering Practices	– Asking Questions and Defining Problems: Learners generate questions for the Driving Question Board after observing the anchoring phenomenon (surprising colour changes of a universal indicator in everyday Kenyan drinks and foods), driving the investigation forward. – Planning and Carrying Out Investigations: Learners design and conduct three structured experiments — testing commercial indicators, performing acid reactions (with metals, bases, carbonates, hydrogen carbonates), and measuring pH — following safe laboratory procedures. – Analysing and Interpreting Data: Learners record, tabulate, and interpret colour-change data from indicator experiments and numerical pH data, identifying patterns that distinguish acids from bases and strong from weak. – Constructing Explanations: Learners apply the Arrhenius definition and ionic equations to explain the results of all three experiments, connecting macroscopic observations to particle-level reasoning. – Obtaining, Evaluating, and Communicating Information: Learners extract key information from two reading texts and two videos on acids, bases, and pH, critically evaluate the sources, and communicate findings through DQB entries, written equations, and model diagrams.
Pertinent & Contemporary Issues (PCIs)	– Health Education: Learners relate pH to human health — stomach acidity (HCl in gastric juice), antacid action (bases neutralising excess acid), and safe handling of household chemicals such as bleach and battery acid common in Kenyan homes. – Environmental Education: Learners investigate the effects of acid rain (linked to vehicle emissions in Nairobi) and acidic soils on ecosystems and crops, understanding how chemistry underpins environmental stewardship. – Agriculture and Nutrition: Learners connect soil pH management — liming acidic soils in tea-growing regions like Kericho — to crop productivity and food security, integrating chemistry into Kenya's agricultural context. – Financial Literacy / Career Awareness: Learners explore how industries (e.g., Kenyan soda ash extraction at Lake Magadi, battery manufacturing, food preservation) rely on controlled acid-base chemistry, linking school chemistry to economic productivity. – Safety: Laboratory safety protocols for corrosive substances are explicitly taught and practised across all three experiments, reinforcing lifelong safe-handling habits.
Career Connections	– Medical Doctor / Clinical Officer: Understanding gastric acid, antacids, and blood pH buffers is foundational for diagnosing and treating conditions like ulcers and acidosis — relevant to Kenya's growing healthcare sector.

	– Agricultural Scientist / Agronomist: Soil pH testing and amendment (liming) are daily tasks for agronomists advising smallholder farmers across Kenya's highlands; this sub-strand provides the chemical basis for those practices. – Chemical / Process Engineer: Industries such as EABL (brewing), Bamburi Cement, and the Magadi Soda Company rely on acid-base chemistry in manufacturing — engineers must control reaction conditions precisely. – Environmental Scientist: Monitoring river and lake pH (e.g., Lake Victoria, Nairobi River) for pollution assessment and acid-rain impact studies requires the foundational knowledge built in this sub-strand. – Pharmacist: Formulating antacids, pH-sensitive drug coatings, and intravenous fluids requires deep understanding of acid-base equilibria introduced here. – Food Scientist / Technologist: Preservation techniques (pickling with acetic acid, fermentation producing lactic acid in mursik) and quality control in Kenyan food processing depend on pH measurement and control.
Focus for Lessons	This unit anchors learning in a puzzling everyday phenomenon — the dramatic colour changes produced when a universal indicator is added to common Kenyan foods and drinks (lemon juice, milk, black tea, soda, soap solution, baking soda mixture) — to drive students to investigate what makes substances acidic or basic. Using a carefully sequenced combination of videos, readings, and three hands-on experiments, learners build from simple observable classification toward the Arrhenius particle-level definition, the chemistry of acid reactions, and the quantitative pH scale. Throughout, Kenyan agricultural, industrial, and health contexts ensure that abstract chemistry concepts are grounded in students' lived realities and future career pathways.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: "What invisible properties of everyday Kenyan liquids cause them to turn such different colours — and why does it matter for our health, our farms, and our environment?" KICD KEY INQUIRY QUESTIONS: 1. What are acids and bases? 2. What are the chemical properties of acids and bases? 3. How is the strength of acids and bases determined?
Anchoring Phenomenon	ANCHORING PHENOMENON — "The Colour-Changing Drinks of the School Canteen" A Chemistry teacher at a school in Nairobi brings eight small cups to the front of the class. Each cup contains a clear or lightly coloured liquid that students recognise from daily life: freshly squeezed lemon juice, black tea (chai), cow's milk, a carbonated soft drink (e.g., Stoney Tangawizi), clean tap water, a baking soda solution, liquid soap (the kind used for handwashing in the school), and a dilute solution of "Jik" bleach. The teacher adds five drops of universal indicator solution to each cup and swirls gently. In seconds, the cups explode into a rainbow of colours — brilliant reds and oranges in some, a stable green in one, and striking blues and purples in others. Students are visibly surprised: all the liquids looked ordinary, yet they behave so differently. The teacher asks: "These are all liquids you encounter every day — some you even drink. Why do they turn such different colours? What is in them that causes this? And does it matter?" This puzzle — that invisible dissolved particles in familiar substances can produce such dramatic, ordered colour patterns — anchors the entire unit. Students cannot yet explain it, but every lesson peels back another layer of understanding until they can account for every colour in every cup using the Arrhenius definition, the pH scale, and the chemical properties of acids and bases.
Storyline Thread	Lesson 1: ANCHORING PHENOMENON + PREDICT — Students observe the universal indicator colour-change demonstration with eight Kenyan everyday liquids; they record predictions and generate initial questions for the Driving Question Board. Introduction to the concept of classifying substances as acids or bases based on observable properties (taste, feel, litmus colour — safety-discussed only, not tasted in lab). Lesson 2: OBSERVE + EXPLAIN — Students watch the introductory video on acids and bases and complete the paired reading on

	acids and bases. They use the new vocabulary (acid, base, alkali, neutral, indicator) to begin revising their Lesson 1 predictions and update DQB entries with new questions about what acids and bases are made of at the particle level. Lesson 3: EXPLAIN — Arrhenius Definition of Acids and Bases. Teacher-led sensemaking: acids produce H^+ ions in aqueous solution; bases produce OH^- ions. Students write ionisation equations for HCl, H_2SO_4, NaOH, and $Ca(OH)_2$. They connect the Arrhenius model to the colour patterns observed in Lesson 1 and revisit the anchoring phenomenon with new explanatory power. Lesson 4: OBSERVE (Experiment 1) — Differences Between Acids and Bases Using Commercial Indicators. Students test a range of solutions (including lemon juice, vinegar, baking soda solution, soap solution, tap water) with litmus (red and blue), phenolphthalein, and methyl orange. They tabulate colour changes, classify each substance, and identify patterns. DQB updated: "We can now tell acids from bases — but what makes acids chemically reactive?" Lesson 5: OBSERVE (Experiment 2a & 2b) — Chemical Properties of Acids: Acid + Metal and Acid + Base (Neutralisation). Students react dilute HCl and H_2SO_4 with zinc/magnesium ribbon and observe gas production (hydrogen — tested with a lit splint). They then perform neutralisation reactions (HCl + NaOH with phenolphthalein indicator). Balanced and ionic equations written. Connected to the corroded battery supporting phenomenon. Lesson 6: OBSERVE (Experiment 2c & 2d) — Acid + Metal Carbonate and Acid + Metal Hydrogen Carbonate. Students react dilute HCl with marble chips ($CaCO_3$) and sodium hydrogen carbonate ($NaHCO_3$/baking soda), testing the gas produced with limewater (CO_2). Balanced equations written for all four acid reaction types. Connected to the Kericho soil and Mombasa battery supporting phenomena. DQB updated with a summary of all acid reactions. Lesson 7: EXPLAIN + MODEL BUILDING (Mid-Unit) — Consolidation of chemical properties of acids and bases. Students construct a summary graphic model showing the four types of acid reactions with generalised equations, products, and gas tests. Class discussion connects all reactions to the Arrhenius definition: H^+ ions as the active agent. Comparison of base properties (reaction with acids, effect on indicators) completed. Lesson 8: OBSERVE — Introduction to pH. Students watch the introductory video on pH, then conduct Experiment 3: testing the pH of a range of solutions (including those from Lessons 4–6) using universal indicator/pH chart and pH indicator paper. They construct a pH number line (0–14) populated with Kenyan real-world examples (stomach acid, black tea, milk, blood, soap, bleach, Lake Magadi water). DQB updated: "What does the pH number actually tell us about the strength of the acid or base?" Lesson 9: EXPLAIN — Strong vs. Weak Acids and Bases. Teacher-guided sensemaking using pH data from Lesson 8: degree of ionisation explains pH differences. Students compare HCl (strong, fully ionised) vs. ethanoic acid/citric acid (weak, partially ionised) using particle diagrams. Agricultural connection: why soil pH 4.2 (strong acidity) damages tea plants but pH 5.5–6.0 is ideal. DQB updated with refined model entries. Lesson 10: USES OF ACIDS AND BASES + FINAL MODEL BUILD — Students watch the video and complete the reading on uses of acids and bases (industrial, agricultural, domestic, medical Kenyan contexts). They revise and finalise their cumulative particle/concept model of acids and bases built across the unit. A Driving Question Board closure activity has students match every original question to an evidence-based answer. Final reflection: students return to the eight cups from Lesson 1 and write a full scientific explanation of every colour change, demonstrating mastery of the sub-strand learning outcomes.
Supporting Phenomena	– SUPPORTING PHENOMENON 1 — "The Fizzing Antacid Tablet": A student who complains of stomach pain after eating too

	much ugali and nyama choma takes a Gelusil or Eno antacid tablet dissolved in water. The fizzing and rapid relief is puzzling — what is in the tablet, why does it fizz, and how does it stop the pain? (Connects to acid + base neutralisation and acid + carbonate reactions.) – SUPPORTING PHENOMENON 2 — "The Dying Tea Farm": A farmer in Kericho notices that one section of her tea bushes has yellowing leaves and stunted growth despite regular watering and fertiliser. A soil test reveals the soil pH is 4.2. After lime (calcium carbonate) is applied, the plants recover within weeks. Why does the number "4.2" explain sick plants, and why does a white powder fix it? (Connects to pH scale, strong/weak acids in soil, and acid + carbonate reaction.) – SUPPORTING PHENOMENON 3 — "The Corroded Matatu Battery": A mechanic in Mombasa shows students a car battery terminal covered in a blue-white crusty deposit and explains that the battery contains sulfuric acid. He uses baking soda solution to clean the terminal, and it fizzes vigorously. Why does acid leak from the battery, why does the fizzing happen, and why does baking soda work? (Connects to acid properties, acid + carbonate/hydrogen carbonate reaction, and uses of acids.) – SUPPORTING PHENOMENON 4 — "Lake Magadi's Strange Pink Water": Students view an image of Lake Magadi in the Rift Valley, whose water appears pink-red and kills most animals that enter it. Local communities harvest trona (sodium carbonate) from it commercially. Why is this lake so different from Lake Victoria? What makes the water so harmful, and yet so economically valuable? (Connects to bases, pH extremes, industrial uses of bases/salts.)

LESSON 1 (80 minutes): The Colour-Changing Drinks of the School Canteen — Launching the Phenomenon	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To introduce the anchoring phenomenon and establish that familiar Kenyan everyday liquids have invisible properties that cause them to produce dramatically different colours with universal indicator, launching student curiosity about acids and bases.
Knowledge	Students will know that substances can be classified as acids or bases based on observable properties such as taste (sour for acids, bitter for bases), feel (slippery for bases), and effect on litmus paper (acids turn litmus red; bases turn litmus blue). Students will know that universal indicator produces a range of colours corresponding to different substances, and that these differences reflect something invisible dissolved in the liquid.
Skills	Students will be able to record careful observations of the universal indicator colour-change demonstration, make and justify written predictions about the acidity or basicity of familiar Kenyan liquids, generate testable questions from an anchoring phenomenon, and organise initial questions onto a Driving Question Board.
Attitudes	Students will demonstrate curiosity and open-mindedness by committing predictions to paper before instruction, and show respect for evidence by distinguishing between what they observed and what they merely assumed.
Key Inquiry Question	What are acids and bases?
Purpose in Storyline	This is the opening lesson of the unit. It launches the anchoring phenomenon — the colour-changing drinks demonstration — and gives students their first unresolved puzzle: eight familiar Kenyan liquids, all looking ordinary, produce a rainbow of colours when universal indicator is added. Students cannot yet explain the pattern, but they generate their first questions for the Driving Question Board, which will be revisited and updated in every subsequent lesson. Every later lesson on chemical properties, pH, the Arrhenius definition, and indicator behaviour is motivated by the questions students raise today.
Safety Notes	The Jik bleach solution must be prepared by the teacher as a heavily diluted solution (no more than 1 part Jik to 50 parts water) before the lesson. Students must NOT taste any liquid in the laboratory at any time. The teacher explicitly discusses the safety rule: we can talk about taste as a property scientists have historically described, but we never taste substances in the lab. Students should not handle the Jik or liquid soap cups directly. The teacher conducts the universal indicator addition as a demonstration. If universal indicator solution is not available, red and blue litmus paper may be substituted for the formative check at the end of the lesson, but the dramatic rainbow effect is best achieved with universal indicator.

B. LESSON OVERVIEW

The teacher opens the unit with a vivid demonstration: eight small cups containing familiar Kenyan liquids (lemon juice, black chai, cow's milk, Stoney Tangawizi, tap water, baking soda solution, liquid handwash soap, and dilute Jik bleach) are displayed at the front of the class. Before any drops are added, students record predictions about which liquids they think are similar or different in some invisible way, and why. The teacher then adds five drops of universal indicator to each cup, swirling gently, producing a dramatic rainbow — reds, oranges, yellow-green, green, blues, and purples. Students record observations, compare with predictions, and generate questions they cannot yet answer. The lesson closes by transferring the most compelling student questions onto the class Driving Question Board, establishing the intellectual agenda for the entire unit. Observable properties of acids and bases (taste, feel, litmus) are introduced through guided discussion — with an explicit safety conversation about why we discuss taste as a concept but never practise it in the lab.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Experimental probability Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students are seated in groups of four. On each desk is a printed A5 'Prediction Card' listing the eight liquids with a blank column for 'My Predicted Colour' and a column for 'My Reasoning'. Before the teacher adds any indicator, students study the eight labelled cups at the front (all still uncoloured) and silently complete their Prediction Cards individually. They then share predictions within their group for two minutes, noting where they agree or disagree. Each group nominates one spokesperson to share one surprising prediction with the class. Students also answer the written anchor question at the bottom of the Prediction Card: 'These eight liquids all look similar — clear or very lightly coloured. Do you think they are all the same kind of substance? Write one sentence explaining your reasoning.'	Before distributing Prediction Cards, the teacher holds up each cup in turn, names the liquid, and connects it to student life: 'This is lemon juice — the kind you might squeeze into your chai or onto your nyama choma. This is Stoney Tangawizi — some of you had one at break. This is the liquid soap at the handwashing station by the canteen.' Allow 30 seconds per cup for recognition. Then say exactly: 'Before I add anything to these cups, I want you to think carefully. You know these liquids. Based on everything you already know — how they taste, how they feel, what they are used for — predict what colour you think each one will turn. There are no wrong answers right now. This is your honest thinking before we see any evidence.' Distribute Prediction Cards. Allow 4 minutes of silent individual thinking. Circulate and scan cards — do NOT correct any prediction. After individual time, say: 'Now share with your group for two minutes. Find one prediction where your group completely disagrees — you will share that one.' After group sharing, cold-call FOUR group spokespeople (choose groups with visibly different predictions). For each, ask: 'What is your group's most disagreed-upon prediction, and what is the reasoning behind each side?' Allow 10 seconds of WAIT TIME after each group responds before moving on. Record three or four contrasting predictions on the board under the heading 'Our Predictions — Before	Strategy: Prediction Commitment with Justified Reasoning. By requiring students to write down predictions with explicit reasoning before any evidence is shown, the teacher activates prior knowledge and creates a productive cognitive tension. When observations later contradict predictions, students experience the surprise as personally meaningful — their own mental model was challenged — which is a far more powerful motivator for learning than simply being told information. The Prediction Card also creates a written artefact that students can revisit at the end of the unit to measure their own conceptual growth.	Circulate during the 4-minute individual prediction phase and look for two observable indicators: (1) At least 80% of students are writing in the 'My Reasoning' column, not just guessing colours randomly — this shows they are drawing on prior knowledge rather than guessing blindly. (2) Listen for the reasoning students give during group sharing — specifically note whether any student spontaneously uses words like 'sour', 'bitter', 'burns', 'cleaning', or 'stomach' — these indicate prior conceptual resources that the teacher can build on during the Explain phase. Flag any group that claims ALL eight liquids will turn the same colour — this is a key misconception to address publicly during the Observe phase debrief.

		Evidence'.		
Observe Phase 📹 **VIDEO:** [Experimental probability](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Properties of Acids (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	The teacher conducts the universal indicator demonstration at the front of the class. Students watch in silence for the first 30 seconds, then are invited to call out colour observations as each cup changes. Students record final colours on the second column of their Prediction Card ('Actual Colour Observed'). After all eight cups are coloured, students complete a structured Observation Table in their exercise books: three columns — Liquid, Colour with Indicator, Similar to Which Other Liquid? They compare their predictions with observations and place a tick or cross against each prediction. At the bottom of the table they answer: 'Which result surprised you the most? Write two sentences explaining why it was unexpected.' Students share their most surprising result with a partner for 90 seconds.	Stand at the demonstration table. Say: 'I am now going to add five drops of universal indicator solution to each cup. I want complete silence for the first 30 seconds so you can see exactly what happens. Watch carefully — you are scientists gathering evidence.' Add indicator to lemon juice first — the dramatic red colour change tends to produce audible reactions. Pause after each colour change for 5 seconds before moving to the next cup. After all eight are coloured, say: 'Now, quietly and individually, record the actual colour of each cup in your Prediction Card. Do not discuss yet.' Allow 2 minutes. Then ask: 'By a show of hands — how many of you got at least three colours exactly right?' (Count and note the number.) 'How many got fewer than two correct?' (Count and note.) Then cold-call THREE students who had unexpected results: 'Tell us which liquid surprised you and what you had predicted instead.' Allow 10 seconds of WAIT TIME after each. Then ask the whole class: 'Look at the eight colours. Can you see any groupings — any cups that seem to belong together? Do not use any chemistry words yet — just describe what you see.' Cold-call TWO students. Probe with: 'What makes you group those two together?' Write the groupings students suggest on the board. Do not yet confirm or name them as 'acids' and 'bases' — preserve the puzzle.	Strategy: Structured Observation with Anomaly Identification. The Observation Table is structured so that students must actively compare and group, not merely copy colours. The 'Similar to Which Other Liquid?' column forces pattern recognition before any vocabulary is introduced — students discover the acid/base grouping themselves through colour similarity. The 'most surprising result' prompt targets the moments of cognitive disequilibrium that are most likely to generate genuine questions for the DQB. The teacher deliberately withholds the vocabulary of 'acid' and 'base' at this stage, preserving student ownership of the groupings they have constructed from evidence.	As students complete Observation Tables, circulate and look for two specific indicators: (1) Students correctly identify at least two groupings of similar colours (e.g., the red/orange group: lemon, Stoney, chai; the blue/purple group: baking soda, soap, Jik; and green for water) — this demonstrates the capacity to organise evidence. (2) Students' 'most surprising result' responses reveal specific misconceptions — for example, students who are surprised that chai turned orange (expecting it to be neutral because it is a drink) reveal the misconception that 'safe to drink = neutral'. Record these misconceptions on a sticky note for targeted addressing in the Explain phase. Any student who states that the colours 'don't mean anything' or 'are just the indicator's colour' should be noted for a Socratic follow-up question: 'If the indicator caused all the colours, why is each cup a different colour?'
Explain Phase 📹 **VIDEO:**	The teacher introduces the vocabulary and concepts needed	Begin with a direct connection to the observation: 'You noticed that	Strategy: Concept Introduction Anchored to Phenomenon	Collect two observable indicators: (1) In the annotation task, the key

[Experimental probability](#) Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Properties of Acids (Flexbook)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	to begin making sense of the colour pattern: acids and bases as two categories of substances, the observable properties that distinguish them (taste — sour vs. bitter; feel — slippery for bases; effect on litmus — red for acids, blue for bases), and the fact that the universal indicator is showing us the same property that litmus paper would show, but across a spectrum of colours. Students annotate their Observation Table by labelling each liquid 'A' (acid), 'B' (base), or '?' using the new property descriptions as evidence. They then read a short illustrated text card (provided by the teacher) that lists examples of acids and bases in Kenyan daily life — referencing lemon juice in coastal cooking, the carbonic acid in Stoney Tangawizi, the hydrochloric acid that helps digest ugali and githeri in the stomach, and the bases in toothpaste, washing soap, and antacid tablets sold at Nairobi pharmacies. Students answer two questions in their exercise books: (1) 'Using the properties described, explain why lemon juice turned red and baking soda solution turned blue.' (2) 'Name one acid and one base that you encounter in your daily life outside this classroom. How do their properties affect how they are used?'	the cups fell into groups — reds and oranges on one side, blues and purples on the other, with green in the middle. Those groups already have names in chemistry. Let me introduce them.' Write 'ACIDS' and 'BASES' on the board as two columns. Say: 'Chemists have been studying these two groups for centuries. Before modern labs, they identified them using properties they could observe directly. I am going to describe three of those properties — and after I describe each one, I want you to look back at your eight cups and see if the property matches the colour you saw.' Describe taste (acids are sour — think of the taste of lemon; bases are bitter — think of the taste of unsweetened baking soda if you have ever accidentally tasted it). STOP and say: 'Important safety rule, and I need everyone to hear this: In this classroom and in any chemistry laboratory, we NEVER taste substances to test them. We describe taste as a historical property that scientists have documented — we do not practise it. The consequences of tasting an unknown substance can be severe.' Pause for 5 seconds. Continue with feel (bases feel slippery — think of how soap feels on your hands) and litmus (acids turn red litmus paper red; bases turn red litmus paper blue). After all three properties, say: 'Now annotate your table — write A, B, or ? next to each liquid based on these properties. You have 3 minutes.' Circulate and cold-call FOUR students to share their annotations, choosing students who labelled something as '?'. For each: 'Why were you uncertain	Evidence. Rather than introducing acids and bases as abstract definitions, the teacher uses the eight cups — already coloured and sitting at the front of the class — as the concrete referent for every new property described. The annotation task ensures that each new property is immediately applied to familiar evidence rather than stored as inert vocabulary. The Kenyan-context text card ensures that the abstract category 'acid' is immediately populated with personally meaningful examples (lemon, Stoney, stomach acid during digestion of githeri), preventing the common misconception that acids and bases are only found in industrial or laboratory settings.	diagnostic is whether students correctly label tap water as neither A nor B (or as '?') rather than forcing it into one category — this tests whether they have understood that not all substances are strongly acidic or basic, setting up the pH scale in later lessons. (2) In the written question responses, look for students who can give a property-based explanation for why lemon turned red ('lemon juice is an acid, and acids turn the indicator red') rather than a circular explanation ('it turned red because it is a red-coloured liquid'). Students still giving circular explanations need targeted follow-up in Lesson 2. Note: any student who spontaneously mentions 'concentration' or 'strength' of acid is ready for extension work and should be flagged for deeper engagement with the pH scale in Lesson 3.

		about this one? What property was ambiguous?' Allow 10 seconds WAIT TIME. Distribute the text card on Kenyan acids and bases. After 3 minutes of reading, say: 'Using what you just read, answer the two questions in your exercise book. I want full sentences — not bullet points.' Allow 5 minutes. Cold-call TWO students to share question 2 answers — look for responses that go beyond the text card and connect to personal experience.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Experimental probability] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Properties of Acids (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each student receives two sticky notes. On the first, they write one question that today's lesson has left unanswered for them. On the second, they write one connection between what they saw today and something in their life outside school — a food, a farm input, a cleaning product, a health experience, or an environmental observation. Students post their sticky notes on a large sheet of paper fixed to the classroom wall under the heading: 'DRIVING QUESTION BOARD — What invisible properties of everyday Kenyan liquids cause them to turn such different colours — and why does it matter for our health, our farms, and our environment?' The teacher reads several questions aloud and clusters them into emerging themes with the class: questions about 'what causes the colours', questions about 'which is stronger', questions about 'how this affects health', and questions about 'farming and environment'. The DQB remains on the wall for the entire unit.	Distribute two sticky notes per student. Say: 'Before you write, think about what genuinely puzzles you — not what you think the teacher wants to know, but what YOU want to understand after today. Write your question in the form of a full question sentence.' Allow 3 minutes of silent writing. Then say: 'On your second sticky note, write one real connection to your life. It could be: something you eat at home, something your parents use on the farm, something from a pharmacy, or something in the environment near where you live — like the water in Lake Victoria or the soil in Rift Valley farms. Be specific.' Allow 2 minutes. Invite all students to come up in two waves and post their sticky notes. Read FIVE questions aloud, choosing a variety — one that is answerable in the next lesson, one that requires several lessons, one that connects to health, one to farming, and one that is unexpectedly deep. For each say: 'This is an excellent question because it asks about something our observation today cannot yet explain. We will return to this question in Lesson [X].'	Strategy: Driving Question Board as a Living Epistemic Artefact. The DQB externalises student thinking and makes the class's collective intellectual agenda visible and permanent. By requiring both a question AND a real-world connection on separate sticky notes, the teacher ensures that the DQB captures both the cognitive puzzle (what students don't know) and the motivational stake (why it matters to their lives). Clustering questions by theme previews the structure of the upcoming lessons without pre-teaching content, giving students a sense of the unit's arc. Returning to the DQB in every lesson builds the habit of checking growing understanding against original questions — a core scientific practice.	The DQB sticky notes are a direct window into student thinking. Look for three specific indicators: (1) At least 60% of questions are about mechanism ('why' or 'how') rather than just identification ('what is') — this shows students are already thinking causally. (2) At least one-third of the real-world connection notes reference a specific Kenyan context (e.g., 'the fertiliser my father uses in Eldoret', 'the antacid my grandmother takes for stomach pain') — this indicates that the phenomenon has landed as personally relevant, not just as a classroom exercise. (3) Flag any question that directly asks about 'strength' of acids and bases (e.g., 'Why is Jik more dangerous than lemon juice?') — this is the perfect launch point for the pH scale lessons and should be highlighted on the DQB with a star sticker.

		Cluster the remaining questions visibly into three or four theme groups using different coloured markers to draw circles. Say: 'This board belongs to you. Every lesson in this unit, we will come back here. Some questions will be answered. New ones will appear. By the end of the unit, you should be able to answer the driving question completely — using chemistry.'		
Model Building Phase 📹 **VIDEO:** Experimental probability Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students create their first 'Initial Model' in their exercise books — a simple hand-drawn diagram or annotated table that represents their current best explanation for why the eight liquids turned different colours. The model must include: (1) the eight liquids labelled with their observed colour; (2) arrows or brackets grouping them; (3) a label or short phrase explaining what the student thinks is 'in' the liquids that causes each group's colour. Students are explicitly told that this model is expected to be incomplete and possibly wrong — it represents their thinking right now, before they have learned the full chemistry. At the bottom of their model, students write: 'My model cannot yet explain…' and complete the sentence honestly. Models are NOT collected — they remain in students' exercise books as a baseline that they will revise and annotate after each subsequent lesson.	Say: 'Scientists do not just observe and describe — they build models to explain what they cannot directly see. A model is your best current attempt to explain the evidence. Today's model will be incomplete — that is expected and completely fine. What matters is that your model honestly represents what you currently understand.' Draw a rough example model on the board showing two circles labelled 'Group 1 — Red/Orange' and 'Group 2 — Blue/Purple' with a question mark in the middle of each circle where the 'cause' would go. Say: 'Your model should show more than mine — you now know the words acid and base, and you know some of their properties. Use that knowledge.' Allow 6 minutes for individual model drawing. Circulate and look at three or four models — do NOT correct, but ask one probing question per model: 'What is inside the lemon juice that causes the red colour? Can you draw that somehow?' or 'Your model shows two groups — what about the water? Where does it fit?' After 6 minutes, ask TWO volunteers to share their models using the document camera or by holding	Strategy: Initial Baseline Model with an Explicit Incompleteness Prompt. Requiring students to articulate what their model cannot yet explain ('My model cannot yet explain…') is a metacognitive move that prevents the false closure of thinking a partial explanation is a complete one. It also creates a built-in self-assessment tool: as the unit progresses, students revisit this sentence and find that each lesson answers a piece of it. The shared model gallery (two volunteers presenting) introduces the norm of public model critique — assessing what a model explains vs. what it leaves unexplained — which is a core NGSS Science and Engineering Practice (Developing and Using Models) that will deepen throughout the unit.	Scan the Initial Models for three specific indicators: (1) Students correctly use the terms 'acid' and 'base' as labels for at least two of their groupings — confirming that the Explain phase vocabulary has been retained. (2) Students' 'My model cannot yet explain…' sentences reveal the most productive unresolved puzzles — look especially for references to 'why different shades of red' or 'what exactly is in the liquid' or 'why water is green not red or blue' — these map directly onto the pH scale and Arrhenius definition content in Lessons 2–4. (3) At least two-thirds of students place water in a separate category or mark it with uncertainty — confirming understanding that not every liquid is a strong acid or base. Any student whose model treats all eight liquids as equivalent (no grouping at all) should be paired with a peer during the next lesson's opening review.

		them up. For each, ask the class: 'What does this model explain well? What does it not yet explain?' Allow 10 seconds WAIT TIME after each response. Close by saying: 'Every lesson from now on, we will come back to this model and improve it. By Lesson 8, your model should be able to explain every colour in every cup — using the Arrhenius definition, the pH scale, and the chemical properties of acids and bases. Keep your exercise book safe.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did the demonstration produce the intended dramatic visual effect — specifically, were the reds and purples sharp and distinct enough to generate visible student surprise? If not, check the concentration of the universal indicator solution and the freshness of the Jik for the next class. (2) How many DQB questions were mechanistic ('why' or 'how') vs. purely descriptive ('what')? If fewer than half were mechanistic, consider adding a sentence frame to the sticky note task in future: 'I wonder WHY… / I wonder HOW…' to scaffold deeper questioning. (3) Were students able to identify water as a distinct, middle-category substance, or did most force it into acid or base? If water's 'green' colour was not sufficiently discussed during the debrief, open Lesson 2 with a direct revisit of the water cup before introducing the pH scale. (4) Identify one student question from the DQB that was unexpectedly sophisticated — plan to reference this student by name when the relevant lesson is reached, as a way of affirming student intellectual contribution. (5) Note any student who drew a molecular-level diagram in their Initial Model (showing particles rather than just group labels) — this student may be ready for extension work connecting the observable properties of acids to the presence of hydrogen ions (H^+), which will be formally introduced via the Arrhenius definition in a later lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Eight familiar Kenyan liquids — lemon juice, black chai, cow's milk, Stoney Tangawizi, tap water, baking soda solution, liquid handwash soap, and dilute Jik bleach — each turned a distinctly different colour when five drops of universal indicator solution were added: vivid reds and oranges for lemon juice, chai, and Stoney; green for tap water; and deep blues and purples for baking soda, soap, and Jik.
What did I learn?	Substances can be classified as acids (sour taste, turn litmus red, turn universal indicator red/orange) or bases (bitter taste, slippery feel, turn litmus blue, turn universal indicator blue/purple). The colour differences produced by universal indicator are not random — they reveal a real, invisible property of each liquid. Many everyday Kenyan substances — lemon juice, Stoney Tangawizi, stomach acid (which helps digest ugali and githeri), toothpaste, soap, and antacids — are either acids or bases, and their uses depend on these properties. We never taste substances in a chemistry laboratory to test these properties.
How does this explain the phenomenon?	The eight liquids turned different colours because they belong to two different categories of substances — acids and bases — which have different invisible properties that interact with the universal indicator dye to produce different colours. Liquids in the acid group (lemon juice, Stoney Tangawizi, black chai) turned red or orange because acids share a common property that triggers the red end of the indicator's colour range. Liquids in the base group (baking soda, liquid soap, Jik) turned blue or purple because bases share a different property that triggers the blue-purple end. Tap water, which is neither a strong acid nor a strong base,

	produced a stable green — the middle of the indicator's range. The model students built today can explain the groupings but not yet the differences in shade within each group, nor what exactly is dissolved in each liquid that causes it to behave as an acid or a base — those questions are what the rest of this unit will answer.

LESSON 2 (80 minutes): What Are Acids and Bases? — Naming the Invisible

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Students watch an introductory video and complete a paired reading to build foundational vocabulary and conceptual understanding of acids and bases, enabling them to begin explaining the colour patterns observed in the anchoring phenomenon using scientific language.
Knowledge	Students will know: (1) the definitions of acid, base, alkali, and neutral substance; (2) characteristic properties of acids (sour taste, turn litmus red, turn phenolphthalein colourless, turn methyl orange red) and bases (bitter taste, slippery feel, turn litmus blue, turn phenolphthalein pink, turn methyl orange yellow); (3) common Kenyan examples of acids and bases encountered in daily life (lemon juice, chai, Stoney Tangawizi, Jik bleach, liquid soap, baking soda, milk, tap water); (4) that indicators change colour in a predictable, ordered way depending on whether a substance is an acid or a base.
Skills	Students will be able to: (1) use new scientific vocabulary (acid, base, alkali, neutral, indicator, litmus, phenolphthalein, methyl orange) accurately in written and oral explanations; (2) match familiar Kenyan substances to acid or base categories using indicator evidence from Lesson 1; (3) revise Lesson 1 written predictions using new evidence from the video and reading; (4) formulate at least one new particle-level question for the Driving Question Board.
Attitudes	Students demonstrate curiosity and open-mindedness by willingly revising their Lesson 1 predictions in light of new evidence; they show responsibility by engaging honestly with texts and peer discussions without copying.
Key Inquiry Question	What are acids and bases?
Purpose in Storyline	Lesson 2 is the first OBSERVE + EXPLAIN session in the evidence-gathering sequence. Students move from pure wonder (Lesson 1 phenomenon) to organised vocabulary and conceptual categories. By naming what they saw — and beginning to ask WHY at the particle level — they lay the linguistic and conceptual foundation that every subsequent lesson will build upon to fully explain the colour-changing cups.
Safety Notes	No hazardous chemicals are handled in this lesson. If physical samples of Lesson 1 cups are revisited for reference, remind students not to taste any laboratory solutions. Ensure devices (tablets/laptops) are handled carefully and screens are not shared in ways that prevent individual reading engagement.

B. LESSON OVERVIEW

Lesson 2 opens by returning to the eight colour-changing cups from the anchoring phenomenon. Students review their Lesson 1 prediction cards and the teacher reveals the colour results again (via photograph or retained cups). Students then watch a short curated video introduction to acids and bases, followed by a structured paired reading. Using the Claim–Support–Question (CSQ) reading strategy, pairs extract key vocabulary and properties, then apply them to re-label the phenomenon cups. The lesson closes with a whole-class DQB update where students add new particle-level questions, and individuals revise their prediction cards in a different ink colour to show intellectual growth. The lesson advances the driving question by giving students the scientific language to describe WHAT the colour differences mean, setting up Lesson 3's deeper investigation into WHY (chemical properties and reactions).

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Experimental probability] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Properties of Acids (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Students retrieve their Lesson 1 prediction cards (kept in their science journals). Working individually for 3 minutes, they re-read their original predictions about WHY the cups turned different colours and annotate with a star (★) anything they are now unsure about. They then share with a shoulder partner for 2 minutes. The class does a quick show-of-hands poll: 'How many of you used the word ACID in your prediction? How many used BASE? How many feel your prediction still fully holds up?' This surfaces prior knowledge gaps and creates a felt need for the video and reading to follow.	Return prediction cards and say: 'Take 3 minutes — read your own words from last lesson. Put a star next to anything you are now unsure about or want to change. Do this alone first.' Set a visible 3-minute timer. After timer: 'Now share your starred items with your partner. 1 minute each — go.' After sharing, conduct the show-of-hands poll: 'Hands up if you used the word acid anywhere in your prediction… now hands up if you used the word base…' Cold-call 3 students whose hands were NOT up: 'Tell me — if you didn't use those words, what words did you use to explain the colours?' Allow 10–12 seconds WAIT TIME after each cold-call. Close by saying: 'By the end of this lesson, you will have the exact scientific vocabulary to upgrade every single one of these predictions. Keep your cards out — you will revise them in Phase 4.'	Prediction Revision Protocol — students physically annotate their own prior-knowledge text with stars, making metacognitive gaps visible and creating a personal motivation to engage with the new evidence. This is a low-stakes self-assessment that drives authentic reading engagement rather than passive viewing.	Observable indicator: teacher circulates during the 3-minute individual annotation and notes whether students are engaging critically (multiple stars, crossing out ideas) or superficially (no stars, no changes). Cold-call responses reveal which scientific vocabulary (if any) students already own. Students who used no science vocabulary are identified for targeted support during the paired reading phase.
Observe Phase 📹 **VIDEO:** [Experimental probability] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Properties of Acids (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	PART A — VIDEO (12 minutes): Students watch the curated ARES introductory video on acids and bases. Before pressing play, each student writes three 'watching questions' in their journal — things they hope the video will answer. After the video, students do a 90-second 'brain dump' in their journals: every word, fact, or image they remember, written as fast as possible without editing. PART B — PAIRED READING (20 minutes): Pairs receive the ARES reading: 'Acids, Bases and Your Daily Life in Kenya.' They use the Claim–Support–Question (CSQ) strategy: as they read each section, they record (i) one CLAIM	Before video: 'Open your journals to a clean page. Write THREE questions — things you are hoping this video will explain. You have 90 seconds.' Start timer. Play video without pausing. After video: 'Close your eyes for 10 seconds — picture the main ideas — now open and brain-dump everything you remember. Go. 90 seconds.' Collect brain-dump energy with: 'Hands up — who remembered something a partner did not?' Cold-call 2 pairs. Transition to reading: 'Now you will read with your partner. Use the CSQ table in your journal. Every section: one Claim, one Support, one Question. Underline every bold word and	Claim–Support–Question (CSQ) Reading Strategy — this disciplinary literacy structure trains students to read like scientists: extracting propositions (claims), identifying evidential backing (support), and generating further inquiry (questions). By anchoring the CSQ to the phenomenon cups (lemon juice, Jik, soap, Stoney), the reading is not abstract but directly explanatory of observable evidence students have already encountered. The brain-dump after video activates free recall and makes the subsequent reading feel confirmatory and deepening rather than redundant.	Observable indicator: teacher reviews CSQ tables during circulation — specifically checking whether 'Questions' column entries are surface-level ('What is an acid?') or mechanistic/particle-level ('Why do acids make litmus turn red at the particle level?'). The latter indicates readiness for Lesson 3's deeper investigation. Students generating only surface questions receive a prompt card: 'Your question is a good START — now ask WHY or HOW at the level of tiny particles you cannot see.'

	the text makes, (ii) one piece of SUPPORT (evidence or example) from the text, and (iii) one QUESTION the text raises for them. Key vocabulary terms — acid, base, alkali, neutral, indicator, litmus, phenolphthalein, methyl orange — are printed in bold in the reading and students underline each one and write a brief personal definition in the margin using their own words. The reading includes Kenyan examples: lemon juice (acidic, used in cooking), Stoney Tangawizi (carbonic acid), Jik bleach (alkaline, used for cleaning), liquid soap (alkaline, used for handwashing), baking soda/bicarbonate (alkaline, used in mandazi preparation), cow's milk (nearly neutral), and hydrochloric acid (produced in the stomach for digestion of ugali and other foods).	write what YOU think it means — not the dictionary, YOUR words.' Circulate during reading, crouching to pair level. Prompt stuck pairs: 'What does the text say lemon juice does to litmus? Is that a Claim or Support?' Use targeted questioning for fast finishers: 'Your CSQ table says acids are sour — can you connect that to which cup in our canteen turned RED?' Allow 10 seconds WAIT TIME before providing any scaffold. In the final 3 minutes of reading, say: 'Finish your last CSQ row and circle the one QUESTION you most want answered about particles.'		
Explain Phase 📹 **VIDEO:** Experimental probability Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each pair receives a 'Phenomenon Match Card' — a printed A5 card showing images of the eight canteen cups with their Lesson 1 colours (red, orange, yellow, green, blue, violet, etc.). Using ONLY their new vocabulary and the properties from the reading and video, pairs must: (1) label each cup as ACID, BASE/ALKALI, or NEUTRAL; (2) write one sentence per cup explaining the evidence (e.g., 'Lemon juice turned red/orange with universal indicator — this is characteristic of an acid, which also turns litmus red and tastes sour'). Pairs then compare with another pair and resolve any disagreements using the reading as their referee — they must quote the text. One spokesperson per group of four shares one cup explanation with the class.	Distribute Phenomenon Match Cards and say: 'No new information allowed — only what is in your CSQ table and your brain-dump. Label each cup and write one evidence sentence. 8 minutes. Go.' Circulate and listen for misconceptions — particularly students labelling milk as strongly acidic or tap water as a base. After 8 minutes: 'Compare with the pair next to you. If you disagree on any cup, open the reading and find the sentence that settles it — quote page and paragraph.' After comparison: cold-call one spokesperson per group of four for THREE cups (choose cups that generated most disagreement during circulation — likely milk, tap water, and liquid soap). For each: 'Tell me your label AND your evidence sentence AND which	Phenomenon Re-labelling — by applying new vocabulary directly to the original phenomenon evidence (the cups), students perform the core scientific practice of using models and definitions to explain observations. The use of a printed card (rather than recollection alone) reduces cognitive load and keeps attention on the explanation rather than memory retrieval. The peer-disagreement–resolution–via–text protocol models scientific argumentation from evidence.	Observable indicator: the Phenomenon Evidence Table on the board serves as a public, co-constructed formative artefact. Teacher checks for three specific errors: (1) confusing alkali with acid (reversal of properties); (2) labelling neutral substances (tap water, milk) as one extreme; (3) evidence sentences that use everyday language only ('lemon is sour') without scientific vocabulary ('acids are sour in nature and turn litmus red'). Any of these triggers a whole-class pause-and-correct moment before Phase 4.

	Teacher records class consensus labels on the board in a running 'Phenomenon Evidence Table.'	property from the reading supports it.' Use WAIT TIME of 12 seconds before accepting the answer. After each response, ask the class: 'Agree? Thumbs up, down, or sideways.' Address thumbs-down by asking: 'What in the reading makes you disagree? Quote it.' Build the Phenomenon Evidence Table on the board collaboratively, ensuring all eight cups are labelled and evidenced by end of phase.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Experimental probability] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Properties of Acids (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	Students individually write on a sticky note (or the DQB section of their journal if physical board is unavailable) ONE new question that has emerged from today's video and reading — specifically a question about PARTICLES: what is inside acids and bases at a level too small to see, that causes indicator colours to change? Students post their sticky notes grouped under two column headers the teacher has drawn on the DQB: 'WHAT acids/bases ARE' and 'WHY they behave this way.' The class does a 3-minute silent gallery walk past the DQB. Each student places a small dot sticker on the ONE question they most want the unit to answer. The top three most-dotted questions are read aloud by the teacher and marked as 'Priority Questions' on the board. Students then return to their prediction cards and — using a different colour pen — revise or add to their original prediction, incorporating at least THREE vocabulary terms from today's lesson.	Say: 'You now know the WHAT — acids, bases, neutral, indicators. But look at the Phenomenon Evidence Table — it still doesn't tell us WHY lemon juice makes a red colour and soap makes a blue one. What is happening at the level of particles too tiny to see? Write that question on your sticky note. One question only — make it your best one. 2 minutes.' After posting: 'Gallery walk — silent — 3 minutes. Put your dot on the question you most want answered.' After gallery walk, read the top three questions aloud slowly and say: 'These are the questions that will drive Lessons 3 through 10. Notice that they all require us to think about particles.' Connect explicitly: 'In Lesson 3 we will begin to look inside acids at the particle level — and you will start to see WHY the lemon juice cup turned that brilliant red.' For prediction card revision: 'Use a different colour. Add at least THREE new vocabulary words. Show me that your thinking has grown since Lesson 1.'	Driving Question Board — Priority Dotting — the dot-voting protocol makes student curiosity visible and democratic. By sorting new questions into 'WHAT' vs 'WHY' columns, the teacher scaffolds students' understanding of the difference between descriptive and mechanistic scientific explanation — a key epistemological move in the progression from Lesson 2 to Lesson 3. The prediction card revision in a second colour is a metacognitive growth-tracking tool that builds scientific identity ('I am someone whose thinking changes with evidence').	Observable indicator: teacher reads all sticky-note questions during the gallery walk and categorises them mentally. Questions that are still descriptive ('What colour does litmus turn?') indicate students have not yet moved to mechanistic thinking — these students are flagged for a targeted prompt in Lesson 3. Questions that reference particles, ions, or invisible structure ('What is inside the acid that pushes the indicator to change colour?') indicate readiness for the Arrhenius definition in Lesson 3. Prediction card revisions are collected at the end of class and scanned for presence/absence of vocabulary terms — minimum threshold is 3 terms used correctly in context.
Model Building Phase 📹 **VIDEO:**	Students draw or refine a 'Concept Map for the Phenomenon' in their journals. The central node is the anchoring phenomenon image	Say: 'Your last task today is to build a concept map that connects everything you know so far — but with an honest gap. Draw the eight	Concept Mapping with an Explicit 'Not-Yet' Gap — the concept map consolidates today's vocabulary and property knowledge into a	Observable indicator: concept maps are scanned by teacher in the final 2 minutes of class using a 3-point rubric: (1) EMERGING —

[Experimental probability](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Properties of Acids (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	(eight coloured cups). Branching from the centre are two major nodes: ACID and BASE/ALKALI. From each node, students add at least four property branches using today's vocabulary (e.g., from ACID: sour taste → lemon juice, turns litmus red, turns phenolphthalein colourless, turns methyl orange red, examples: Stoney Tangawizi/stomach HCl). They add a NEUTRAL node and link tap water and milk with appropriate justification. A special dashed-line branch labelled 'NOT YET EXPLAINED' points from both ACID and BASE nodes to a cloud shape labelled 'Particle-level cause — to be investigated in Lesson 3.' This 'not-yet' cloud keeps the inquiry open and models scientific humility.	cups in the centre. Branch out to ACID, BASE, NEUTRAL. Add the properties and Kenyan examples. Then — and this is the most important part — draw a dashed line from both ACID and BASE to a cloud you label: NOT YET EXPLAINED — particle level.' Model the first two branches on the board as a worked example, then step back: 'Finish it on your own. 8 minutes.' Circulate and check for the 'not-yet' cloud — if missing, say quietly: 'Your map looks complete — but is it? What question from the DQB is not yet answered in your map?' Allow 10 seconds WAIT TIME. In the final 2 minutes: cold-call one student to describe their map aloud without showing it — classmates draw what they hear, then compare. Close with: 'The NOT YET cloud is the engine of science. Every lesson from here peels back one more layer of that cloud. See you in Lesson 3 — we go particle-level.'	relational structure, making links between ideas visible. The mandatory 'NOT YET EXPLAINED' cloud is a deliberate epistemic move: it models scientific practice (acknowledging the limits of current understanding) and sustains the inquiry narrative into subsequent lessons. This prevents premature closure of the phenomenon and keeps students in productive intellectual tension.	map has acid and base nodes but fewer than 3 property branches and no 'not-yet' cloud; (2) DEVELOPING — map has both nodes, 3+ property branches, Kenyan examples, but 'not-yet' cloud is present without a specific question; (3) SECURE — map has both nodes, 4+ property branches per node, multiple Kenyan examples, 'not-yet' cloud labelled with a specific particle-level question connected to the DQB. Teacher notes names of EMERGING students for a 5-minute check-in at the start of Lesson 3.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal: (1) VOCABULARY UPTAKE — When students completed the Phenomenon Match Cards, were they using scientific vocabulary accurately in their evidence sentences, or defaulting to everyday language? Which cups generated the most disagreement, and what does that reveal about persistent misconceptions? (2) PARTICLE-LEVEL READINESS — Review the DQB sticky notes: what proportion of student questions were mechanistic (particle-level) versus descriptive? If fewer than 40% were mechanistic, plan an explicit 'zoom in' provocation at the start of Lesson 3 — e.g., 'Imagine you could shrink to the size of one molecule of lemon juice. What would you see around you?' (3) PREDICTION CARD REVISIONS — Did students revise substantively (new ideas, new vocabulary, crossed-out errors) or superficially (a few added words)? Superficial revision may indicate students are not yet experiencing cognitive dissonance between their prior ideas and the new evidence. Consider using one student's authentic revised prediction (anonymised) as a model at the start of Lesson 3. (4) KENYAN CONTEXT CONNECTIONS — Did the Kenyan examples in the reading (mandazi/baking soda, Stoney Tangawizi, Jik, stomach acid/ugali digestion) genuinely resonate with students, or did they feel tokenistic? Adjust Lesson 3 examples based on student energy and engagement during the Explain phase. (5) EQUITY CHECK — Were all students able to access the reading, or did some struggle with vocabulary load? Identify 2–3 students who may need a simplified glossary card or L1 (mother tongue) vocabulary bridge for Lesson 3.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students watched an introductory video and completed a paired reading on acids and bases. They observed that the eight canteen cups from the anchoring phenomenon (lemon juice, chai, Stoney Tangawizi, milk, tap water, baking soda solution, liquid soap, Jik bleach) can be systematically categorised as acid, base/alkali, or neutral based on their colours with universal indicator, litmus, phenolphthalein, and methyl orange.
What did I learn?	Acids are sour, turn litmus red, turn phenolphthalein colourless, and turn methyl orange red. Bases (alkalis in solution) are bitter and slippery, turn litmus blue, turn phenolphthalein pink, and turn methyl orange yellow. Neutral substances (like clean tap water) do not change indicator colours in either direction. Common Kenyan substances — including Stoney Tangawizi, lemon juice, and stomach acid (which helps digest ugali and other foods) — are acids, while liquid soap, Jik bleach, and baking soda (used in mandazi) are bases. Indicators are tools that reveal whether a substance is acidic, basic, or neutral through predictable colour changes.
How does this explain the phenomenon?	Students cannot yet explain WHY acids and bases cause indicator colour changes at the particle level — this requires understanding what dissolved particles in acids and bases actually are, which will be investigated beginning in Lesson 3 through the Arrhenius definition and the concept of ions.

LESSON 3 (40 minutes): What Is Inside an Acid or a Base? — The Arrhenius Definition	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students will be able to explain what makes a substance an acid or a base at the particle level using the Arrhenius definition, and will connect this model to the colour patterns they observed in Lesson 1.
Knowledge	Students will state the Arrhenius definitions of acids (substances that produce H+ ions in aqueous solution) and bases (substances that produce OH- ions in aqueous solution); write ionisation equations for HCl, H2SO4, NaOH, and Ca(OH)2; and distinguish between monoprotic and diprotic acids using these equations.
Skills	Students will write and balance ionisation equations, use correct state symbols in equations, interpret particle-level representations of ionisation, and apply the Arrhenius model to classify the eight liquids from Lesson 1.
Attitudes	Students will appreciate that scientific models — even invisible ones about particles — give us real explanatory power over everyday phenomena, and will develop confidence in using symbolic chemical language to describe what cannot be seen.
Key Inquiry Question	What are acids and bases? (Key Inquiry Question 1) — specifically, what are they made of at the particle level that causes all the observable properties identified in Lesson 2?
Purpose in Storyline	This is the first major EXPLAIN lesson. Students have now observed the colour-change phenomenon (Lesson 1) and named acids and bases using macroscopic properties (Lesson 2). Lesson 3 gives them the particle-level model — the Arrhenius definition — that begins to explain WHY those colour changes occur and WHY acids and bases behave differently. This model will be the conceptual engine for every subsequent lesson in the sequence.
Safety Notes	No chemicals are handled by students in this lesson. All ionisation equations use substances students will later encounter in supervised lab work. Teacher should remind students that HCl and H2SO4 are corrosive acids and NaOH is a corrosive base — these are not to be handled outside a supervised lab context. Jik (sodium hypochlorite solution) from Lesson 1 is referenced but not used.

B. LESSON OVERVIEW
Lesson 3 is a teacher-led sensemaking lesson in which students move from the macroscopic world of colour changes and observable properties (Lessons 1 and 2) into the invisible particle world of ions. The teacher introduces the Arrhenius definitions of acids and bases, guides students through writing ionisation equations for four key substances (HCl, H2SO4, NaOH, Ca(OH)2), and then returns students to the eight cups from the anchoring phenomenon to apply their new model. Students predict and explain why lemon juice turned red and liquid soap turned blue using the language of H+ and OH- ions. The lesson advances the Driving Question Board by adding a new layer of explanation: colour changes happen because of specific ions released into solution.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Arrhenius Acids](#)	Students open their notebooks to their Lesson 1 phenomenon observations and their Lesson 2	Teacher circulates for 90 seconds reading over shoulders without commenting. After 2 minutes	Activating prior knowledge from Lessons 1 and 2 through written prediction. Connecting	Teacher scans student predictions during circulation. Look for: students who already use the word

and Bases - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Arrhenius Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	updated DQB entries. The teacher displays an image of the eight cups from Lesson 1 alongside their indicator colours. Students are given 2 minutes to silently read this prompt written on the board: 'In Lesson 2 you learned that lemon juice is an acid and liquid soap is a base. Both dissolve in water. What do you think is DIFFERENT about what they release into the water — at the level of tiny particles — that makes one turn red and the other turn blue?' Students write a one-sentence prediction in their notebooks using the sentence stem: 'I think the difference is...' They may refer to their Lesson 2 vocabulary list. This prediction is deliberately speculative — students are not expected to know the answer yet.	teacher says: 'I am not going to ask you to share your predictions yet — I want you to hold them. By the end of this lesson, you will be able to look back at what you wrote and judge for yourself whether your particle-level thinking was on the right track. That is what real scientists do — they make predictions and then test them against new information.' Teacher then says: 'Let us now build the model that will let you answer that question with precision.' WAIT TIME: Teacher pauses for 5 full seconds after posing the prompt before allowing any student to begin writing, to signal that thinking time is expected.	macroscopic observations (colour) to the need for a particle-level explanation. Creating cognitive anticipation that motivates engagement with the Arrhenius definition.	'ions' or 'particles' (from Lesson 2 reading), students who conflate colour change with taste or texture (misconception to address in Explain phase), and students who write 'I do not know' — these students need additional scaffolding during the Explain phase. No formal collection — this is a private thinking record.
Observe Phase 📹 **VIDEO:** Arrhenius Acids and Bases - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Arrhenius Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Teacher writes on the board: 'Svante Arrhenius — Swedish chemist, 1887.' Teacher explains briefly: 'More than 130 years ago, a scientist named Svante Arrhenius proposed something radical — that when certain substances dissolve in water, they split apart into charged particles called IONS. This was controversial at the time because no one could see ions. But this model explained so many observations that it became the foundation of acid-base chemistry.' Teacher then shows a simple visual (drawn on the board or projected) of an HCl molecule dissolving in water and splitting into H+ and Cl- ions — represented as labelled circles separating in a beaker of water molecules. Students observe and copy the diagram. Teacher then	Teacher draws both diagrams slowly and deliberately, narrating each step: 'HCl is hydrogen chloride. When I place it in water — aqueous solution — it does not stay as HCl. It ionises. It splits. One part is the hydrogen ion, H+. One part is the chloride ion, Cl-.' Teacher writes H+ and OH- in large text in two separate columns on the board labelled ACID and BASE. Teacher then says: 'Before I give you the formal definition, I want you to look at these two diagrams for 30 seconds and write in your own words what you think the difference between an acid and a base is at the particle level.' WAIT TIME: 30 seconds of writing time before teacher continues. This is not collected — it is a private formulation moment. Teacher then reveals the Arrhenius definitions formally,	Visual particle diagrams make the abstract concept of ionisation concrete and observable. The Kenyan classroom context is honoured by keeping diagrams simple and reproducible without projector dependency — chalk drawings are equally effective. The contrast-and-compare structure (HCl vs NaOH) scaffolds students toward independently constructing the definition before it is stated formally.	Teacher uses the 'What do you NOTICE / What is DIFFERENT' questioning sequence as a live comprehension check. Student responses reveal whether they can identify H+ and OH- as the distinguishing particles. If students say 'one is positive and one is negative' rather than naming the specific ions, teacher redirects: 'Yes — and what are the NAMES of those charged particles? Look at your diagram.' This is a targeted formative probe, not a summative judgment.

	shows a parallel diagram for NaOH dissolving and splitting into Na+ and OH- ions. Students observe and copy. Teacher asks: 'What do you NOTICE is the same about both diagrams?' WAIT TIME: 8 seconds of silence before accepting responses. Students observe that both show a substance splitting into two ions in water. Teacher asks: 'What is DIFFERENT between the two?' WAIT TIME: 8 seconds. Students observe that HCl produces an H+ ion while NaOH produces an OH- ion. Teacher validates: 'That difference — H+ versus OH- — is the entire Arrhenius definition. Let us write it precisely.'	written on the board word by word.		
Explain Phase 📹 **VIDEO:** Arrhenius Acids and Bases - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Arrhenius Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Teacher writes the formal Arrhenius definitions on the board: 'An ACID is a substance that produces hydrogen ions (H+) when dissolved in water (aqueous solution). A BASE is a substance that produces hydroxide ions (OH-) when dissolved in water.' Students copy these into their notebooks under a heading: 'The Arrhenius Model.' Teacher then guides students through writing four ionisation equations step by step. EQUATION 1 — HCl(aq): Teacher says: 'HCl dissolved in water. It gives one H+ and one Cl-. We write: HCl(aq) → H+(aq) + Cl-(aq). This is a monoprotic acid — it gives only ONE H+ per molecule.' Students copy and annotate: monoprotic. EQUATION 2 — H2SO4(aq): Teacher says: 'Sulphuric acid. Look at the formula — H2SO4. How many hydrogen atoms?' Students respond: two. Teacher: 'So it can release TWO H+ ions. We call this DIPROTIC. We write: H2SO4(aq)	Teacher writes each equation on the board slowly, checking charge balance aloud at each step. After writing each equation, teacher says: 'Check your charge balance — the total charge on the left must equal the total charge on the right.' For H2SO4, teacher explicitly models charge counting: 'Left side: H2SO4 is neutral — charge zero. Right side: 2 times H+ is +2, plus SO42- is -2. Total: zero. Balanced.' Teacher uses the Kericho/lime connection deliberately: 'Farmers in Kericho use Ca(OH)2 to reduce soil acidity. Now you can explain what is actually happening — the OH- ions are entering the soil. We will explore that more in Lesson 9.' After the 4-minute writing period, teacher cold-calls two students: one who predicted correctly in the Predict phase, one who did not. Teacher says: 'Compare what you predicted at the start of the lesson with what you can now explain. What changed in your thinking?' WAIT TIME: 6 seconds after each	Guided equation writing builds procedural fluency and conceptual understanding simultaneously. The Kericho agricultural connection grounds an abstract chemical concept in a real Kenyan land-use context that students may have heard of from geography or family experience. The return to the eight cups closes the predict-observe-explain cycle and gives students immediate evidence of their own learning — a metacognitive moment that builds scientific identity. Charge balance as a self-checking routine establishes a scientific habit of mind.	Teacher scans student-written explanations of the lemon juice and soap colour changes during the 4-minute writing period. Success criteria: explanation must name H+ ions (for lemon juice) or OH- ions (for soap) as the cause of the indicator colour change. A response such as 'lemon juice is acidic so it turns red' is insufficient — teacher prompts: 'Can you add the particle-level reason? What is in the lemon juice that makes it acidic?' Students who successfully include H+ and OH- are ready for the Model Building phase. Students who cannot yet make the connection need the sentence frame provided in the Model Building phase.

	→ 2H+(aq) + SO42-(aq). Notice the 2 in front of H+ and the 2- charge on the sulphate ion — the charge must balance.' Students copy, check charge balance, and annotate: diprotic. EQUATION 3 — NaOH(aq): Teacher says: 'Sodium hydroxide — the base in many household cleaners. It releases one OH-. We write: NaOH(aq) → Na+(aq) + OH-(aq).' Students copy. Teacher connects: 'This is why NaOH turns litmus blue — the OH- ion is responsible for basic behaviour.' EQUATION 4 — Ca(OH)2(aq): Teacher says: 'Calcium hydroxide — slaked lime, used by farmers in Kenya to treat acidic soil in tea-growing areas like Kericho. Look at the formula — Ca(OH)2. How many OH- groups?' Students respond: two. Teacher: 'Correct — so it releases TWO OH- ions. We write: Ca(OH)2(aq) → Ca2+(aq) + 2OH-(aq). Notice the 2- charge on calcium balances the two OH- ions.' Students copy and check. Teacher then says: 'Now return to the eight cups. I want you to use the Arrhenius model to explain — in writing — why lemon juice turned RED and why liquid soap turned BLUE. You have 4 minutes. Use the word H+ or OH- in your explanation.' WAIT TIME: 4 full minutes of independent writing before teacher asks for responses.	student speaks before responding or redirecting.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Arrhenius Acids and Bases - Overview] Source: CK-12	Teacher directs students to the class Driving Question Board (physical sticky-note board or designated section of the chalkboard). Teacher reads aloud the driving question: 'What invisible properties of everyday Kenyan liquids cause them to turn such different colours — and why	Teacher models the distinction between 'answerable now' and 'still open' questions by demonstrating with one example of each type before students write. Teacher says: 'Here is an example of a question I can now answer using Arrhenius: Why does HCl make litmus turn red? Answer: because	The DQB update ritual reinforces the storyline structure and gives students agency over their own inquiry trajectory. Categorising questions publicly (answerable now vs. still open) models scientific metacognition and previews the remainder of the lesson sequence. The new header	Teacher reads student DQB entries to gauge the quality of Arrhenius understanding. Well-formed entries that correctly use H+ or OH- show consolidation. Entries that conflate the Arrhenius model with macroscopic properties (e.g., 'Why does H+ taste sour?') indicate that students need the

🔍 Search ARES for similar videos 📖 **HTML:** Arrhenius Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	does it matter for our health, our farms, and our environment?' Teacher says: 'In Lesson 1 you asked questions you could not yet answer. In Lesson 2 you added questions about what acids and bases are made of. Today we have a new tool — the Arrhenius model. I want each of you to take one sticky note (or a small piece of paper) and write ONE of the following: either (A) a question you can now ANSWER using the Arrhenius model, or (B) a new question the Arrhenius model has raised for you that we have not yet answered.' Students write individually for 2 minutes, then one representative from each pair posts their note on the board. Teacher quickly reads three or four posted notes aloud and categorises them: 'This one — Can all acids produce H+ ions? — is a question about ALL acids, not just HCl and H2SO4. We will answer that in Lesson 4. This one — Why does H+ make indicators change colour? — is a great mechanism question we will explore more deeply across the unit.' Teacher adds a new header strip to the DQB: 'Lesson 3: We know H+ and OH- ions cause the colour changes. But what else do these ions DO chemically?'	HCl produces H+ ions in solution and H+ ions cause acidic behaviour. Here is a question the Arrhenius model raises but does not yet answer: Do ALL acids produce exactly the same amount of H+ ions? That is something we do not yet know — and it will matter a lot in Lesson 8 and 9.' Teacher uses the DQB update as a public record of the class thinking trajectory — not as an assessment. Teacher says: 'Every question on this board is evidence that your thinking is growing. Scientists who stop asking questions have stopped doing science.'	strip physically marks progress on the board, making the learning journey visible.	particle-macro link made more explicit in Lesson 4. Teacher notes these students for targeted support. No marks are given — the DQB is a thinking tool, not a test.
Model Building Phase 📹 **VIDEO:** Arrhenius Acids and Bases - Overview Source: CK-12 🔍 Search ARES for similar videos	Students turn to a fresh notebook page headed: 'My Particle Model — Lesson 3.' Teacher provides the following scaffold on the board (students copy and complete): A two-column table with headers ACIDS and BASES. Row 1: 'Arrhenius definition in words.' Row 2: 'Ion produced in water.' Row 3: 'Symbol of ion.' Row 4: 'Example substance from Lesson 1 canteen	Teacher circulates during table completion and uses targeted questioning: 'What ion does NaOH produce? So what would you write in Row 3 for bases?' For students who are stuck on Row 5 (ionisation equation for lemon juice), teacher provides a sentence frame: 'Citric acid produces H+ ions in water, just like HCl. The key Arrhenius idea is not which acid — it is that	The two-column scaffold externalises the conceptual structure of the Arrhenius model and makes the contrast between acids and bases visually explicit. Connecting the abstract model back to the physical cups from Lesson 1 closes the sensemaking loop and rewards students who have been patient with abstract definitions. The explicit framing of	Teacher checks completed tables during circulation using a mental rubric: Does the student correctly place H+ under ACIDS and OH- under BASES? Does the ionisation equation show correct ion separation with state symbols? Does the student connect the ion to the indicator colour from Lesson 2? Students whose tables are complete and accurate are ready

📖 **HTML:** [Arrhenius Acids (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	cups.' Row 5: 'Ionisation equation for example.' Row 6: 'Indicator colour (from Lesson 2).' Students complete the table independently, using their notes. For the ACIDS column, they should use lemon juice (modelled as containing H+ ions, represented by citric acid — teacher notes that lemon juice contains citric acid, not HCl, but both are acids that produce H+ ions). For the BASES column, they should use liquid soap (modelled as containing OH- ions). Teacher circulates and checks. Students who complete the table early are invited to add a third column: NEUTRAL, and to explain what ions are present in tap water (neither H+ nor OH- in excess — this previews the concept of neutrality). Teacher closes the Model Building phase by saying: 'This table is the beginning of your cumulative model of acids and bases. You will add to it in every lesson. By Lesson 10, it will be detailed enough to explain every colour in every cup from Lesson 1 — completely and scientifically.'	H+ is released.' Teacher does not insist on students writing the full ionisation equation for citric acid (which is triprotic and complex) — the conceptual point is sufficient at this stage. Teacher says: 'The Arrhenius model is a simplification — a useful model. Real lemon juice is more complex. But the H+ ion idea is real and powerful. That is what models do: they simplify just enough to give us explanatory power.' WAIT TIME: After posing the extension question about tap water, teacher waits 10 seconds before offering any hint, to allow genuine thinking.	the model as cumulative — growing across 10 lessons — builds metacognitive awareness of the learning journey and motivates continued engagement. The honest acknowledgement that citric acid is more complex than HCl models scientific honesty and appropriate use of simplified models.	for Lesson 4 laboratory work. Students with errors in charge or ion identity receive immediate verbal correction with explanation. Teacher mentally notes the proportion of the class achieving full table completion as a readiness indicator for Lesson 4.

D. TEACHER REFLECTION

After the lesson, consider the following questions: Did students successfully write all four ionisation equations with correct charges and state symbols, or were there systematic errors (e.g., missing the 2 in front of H+ for H2SO4)? Were students able to connect H+ and OH- ions to the indicator colour changes from Lesson 1, or did they treat the Arrhenius definition as an isolated vocabulary exercise? Did the Kericho Ca(OH)2 connection land as meaningful — did any students mention farms or soil from their own experience? Were the DQB entries showing genuine new questions, or were students repeating previously posted questions? If the ionisation equations proved difficult, consider beginning Lesson 4 with a 5-minute equation practice warm-up before the laboratory work. If students showed confusion between the ion produced and the other ion in the formula (e.g., writing Cl+ instead of H+ for HCl), address this explicitly in Lesson 4 by re-emphasising: it is always the H that becomes H+ in an acid, and always the OH that becomes OH- in a base. If time was short, the extension NEUTRAL column in the Model Building phase can be carried forward as the opening warm-up of Lesson 4.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed — using diagrams and equations — that when HCl and H2SO4 dissolve in water they release H+ ions, and when NaOH and Ca(OH)2 dissolve in water they release OH- ions. We revisited the eight cups from Lesson 1 and connected their colours to the presence of these specific ions.
What did I learn?	We learned the Arrhenius definitions: an acid produces H+ ions in aqueous solution; a base produces OH- ions in aqueous solution. We learned to write ionisation equations for HCl, H2SO4, NaOH, and Ca(OH)2, and we learned the terms monoprotic and diprotic.
How does this explain the phenomenon?	We can now explain why the cups in Lesson 1 turned different colours: the cups that turned red or orange contained substances that release H+ ions (like lemon juice and Stoney Tangawizi), while the cups that turned blue or purple contained substances that release OH- ions (like liquid soap and Jik). The Arrhenius model gives us particle-level explanatory power over the phenomenon we observed on Day 1.

LESSON 4 (80 minutes): Differences Between Acids and Bases Using Commercial Indicators	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate and classify acids and bases using commercial indicators (litmus, phenolphthalein, and methyl orange) and to identify patterns in colour changes that distinguish acidic from basic solutions.
Knowledge	Students will know that: (1) Acids turn red litmus paper red and blue litmus paper red, phenolphthalein colourless, and methyl orange red. (2) Bases turn red litmus paper blue and blue litmus paper blue, phenolphthalein pink/magenta, and methyl orange yellow. (3) Neutral substances produce no colour change in litmus and leave phenolphthalein colourless and methyl orange orange-yellow. (4) Common Kenyan everyday liquids — lemon juice, vinegar, baking soda solution, liquid soap, tap water, black tea, milk — can be classified as acidic, basic, or neutral using indicators.
Skills	Students will be able to: (1) Safely set up and conduct an indicator experiment using at least three commercial indicators. (2) Accurately observe, record, and tabulate colour changes for each test solution. (3) Classify solutions as acidic, basic, or neutral based on evidence from multiple indicators. (4) Identify repeating patterns across indicators to draw justified conclusions.
Attitudes	Students will demonstrate: (1) Curiosity and open-mindedness when observing unexpected colour changes in familiar substances. (2) Honesty and accuracy in recording observations without altering results to fit expectations. (3) Responsibility and care during practical work, including safe handling of chemicals. (4) Collaboration and respect for peers' contributions during group investigation.
Key Inquiry Question	What are acids and bases, and what are their chemical properties?
Purpose in Storyline	In Lessons 1–3 students observed the phenomenon (colour-changing cups), made predictions, and built initial models about what might be dissolved in familiar liquids. In Lesson 4 — the first hands-on experiment in the evidence-gathering sequence — students gather direct physical evidence by testing the same category of familiar liquids with three commercial indicators. This is the first lesson where students move from wondering to measuring, replacing guesses with observable data. By the end, the DQB advances: students can now reliably tell acids from bases, which sharpens the next question — what makes acids chemically reactive?
Safety Notes	SAFETY NOTES: (1) Phenolphthalein is prepared in ethanol — keep away from open flames; use dropper bottles only. (2) Vinegar and lemon juice are mild irritants — avoid contact with eyes; wash hands after handling. (3) Liquid soap and baking soda solution are low hazard but should not be ingested. (4) Remind students never to taste any laboratory solution, even if the substance is familiar from home. (5) Dispose of all tested solutions by rinsing down the sink with plenty of water. (6) Broken glassware (spot tiles, test tubes) must be reported immediately and swept up by the teacher — not students.

B. LESSON OVERVIEW

In this 80-minute OBSERVE lesson, student groups of four conduct Experiment 1: they test seven familiar Kenyan liquids — freshly squeezed lemon juice, white vinegar (siki), baking soda solution, liquid handwashing soap solution, clean tap water, black tea (chai), and cow's milk — against three commercial indicators: red litmus paper, blue litmus paper, phenolphthalein solution, and methyl orange solution. Students record all colour changes in a structured results table, classify each substance, and then analyse patterns across indicators to build an evidence-based classification rule. The lesson opens by briefly revisiting the phenomenon and the DQB questions raised in Lessons 1–3, anchors the practical in a real question ("Can we use colour to reliably sort the cups from the canteen?"), and closes with a class consensus table and a DQB update that propels curiosity toward the chemical reactivity of acids in Lesson 5.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Video: pH Scale Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students individually complete a pre-lab prediction table in their science notebooks. For each of the seven test solutions (lemon juice, vinegar/siki, baking soda solution, liquid soap solution, tap water, black tea/chai, cow's milk), they predict: (a) whether the substance is acid, base, or neutral, and (b) what colour they expect each of the four indicators to turn. Students then share predictions with their table partner (Think-Pair) before the teacher takes a quick class poll. Students are encouraged to connect to prior lessons: 'In Lesson 3 we used universal indicator and saw a rainbow — today we use different tools. Will they tell us the same story?'	BEFORE THE LESSON: Prepare seven labelled dropper bottles or small beakers of each test solution. Pre-cut litmus paper strips (red and blue) into sets of 7 per group. Set up spot tiles or white ceramic tiles at each station. OPENING (2 min): Display the phenomenon image (the eight coloured cups) on the board. Say verbatim: 'Last time we updated our DQB — we said we think some of these cups contain acids and some contain bases. Today we find out if we are right. But first — predictions.' PREDICTION TASK (5 min): Distribute the pre-printed prediction table (or instruct students to draw one). Say: 'You have 3 minutes — silently, on your own — fill in your predicted colour for every cell. Use what you already know. Go.' Set a visible 3-minute timer. Do NOT answer questions about the indicators during this time. THINK-PAIR (2 min): 'Turn to your partner. Compare your predictions. Circle any cell where you disagree — those are the most interesting ones.' COLD-CALL POLL (1 min): Ask 3 randomly selected students: 'Tell me ONE prediction you and your partner disagreed on and why.' Accept all answers without evaluating. Say: 'Hold those	STRATEGY — Prediction Table with Accountability: By committing predictions to paper before the experiment, students activate prior knowledge from Lessons 1–3 (universal indicator colours, the phenomenon) and create a personal cognitive stake in the results. Disagreements between partners surface misconceptions the teacher can monitor during the practical. This strategy ensures the observation phase is driven by genuine intellectual curiosity rather than passive procedure-following.	OBSERVABLE INDICATOR: Circulate and scan prediction tables. Look for: (1) Students who predict litmus will change colour the same way for acids and bases — flag for targeted questioning during the practical. (2) Students who leave the neutral row blank — prompt: 'What do you expect water to do to a colour-changing paper?' (3) Students who correctly predict all four indicators for acids or bases — note them as potential explainers in Phase 3. Record 3–4 names on a sticky note for cold-calling during Phase 3.

		disagreements — the experiment will settle them.'		
Observe Phase 📹 **VIDEO:** [Video: pH Scale] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Properties of Acids (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Groups of four conduct the experiment in two coordinated roles that rotate halfway through: two students are 'testers' (handling indicators and solutions) and two are 'recorders' (observing and entering results in the group results table). PROCEDURE: (1) Place a strip of red litmus and a strip of blue litmus side-by-side on the white tile. Add 2–3 drops of Test Solution 1 (lemon juice) to each strip. Observe and record colour immediately and after 30 seconds. (2) Place 5 drops of Test Solution 1 into a small well on the tile. Add 2 drops of phenolphthalein. Swirl gently. Record colour. (3) Repeat with 2 drops of methyl orange. Record. (4) Repeat steps 1–3 for all seven solutions. (5) At the halfway mark (approx. 15 minutes) — testers and recorders swap roles. AFTER DATA COLLECTION: Each group constructs a consensus results table on a large sheet of manila paper or on the shared class whiteboard section allocated to their group. They add a classification column: Acid / Base / Neutral.	LAUNCH (3 min): Demonstrate the procedure for Solution 1 (lemon juice) at the front bench — narrate every step aloud: 'I place the red strip here, I add exactly three drops — not more — I watch immediately, I record what I see, not what I expected.' Emphasise: 'If the colour is not what you predicted, that is excellent data — do NOT change your observation to match your prediction.' CIRCULATION PROTOCOL — visit every group at least twice during 35 minutes: — FIRST ROUND (min 5–15): Check technique. Ask: 'How many drops did you use? Are you reading the colour before it dries?' Prompt struggling groups: 'Hold the litmus strip up to the light — do you see any colour shift, even a small one?' — SECOND ROUND (min 20–30): Focus on the classification column. Ask targeted groups: 'You classified milk as a base — what evidence from the table supports that?' (WAIT 10 seconds.) 'Which indicator gave you the clearest signal?' (WAIT 12 seconds.) CHECK FOR HONEST RECORDING: If you observe a group changing a result, stop the class briefly: 'I want to remind everyone — in real science, an unexpected result is a discovery, not a mistake. Record what you see.' ROLE SWAP PROMPT (min 15):	STRATEGY — Structured Data Table with Role Differentiation: The two-role system (tester / recorder) ensures every student has an active, defined job and prevents one student from dominating the practical. The large visible results table (manila paper) turns individual group data into class-level evidence, enabling pattern recognition across groups in the next phase. The mid-experiment role swap ensures all students practise both observation and recording — NGSS Science and Engineering Practice 3 (Planning and Carrying Out Investigations) and Practice 4 (Analysing and Interpreting Data).	OBSERVABLE INDICATORS DURING CIRCULATION: (1) Correct technique — drops applied without contamination of dropper tip. (2) Accurate colour vocabulary — students should write specific colours ('turned pale pink', 'no change', 'bright red') not vague terms ('changed'). Redirect: 'Be a scientist, not a poet — what exact colour do you see?' (3) Classification accuracy — groups should classify lemon juice and vinegar as Acid, baking soda and soap as Base, and tap water as Neutral (approximately). Flag any group classifying milk as strongly acidic — prompt re-testing. (4) Role engagement — both recorders should be writing; if one is idle, assign them to write the classification justification column.

		Announce: 'Testers and recorders — swap now. The new testers pick up from Solution 5 — tap water.' FINAL 5 MINUTES OF PHASE: Ask each group to place their manila paper/whiteboard results where the class can see. Say: 'Do not compare yet — we do that together in Phase 3.'		
Explain Phase 📹 **VIDEO:** Video: pH Scale Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class constructs a shared Consensus Results Table on the main whiteboard, compiled from all group data. Students compare results across groups, resolve discrepancies (e.g., disagreement on milk classification), and — guided by the teacher — articulate the pattern rule for each indicator. Students then individually complete a 'Pattern Statement' in their notebooks: for each indicator, they write a rule in the form 'An acid turns [indicator] from ___ to ___ ; a base turns it from ___ to ___.' Finally, students revisit their prediction tables from Phase 1, mark each cell correct or incorrect, and write one sentence explaining the most surprising result.	BUILD CONSENSUS TABLE (8 min): Ask one recorder from each group to call out their results for Solution 1. Enter on the board. Highlight agreements in green, disagreements in yellow. For yellow cells: 'Two groups got different results for milk with blue litmus. What could cause that? What would you do to resolve it?' (WAIT 12 seconds — cold-call 2 students who have not spoken yet.) Guide class to agree on the most reliable result or suggest re-testing. ELICIT PATTERN RULES (7 min): Point to the completed consensus table. Ask: 'Look across the ACID row. What do you notice about red litmus every time?' (WAIT 10 seconds.) Cold-call 3 students. Record their language on the board, then sharpen it: 'Scientists say: Acids turn blue litmus red and leave red litmus red — it is already red, so no change.' Repeat for bases. Ask: 'Which indicator gave the most dramatic, easiest-to-read result for bases?' (WAIT 10 seconds.) Elicit phenolphthalein / pink answer. CONNECT TO PHENOMENON (3 min): Return to the image of the	STRATEGY — Consensus Table + Pattern Statement: Building a class-level consensus table from multiple group datasets mirrors authentic scientific practice (aggregating evidence). The 'Pattern Statement' writing task requires students to convert tabular data into a generalisable rule — this is NGSS Practice 6 (Constructing Explanations). Returning to the prediction table activates metacognition: students see where their prior models were incomplete, creating the cognitive discomfort that motivates deeper inquiry in Lesson 5.	OBSERVABLE INDICATORS: (1) Pattern Statements — collect or photograph notebooks. Correct statements must specify BOTH the starting colour and the final colour for each indicator/acid-base combination. Incomplete statements (e.g., 'acids make litmus red' without specifying which litmus) indicate surface-level understanding — follow up in Lesson 5 warm-up. (2) Phenomenon connection responses — listen for students who can name at least two specific cups from the phenomenon and justify their classification using today's data. (3) Surprise sentence — reveals which misconceptions were held and corrected; review these before Lesson 5 to inform Phase 1 planning.

		eight coloured cups. Ask: 'In our phenomenon, which cups were probably showing us acid solutions? Which were showing base solutions? How does what we just did in the lab support your answer?' (WAIT 15 seconds.) Cold-call 2 students. Say: 'So we now have a reliable, reproducible tool to classify these cups. But here is the next puzzle for our DQB — knowing a liquid is an acid tells us what colour litmus turns. Does it tell us what the acid will DO — will it react? Will it corrode? Will it harm our stomach?' Leave this unanswered — it launches Phase 4.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: pH Scale Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each student writes one new question on a sticky note prompted by today's evidence and adds it to the class DQB. The teacher then organises the new questions into two clusters visible to the whole class: CLUSTER A — 'How do we measure HOW acidic or basic something is?' and CLUSTER B — 'What do acids and bases actually DO — how do they react?' Students also review sticky notes from Lessons 1–3 and physically move any questions that today's experiment has now answered into a 'SOLVED' zone on the DQB, writing a brief evidence-based answer on the back.	INDIVIDUAL WRITING (3 min): Distribute sticky notes. Say: 'Today's experiment gave us a tool to classify. But look at our original driving question — it asks about health, farms, and environment. Does knowing if something is an acid or a base answer that? What do you still need to know? Write your question now.' POST AND CLUSTER (3 min): Students post notes. Teacher reads 4–5 aloud and thinks aloud while clustering: 'This one asks how strong an acid can be — that goes with CLUSTER A. This one asks why acids make fizzing — that sounds like reactivity — CLUSTER B.' SOLVED ZONE (2 min): Ask: 'Which questions from Lessons 1 to 3 can we now answer with our experiment data?' Invite 2 volunteers to move a card and read their evidence answer aloud. Affirm: 'This is how science works	STRATEGY — Driving Question Board Curation: The DQB is a living artefact of the class's collective inquiry. Moving answered questions to the 'SOLVED' zone gives students visible evidence of their own growing understanding — a motivational anchor. Clustering new questions into themes previews the logical structure of upcoming lessons (pH scale in Lesson 5–6; chemical reactions of acids in Lessons 7–8), helping students see the coherent arc of the investigation rather than disconnected activities.	OBSERVABLE INDICATORS: (1) Quality of new sticky-note questions — surface-level questions ('What colour does litmus turn?') suggest the student has not yet advanced beyond today's content; probe these students: 'You answered that today — what do you still wonder?' (2) Accuracy of 'SOLVED' zone answers — students who correctly cite today's experimental evidence (naming specific colour changes and specific solutions) demonstrate consolidation of Phase 3 learning. (3) Students who cannot identify any solved question may need a brief one-on-one prompt before Lesson 5.

		— every experiment answers some questions and opens new ones. Our DQB is growing.'		
Model Building Phase 📹 **VIDEO:** Video: pH Scale Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students retrieve their personal anchor model (begun in Lesson 1 and revised in subsequent lessons) — a diagram of the eight phenomenon cups showing what they believe is inside each liquid. Using today's experimental evidence, they annotate or revise their model: (1) Label each cup as ACID, BASE, or NEUTRAL. (2) Draw or describe what 'signal' the indicator molecules are reading — even if they cannot yet explain WHY, they can represent the idea that 'something in the liquid is causing the colour change.' (3) Write one model limitation: 'My model still cannot explain ___.' This limitation directly seeds the next lesson's inquiry.	DISTRIBUTE MODELS (1 min): Return students' Lesson 1 model diagrams (kept in a class folder). Say: 'Look at what you drew in Lesson 1 — before any experiments. You are about to improve it with real data.' GUIDED REVISION (5 min): Say: 'Step 1 — label every cup Acid, Base, or Neutral using today's evidence. Step 2 — draw an arrow from each cup to the indicator colour it would produce. Step 3 — write one sentence: My model still cannot explain ___.' Circulate. Prompt students who only label without drawing: 'Can you add anything to show HOW the indicator is responding? Even a simple arrow or symbol?' CLOSING SHARE (1 min): Ask 2 students to read their 'limitation sentence' aloud. Do NOT resolve the limitation. Say: 'Perfect — those are exactly the questions that will drive Lessons 5 and 6. Hold that curiosity.' Collect models back into the class folder.	STRATEGY — Cumulative Model Revision: Returning to the same model diagram across lessons makes learning visible and cumulative. Students experience their own intellectual growth concretely — crossing out old ideas, adding new labels, noting remaining gaps. The 'model limitation' sentence is a deliberate metacognitive move (NGSS Practice 2 — Developing and Using Models) that prevents premature closure and sustains inquiry motivation. The unresolved limitation also functions as a teacher diagnostic: the range of limitations written reveals the diversity of readiness entering Lesson 5.	OBSERVABLE INDICATORS: (1) Correct acid/base/neutral labelling on all eight phenomenon cups — check against the consensus table from Phase 3. Common error: labelling milk or black tea as strongly basic — these should be weakly acidic or approximately neutral. (2) Presence of a meaningful 'model limitation' sentence — vague responses ('I don't know everything') need prompting; specific responses ('My model cannot explain why lemon juice and Jik bleach both have strong colours but look completely different') signal high readiness for Lesson 5 content on pH scale and acid strength. (3) Visual complexity of model annotations — increased detail compared to Lesson 1 model is evidence of learning progression.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions to improve future delivery:

1. PRACTICAL MANAGEMENT: Did all groups complete testing of all seven solutions within 35 minutes? If not, which solutions were bottlenecks (most likely phenolphthalein — the pale pink can be hard to read)? How could you adjust drop quantities or lighting for next time?

2. HONEST RECORDING: Did you observe any groups altering their results to match predictions? If so, what was the classroom culture signal that produced this behaviour? How will you reinforce scientific honesty as a value in Lesson 5?

3. INDICATOR CLARITY: Were students more confident reading litmus or phenolphthalein? Methyl orange is often the least familiar — note which groups struggled with it and plan a brief 30-second revision at the start of Lesson 5.

4. PATTERN RECOGNITION: During the Explain phase, could students articulate the pattern rule independently or did they need heavy scaffolding? If heavy scaffolding was needed, consider adding a 'colour pattern sort' card activity to Lesson 5's warm-up.

5. DQB QUALITY: Review the sticky notes posted today before Lesson 5. Are questions clustering strongly around pH/strength (good — Lesson 5 is ready) or around taste/smell/safety (signals students are still thinking about properties rather than mechanisms — build a bridge in Lesson 5's phenomenon revisit)?

6. MODEL LIMITATIONS: Read the 'model limitation' sentences collected. List the top 3 most common limitations — these become the explicit learning targets for Lessons 5 and 6. Share anonymised examples at the start of Lesson 5 to validate student thinking and build continuity.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students tested seven familiar Kenyan liquids — lemon juice, vinegar (siki), baking soda solution, liquid soap solution, tap water, black tea (chai), and cow's milk — against red litmus, blue litmus, phenolphthalein, and methyl orange indicators. They recorded specific colour changes for each solution-indicator combination and classified each liquid as acid, base, or neutral.
What did I learn?	Acids turn blue litmus red (red litmus stays red), turn phenolphthalein colourless, and turn methyl orange red. Bases turn red litmus blue (blue litmus stays blue), turn phenolphthalein pink/magenta, and turn methyl orange yellow. Neutral substances produce no litmus colour change, leave phenolphthalein colourless, and leave methyl orange orange-yellow. Common Kenyan everyday liquids can be reliably classified using these indicator colour rules.
How does this explain the phenomenon?	The colour changes produced by commercial indicators in acidic and basic solutions follow a consistent, reproducible pattern. This pattern allows scientists — and students — to classify any unknown solution as an acid, a base, or neutral without tasting or smelling it. The familiar cups from the school canteen phenomenon can now be classified by their indicator behaviour, revealing that invisible dissolved particles in everyday liquids have measurable, testable properties. However, today's experiment tells us only WHAT a substance is (acid or base), not HOW STRONGLY acidic or basic it is, nor HOW it will react chemically — these questions drive the next phase of our investigation.

LESSON 5 (80 minutes): Chemical Properties of Acids: Reactions with Metals and Neutralisation	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate the chemical properties of acids through reactions with metals and bases (neutralisation), and to write balanced molecular and ionic equations for these reactions.
Knowledge	Students will know that: (1) acids react with metals such as zinc and magnesium to produce a salt and hydrogen gas; (2) hydrogen gas is identified by a 'pop' sound when a lit splint is applied; (3) acids react with bases in a neutralisation reaction to produce a salt and water; (4) phenolphthalein indicator is colourless in acids and pink in bases, and can signal the endpoint of a neutralisation reaction; (5) neutralisation can be represented by full molecular and net ionic equations.
Skills	Students will be able to: (1) safely set up and conduct reactions of dilute HCl and H_2SO_4 with zinc and magnesium ribbon; (2) perform a simple neutralisation reaction between HCl and NaOH using phenolphthalein indicator; (3) write balanced molecular equations and net ionic equations for acid–metal and acid–base reactions; (4) record and interpret experimental observations as evidence linked to the anchoring phenomenon.
Attitudes	Students will appreciate the relevance of acid reactions in everyday Kenyan life — from corroded car batteries and metallic roofing sheets in the Rift Valley to the use of antacids for stomach acidity — and develop careful, safety-conscious laboratory habits.
Key Inquiry Question	What are the chemical properties of acids and bases?
Purpose in Storyline	In Lesson 5, students move from identifying acids and bases using indicators (Lessons 1–4) to investigating what acids actually DO chemically. By reacting dilute HCl and H_2SO_4 with metals and with a base (NaOH), students gather direct experimental evidence of acid reactivity. This evidence explains why the corroded battery terminals and rusted metallic structures seen in the supporting phenomenon look the way they do, and it advances the driving question by showing that the 'invisible properties' of acidic liquids have real, visible, and sometimes destructive chemical consequences.
Safety Notes	Dilute HCl and H_2SO_4 are corrosive — students must wear safety goggles and gloves at all times. No concentrated acids to be used. The hydrogen gas splint test must be performed by the teacher as a demonstration or under strict supervision — keep flames away from reagent bottles. NaOH solution is caustic; avoid skin contact. In case of spillage, flush immediately with large amounts of water. Dispose of all reaction products by neutralising and diluting before pouring down the sink. Ensure the laboratory is well ventilated.

B. LESSON OVERVIEW

Students arrive having spent Lessons 1–4 building understanding of what acids and bases ARE — their definitions, indicator behaviours, and positions on the pH scale. In Lesson 5, the investigation shifts to what acids DO. Working in groups of four, students carry out two paired experiments: (Experiment 2a) reacting dilute hydrochloric acid (HCl) and dilute sulphuric acid (H_2SO_4) with small pieces of zinc and magnesium ribbon, observing gas evolution and testing the gas with a lit splint; and (Experiment 2b) performing a simple acid–base neutralisation by adding dilute NaOH dropwise to dilute HCl in the presence of phenolphthalein indicator and watching the endpoint colour change. Groups then write balanced molecular and net ionic equations for both reaction types and connect their findings to the corroded battery terminal and rusted iron roof supporting phenomena. The lesson closes with a Driving Question Board (DQB) update and a revised class model showing that acid reactivity — not just colour-change — is a core 'invisible property' of the liquids in the school canteen cups.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Acid–base titrations Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students are shown two silent images projected on the board: (A) a close-up photograph of corroded, whitish-grey terminals on a car battery (a matatu battery from a Nairobi garage), and (B) a rusted corrugated iron roofing sheet common in Kenyan rural homesteads. The teacher asks: 'We have been studying acids for four lessons. Look at these two images. In your exercise books, write down: (1) Do you think an acid could be responsible for either of these? Which acid? (2) What do you predict will happen if I drop a small piece of magnesium ribbon into dilute HCl — will anything change? Describe what you expect to see. (3) What do you predict will happen when I mix dilute HCl and dilute NaOH together — two clear, colourless liquids?' Students write individual predictions (3 minutes), then share briefly with a shoulder partner (1 minute).	Display both images clearly and allow 30 seconds of silent observation before issuing the prediction task. Circulate and scan written predictions — look for whether students invoke H^+ ions from prior lessons, whether they predict gas evolution, and whether they expect a colour change in the neutralisation. After partner sharing, cold-call THREE students to share predictions aloud: one who predicts vigorous bubbling, one who is unsure, and one who connects the corroded battery to acid. Use exact language: 'Kamau, share your prediction about the magnesium ribbon — what did you write?' [WAIT 10 seconds]. 'Interesting — Wanjiru, yours was different. Tell us.' [WAIT 10 seconds]. Do NOT confirm or correct predictions yet. Say: 'Hold onto these predictions — by the end of this lesson you will be able to judge them yourself.' Record the three shared predictions visibly on a corner of the board under the heading 'Our Predictions Today.'	Prediction journaling with peer comparison (Elicit–Probe strategy). By committing predictions to writing before any instruction, students activate prior knowledge from Lessons 1–4 (H^+ ions, indicator colours, pH values) and create a cognitive anchor point. The visible board record of class predictions creates accountability and a reference students will return to in Phase 3.	Teacher scans written predictions during circulation. Observable indicator of readiness: students should be able to name at least one acid (HCl, H_2SO_4, lemon juice) as potentially responsible for corrosion. Students who write 'nothing will happen' or 'they will just mix' flag a gap in understanding of acid reactivity — the teacher notes these students for targeted cold-calling during the Explain phase.
Observe Phase 📹 **VIDEO:** Acid–base titrations Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12	Students work in pre-assigned groups of four wearing goggles and gloves. Each group has a prepared tray containing: two test tubes in a rack, small pieces of zinc granules and magnesium ribbon (pre-cut to ~2 cm), dilute HCl (1 mol/L) and dilute H_2SO_4 (1 mol/L) in dropper bottles, a wooden splint, a box of matches (teacher-controlled), dilute NaOH (1 mol/L), phenolphthalein indicator solution in a dropper bottle, and a white tile.	Before experiments begin, the teacher conducts a 3-minute safety briefing: 'Goggles ON before you touch any bottle. HCl and H_2SO_4 are corrosive — if any touches your skin, tell me immediately and go to the tap. You do NOT light the splint — I will come to your group.' Demonstrate the splint test on one tube in front of the class before groups begin, saying: 'Watch carefully — you are listening for a sound, not just looking.' Circulate continuously	Structured Observation with a Results Table (Science and Engineering Practice: Planning and Carrying Out Investigations; Analysing and Interpreting Data). The pre-formatted results table — listing variables to observe — scaffolds students' attention so they notice rate of bubbling, gas identity, and colour change as distinct data points rather than a single vague 'something happened.' This structured recording is essential for the	Teacher checks results tables during circulation. Observable indicators: (1) students correctly record 'vigorous bubbling' or 'slow bubbling' rather than just 'bubbles'; (2) students record the splint test result as 'pop sound' or 'no sound' and connect this to a specific gas; (3) students record the phenolphthalein colour change from colourless → pink at the endpoint. Red flag: any group that records no bubbling for magnesium + HCl — the teacher

🔍 Search ARES for similar readings	EXPERIMENT 2a — Acid + Metal: Students add 2 cm^3 of dilute HCl to one test tube and 2 cm^3 of dilute H_2SO_4 to a second test tube. They drop ONE piece of zinc into each and observe for 90 seconds, recording: colour of solution, rate of bubbling (none/slow/vigorous), any heat change (touch test on glass), and any change to the metal. The teacher (or designated group safety officer) then approaches each group's HCl + zinc test tube, lights the splint, and holds it at the mouth of the tube — students listen for and record the result of the splint test. Groups repeat with magnesium ribbon in fresh acid. Students record all observations in a structured results table in their books. EXPERIMENT 2b — Neutralisation: Students add 5 cm^3 of dilute HCl to a clean test tube and add 3 drops of phenolphthalein — they record the colour. They then add dilute NaOH drop by drop using a dropper, swirling after each drop, and record the drop count and colour at every 5 drops until a permanent pink colour is achieved. They record: the colour in acid alone, colour as NaOH is added, colour at the endpoint, and the number of drops of NaOH required.	during Experiment 2a; visit each group at least twice. Use narrating moves: 'Group 3, I can see vigorous bubbling — write down the RATE. Is it slow or vigorous?' [WAIT 10 seconds]. During Experiment 2b, prompt groups who reach the endpoint: 'You've got a permanent pink colour — what does that tell you about the solution now? Write it down before you do anything else.' [WAIT 10 seconds]. Cold-call TWO groups to describe their neutralisation endpoint aloud before moving to Phase 3. Reconnect to the phenomenon: 'Before we explain this — remember those cups in the canteen? Which cup do you think would react the way your acid just did with the metal?' [WAIT 12 seconds]. Accept two responses.	equation-writing in Phase 3.	checks the acid concentration or metal surface and prompts: 'Is your magnesium ribbon shiny? Use fresh ribbon and try again.'
Explain Phase 📹 **VIDEO:** Acid–base titrations Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook)	Students return to their seats. The teacher facilitates a whole-class debrief using a structured 'Claim–Evidence–Reasoning' (CER) framework written on the board. First, the class pools observations from both experiments. The teacher then introduces the particle-level explanation: (1) For acid + metal: dilute HCl contains H^+ and Cl^- ions; zinc/magnesium donate electrons to H^+ ions,	Open the debrief with: 'Before I tell you anything — what did your data show? Group 2, what happened when magnesium went into HCl?' [WAIT 10 seconds]. Build the molecular equation on the board in stages, pausing at each step: 'What are the reactants? What state symbol does the gas get? Check the charges balance.' Cold-call FOUR different students to supply one part of the equation	Claim–Evidence–Reasoning (CER) Framework combined with Equation Scaffolding. CER ensures students do not jump to memorising equations without grounding them in the experimental evidence just gathered. The step-by-step board equation builds the abstract symbolic representation from the concrete observations (bubbling = H_2 gas; pop = H_2; colour change =	Students write the Mg + H_2SO_4 equation independently — teacher circulates and checks for: (1) correct products ($MgSO_4$ and H_2), (2) correct balancing ($Mg(s) + H_2SO_4(aq) \rightarrow MgSO_4(aq) + H_2(g)$), (3) correct state symbols. Observable indicator: at least 70% of students produce a balanced equation without prompting. For the self-assessment sentence, teacher collects or scans 5–6

Source: CK-12 🔍 Search ARES for similar readings	releasing hydrogen gas (H_2) and forming a dissolved salt. The teacher writes the molecular equation on the board step by step: $Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$, and then guides students to write the ionic equation: $Zn(s) + 2H^+(aq) \rightarrow Zn^{2+}(aq) + H_2(g)$. Students copy, then write the equivalent equations for $Mg + H_2SO_4$ in their books independently. (2) For neutralisation: H^+ ions from HCl combine with OH^- ions from NaOH to form water; Na^+ and Cl^- remain as spectator ions. Molecular equation: $HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$. Ionic equation: $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$. Students annotate both equations, labelling the acid, base, salt, and water. The teacher then reconnects to the phenomenon: 'Look at the corroded battery terminal in the photo. Sulphuric acid from the battery has been reacting with the metal terminals — can you now write an equation for what might be happening there? Discuss in pairs for 2 minutes, then write it in your books.' Students attempt: $Fe + H_2SO_4 \rightarrow FeSO_4 + H_2$ (approximate). Finally, students refer back to their Phase 1 predictions and write a one-sentence self-assessment: 'My prediction was correct / partially correct / incorrect because…'	each — say: 'Akinyi, what is the state symbol for $ZnCl_2$ in solution?' [WAIT 10 seconds]. 'Omondi, what gas was confirmed by the splint test?' [WAIT 10 seconds]. For the ionic equation, say: 'I am going to write the full ionic equation. Your job is to find the spectator ions — ions that appear on both sides unchanged. Tell your partner first, then I'll ask.' [WAIT 12 seconds]. Cold-call TWO students. Reconnect to the phenomenon explicitly: 'We said our driving question is about what INVISIBLE properties cause those colour changes in the canteen cups. Today we found a new invisible property — the ability to react with metals and to neutralise. Where does that fit in our answer?' [WAIT 12 seconds]. Accept THREE student responses, writing key words on the board.	neutralisation endpoint), linking macroscopic, submicroscopic, and symbolic levels of representation — a core chemistry literacy practice.	books to gauge the quality of metacognitive reflection — look for students who can articulate WHY their prediction was right or wrong, not just whether it was.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Acid–base titrations Source: Khan	Each group receives two sticky notes (different colours: one for 'New Evidence We Have' and one for 'New Questions We Now Have'). Groups spend 3 minutes completing both notes based on today's experiments. Sample expected contributions — New	Give groups exactly 3 minutes for the sticky note task — use a visible timer. As groups post notes, stand at the DQB and read each aloud before placing it, saying: 'This group says… — does everyone agree this counts as new evidence from today's	Driving Question Board (DQB) — a living anchor chart that makes the progression of the class's collective understanding visible over time. By categorising contributions as 'evidence' versus 'new questions,' students practise the epistemic practice of	Quality of sticky note entries. Observable indicators: 'New Evidence' notes should reference specific reactants and products (not just 'a reaction happened'); 'New Questions' notes should be testable or connectable to the driving question. The teacher

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Evidence: 'Acids react with metals like zinc and magnesium to produce hydrogen gas and a salt'; 'HCl + NaOH neutralise each other — the H^+ and OH^- cancel out'; 'The corroded battery terminal is caused by H_2SO_4 reacting with the metal.' New Questions: 'Do all metals react the same way with acids?'; 'What happens to the salt that is formed — can we taste it?'; 'If neutralisation removes H^+ ions, does the solution no longer change the indicator colour?' Groups post their notes on the class Driving Question Board under the relevant driving question column. The teacher reads 3–4 notes aloud and briefly acknowledges the quality of thinking, connecting each note back to the phenomenon: 'This group asks whether the salt from neutralisation would still change our universal indicator — that is a powerful question that connects directly back to our colour-changing canteen cups.'	experiment?' [WAIT 8 seconds]. Prompt any group whose 'New Evidence' note is too vague (e.g., 'acids react'): 'Can you be more specific — with WHAT did it react, and what was produced?' [WAIT 10 seconds]. Connect new questions to future lessons: 'Your question about whether all metals react the same — that is exactly what we will investigate in Lesson 6 with a reactivity series.' Point to the original phenomenon image: 'We now know two more invisible properties of the acidic cups — corrosiveness and the ability to neutralise. We are getting closer to answering our big question.'	distinguishing what is known from what is still unknown — a key Science and Engineering Practice (Engaging in Argument from Evidence; Asking Questions and Defining Problems).	photographs the DQB after the lesson to track progression across the unit and identify persistent misconceptions (e.g., a note claiming 'acids destroy everything they touch' flags an over-generalisation to address in Lesson 6).
Model Building Phase 📹 **VIDEO:** Acid–base titrations Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students open their 'Running Model' page — a designated page in their exercise books where they have been building a cumulative diagram since Lesson 1. Today they must add two new features to the model: (1) An arrow from the 'Acidic cup' (e.g., lemon juice, Stoney Tangawizi) labelled 'reacts with metals → salt + $H_2(g)$' with a small sketch of bubbling; and (2) An arrow between the 'Acidic cup' and a newly drawn 'Basic cup' (e.g., liquid soap, Jik) labelled 'neutralisation → salt + H_2O, H^+ + OH^- → H_2O'. Students also annotate the corroded battery image (printed on the model sheet or sketched) with the relevant ionic	Before students begin model revision, display the Lesson 1 model (a simple diagram showing the eight canteen cups with their indicator colours) on the projector and say: 'Your model in Lesson 1 only showed colours. Look at how much more you can now explain. Today, add the two new chemical reaction arrows I have described — your model should be getting more powerful with each lesson.' [WAIT 10 seconds for students to review their own model before writing]. Circulate and prompt students who draw only one arrow: 'You have the acid–metal reaction — now where is the neutralisation arrow? What two cups from the	Cumulative Model Revision (Science and Engineering Practice: Developing and Using Models). The running model functions as a visual knowledge-building record across all 10 lessons. Each revision makes students' evolving understanding externally visible and forces integration of new evidence with prior knowledge. Adding directional reaction arrows to the model — rather than simply listing facts — requires students to think causally: the presence of H^+ ions causes both the colour change AND the reactivity with metals.	Teacher checks that models include: (1) a correctly labelled acid + metal reaction arrow with products; (2) a correctly labelled neutralisation arrow with H^+ + OH^- → H_2O noted; (3) the battery annotation with an appropriate equation. Exit ticket (verbal, cold-call): before students leave, the teacher asks THREE students: 'Name ONE chemical property of acids we proved today and the evidence from your experiment that supports it.' Observable indicator of mastery: the student names a specific property (e.g., 'reacts with metals to produce hydrogen gas') AND cites specific evidence (e.g., 'the pop sound

	equation. Finally, students write one sentence at the bottom of the model: 'The colour change in the canteen cups tells us whether H^+ or OH^- ions are present — and today I found out that these ions also determine how a liquid reacts with metals and with each other.' The teacher invites TWO students to share their updated model with the class under the visualiser or by sketching on the board.	canteen, if mixed together, would neutralise each other?' [WAIT 12 seconds]. For the two sharing students, ask the class: 'Does their model correctly show that BOTH H^+ and OH^- ions are involved in what we observed today? Give them a thumbs up if yes, or point to what is missing.' Close the lesson by pointing to the phenomenon image and saying: 'Five lessons in. We can now explain two new invisible properties of these liquids. Our next lesson will push further — we will look at what acids do to carbonates and explore the pH scale in more depth. Keep your questions on the board — we are going to answer them.'		from the lit splint test confirms H_2 was produced'). Students who can only state the property without the evidence are flagged for follow-up at the start of Lesson 6.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Were students able to correctly identify the pop sound in the splint test as evidence of hydrogen gas — or did some groups report no result due to insufficient gas collection or wet splints? Note which groups struggled and check their technique records. (2) Did students successfully distinguish between the molecular equation and the net ionic equation, particularly in identifying spectator ions? If more than 30% of students could not identify Na^+ and Cl^- as spectator ions in the neutralisation, plan a brief 10-minute ionic equation revision at the start of Lesson 6. (3) How rich were the DQB sticky note entries? Did students generate genuinely testable new questions, or did they repeat observations as questions? (4) Did the connection between the corroded battery terminal and the acid + metal reaction land for students — was it referenced spontaneously in Phase 5 model building, or only when prompted? If the phenomenon connection was weak, consider displaying the battery image more prominently at the front of the class for the remainder of the unit. (5) Were safety procedures followed consistently, particularly regarding the splint test and goggles? Note any safety incidents and adjust group compositions or supervision for Lesson 6.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed: (1) vigorous or moderate bubbling when dilute HCl and H_2SO_4 were added to zinc and magnesium ribbon; (2) a 'pop' sound confirming the production of hydrogen gas when a lit splint was held at the mouth of the test tube; (3) the solution in the acid + metal test tube becoming slightly warm; (4) phenolphthalein indicator remaining colourless in dilute HCl, then turning pink permanently at the endpoint when dilute NaOH was added dropwise; (5) the corroded whitish-grey deposits on a car battery terminal and rust on a corrugated iron roofing sheet — both connected to acid reactivity.
What did I learn?	Students learned that: (1) acids react with metals (zinc, magnesium) to produce a salt and hydrogen gas — the balanced molecular equation for Zn + HCl is $Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$ and the net ionic equation is $Zn(s) + 2H^+(aq) \rightarrow Zn^{2+}(aq) + H_2(g)$; (2) hydrogen gas is identified by the 'pop' sound of a lit splint — not by sight alone; (3) acids react with bases in a neutralisation reaction to produce a salt and water — the net ionic equation $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$ shows it is the H^+ and OH^- ions that

	react; (4) phenolphthalein is a reliable endpoint indicator for acid–base neutralisation; (5) the chemical reactivity of acids with metals explains real-world phenomena such as corroded battery terminals and structural metal degradation.
How does this explain the phenomenon?	The colour-changing canteen cups from the anchoring phenomenon now reveal more than just indicator colours — the cups containing acidic liquids (lemon juice, Stoney Tangawizi, black tea) contain H^+ ions that can react with metals and be neutralised by OH^- ions. The H^+ ion is the active particle responsible for both the colour changes seen in Lessons 1–4 AND the chemical reactivities observed today. The corroded car battery terminal is explained by H_2SO_4 (battery acid) reacting with the metal terminals via the acid + metal reaction. Neutralisation explains why antacids (bases) relieve stomach acidity — the OH^- or carbonate ions in the antacid neutralise the excess HCl produced by the stomach, reducing H^+ concentration and relieving discomfort. Every 'invisible property' we uncover brings us closer to a complete answer to the driving question.

LESSON 6 (80 minutes): Acid Reactions with Carbonates and Hydrogen Carbonates — Completing the Acid Reaction Picture	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate and explain the reactions of acids with metal carbonates and metal hydrogen carbonates, produce and test for carbon dioxide gas, write balanced chemical equations for all four acid reaction types, and connect these reactions to real Kenyan contexts including Kericho tea-farm soils and Mombasa car batteries.
Knowledge	Students will know that: (1) acids react with metal carbonates to produce a salt, water, and carbon dioxide gas; (2) acids react with metal hydrogen carbonates to produce a salt, water, and carbon dioxide gas; (3) carbon dioxide turns limewater milky — a standard confirmatory test; (4) balanced chemical equations can be written for all four acid reaction types (metal, metal oxide, metal carbonate, metal hydrogen carbonate); (5) these reactions have practical applications in agriculture, industry, and everyday Kenyan life.
Skills	Students will be able to: (1) safely conduct experiments mixing dilute HCl with marble chips ($CaCO_3$) and with sodium hydrogen carbonate ($NaHCO_3$); (2) bubble the gas produced through limewater and record observations; (3) write and balance chemical equations for acid + carbonate and acid + hydrogen carbonate reactions; (4) consolidate all four acid reaction types into a single coherent summary; (5) update the Driving Question Board with new evidence.
Attitudes	Students will appreciate the relevance of acid–carbonate chemistry to their daily lives and environment — from the fizz in Stoney Tangawizi, to the liming of acidic Kericho tea-farm soils, to the corrosion risk at Mombasa port — developing curiosity, precision, and a sense of responsibility for applying chemical knowledge to community challenges.
Key Inquiry Question	What are the chemical properties of acids and bases?
Purpose in Storyline	In Lessons 4 and 5, students gathered evidence that acids react with metals to produce hydrogen gas and with metal oxides to produce a salt and water only. Today's lesson completes the acid-reaction evidence set: students discover that acids also react with carbonates and hydrogen carbonates, releasing carbon dioxide — the same gas that makes Stoney Tangawizi fizzy and that baking soda releases in mandazi dough. By the end of this lesson, students hold all four acid-reaction patterns and can begin to explain the full 'personality' of the acidic cups from the canteen phenomenon.
Safety Notes	Dilute HCl is corrosive — students must wear safety goggles and avoid skin contact; wash immediately with water if spilled. Marble chips may have sharp edges — handle carefully. Limewater ($Ca(OH)_2$ solution) is mildly irritating; avoid splashing eyes. No naked flames near the reagents. Dispose of all solutions by diluting with water and pouring down the drain. Supervise gas-delivery tubes closely to prevent suck-back of limewater into the acid flask.

B. LESSON OVERVIEW

This is Lesson 6 of 10 in the evidence-gathering sequence for Sub-Strand 3.1 Acids and Bases. Students build directly on Lessons 4 and 5 (acid + metal → hydrogen gas; acid + metal oxide → salt + water) by conducting two new bench experiments: (2c) dilute HCl reacting with marble chips (calcium carbonate, $CaCO_3$) and (2d) dilute HCl reacting with sodium hydrogen carbonate ($NaHCO_3$ / baking soda). In both cases the gas produced is bubbled through limewater to confirm it is carbon dioxide. Students write and balance equations for both new reaction types, then consolidate all four acid-reaction types onto a class summary chart. Two supporting phenomena anchor the learning: (i) the liming of acidic Kericho tea-farm soils with calcium carbonate to raise pH, and (ii) the corrosive damage of sulphuric acid on carbonate-containing materials at Mombasa port. The Driving Question Board is updated with a full acid-reaction summary, bringing students closer to explaining every colour in the canteen cups. The lesson uses a Predict–Observe–Explain cycle, collaborative group work, equation-writing practice, and whole-class consensus building.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Simple probability: yellow marble Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students recall the two acid reaction types studied so far (acid + metal; acid + metal oxide) and their general products. The teacher displays two new substances: a small beaker of marble chips ($CaCO_3$) labelled 'Limestone — same material as Kericho soil lime' and a sachet of Pembe baking soda ($NaHCO_3$) labelled 'Used in mandazi and chapati dough'. Students are told dilute HCl will be added to each. Working in pairs, students write a prediction in their science notebooks: 'What will you see? What gas, if any, will be produced? Write a word equation for what you think will happen.' After 3 minutes, three pairs share predictions with the class; the teacher records key prediction words on the board without correction — these will be revisited in the Explain phase.	Display marble chips and baking soda prominently. Ask: 'We have now seen two ways acids react. Here are two new substances — a carbonate and a hydrogen carbonate. Before we touch them, predict: what will happen when we add dilute HCl?' Allow 3 minutes of pair writing. Cold-call 3 pairs: 'Pair at the front-left — share your word equation.' Record predictions verbatim on the board under the heading 'Our Predictions'. Say: 'Hold onto these — we will come back and check every single one.' Do NOT correct predictions at this stage. Then ask: 'Has anyone ever seen baking soda added to vinegar at home — maybe when making mandazi?' Allow 2–3 hands; ask one student to describe what happened. This activates prior experiential knowledge and creates cognitive readiness for the experiment.	Elicit-and-hold prediction strategy: by making predictions public and recorded, students are cognitively committed to a hypothesis. This creates productive tension that motivates careful observation and drives sensemaking when results confirm or contradict the prediction. Connecting to mandazi-making activates funds of knowledge from students' home contexts.	Teacher scans written predictions for: (a) whether students correctly identify 'gas' as a likely product; (b) whether any students propose CO_2 specifically (indicating prior knowledge to build on); (c) misconceptions such as 'no reaction because carbonates are not metals'. These inform the level of scaffolding needed during the experiment debrief.
Observe Phase 📹 **VIDEO:** Simple probability: yellow marble Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups of four set up two experiments simultaneously at their benches using pre-prepared equipment. EXPERIMENT 2c (Acid + Metal Carbonate): Each group places 5–6 marble chips ($CaCO_3$) into a boiling tube, then carefully adds 10 mL of dilute HCl using a measuring cylinder. A delivery tube is connected to a second boiling tube containing 5 mL of limewater. Students observe: (i) fizzing/effervescence at the marble chips; (ii) the limewater turning milky/white. They record observations in a structured table with columns: Reactants | Observations at	Before experiments begin, demonstrate the limewater test using CO_2 from a straw blown gently into limewater: 'Watch — I am breathing CO_2 into limewater. What do you see? This is what a positive test looks like.' (Allow 10 seconds observation, then cold-call 2 students to describe it.) Distribute pre-labelled equipment kits. Circulate every 2 minutes. At the marble chips bench ask: 'Is the marble chip getting smaller? What does that tell you about what is happening to it?' WAIT 12 seconds. Cold-call one student. At the baking soda bench ask: 'Which experiment fizzes more vigorously	Parallel experiment design with a shared indicator test (limewater): running both experiments side-by-side allows students to directly compare reaction rates and yields, building pattern recognition. The limewater milkiness is a concrete, unambiguous, shared observable — it anchors the abstract concept of 'carbon dioxide produced' in a physical reality all groups witness simultaneously. The 'hold up your tubes' moment creates a whole-class evidence community.	Teacher checks observation tables for: (a) correct recording of milky limewater in BOTH experiments (confirming CO_2 in both cases); (b) correct inference — 'CO_2 is produced'; (c) comparative observation of reaction rate (baking soda faster than marble chips). Any group that does not observe milky limewater is asked to check their delivery tube connection and repeat — this is a real-time experimental troubleshooting opportunity.

	reactant | Observation in limewater | Inference. EXPERIMENT 2d (Acid + Metal Hydrogen Carbonate): Groups place one spatula of $NaHCO_3$ (baking soda) into a second boiling tube and add 10 mL of dilute HCl. Again, gas is bubbled through limewater. Students observe vigorous effervescence — more immediate than with marble chips — and again the limewater turns milky. Groups record all observations and complete the inference column: 'The gas produced turns limewater milky — this is the test for carbon dioxide.' Teacher circulates, prompting with: 'Compare the rate of fizzing — which is faster, 2c or 2d? Why might that be?' Students also note that marble chips gradually get smaller (dissolving), while the baking soda disappears quickly into solution.	— the marble chips or the baking soda? Can you suggest why?' WAIT 15 seconds. Cold-call one student. After both experiments, ask all groups to hold up their limewater tubes: 'Everyone look around — whose limewater is milkiest? What does that tell us?' Use this as a whole-class moment of shared evidence. Ensure all groups complete the observation table before moving on.		
Explain Phase 📹 **VIDEO:** Simple probability: yellow marble Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students return to their observation tables and use their results to write word equations and then balanced symbol equations for both reactions. The teacher guides the class through a structured equation-writing sequence on the board: Step 1 — Word equation for Experiment 2c: hydrochloric acid + calcium carbonate → calcium chloride + water + carbon dioxide. Step 2 — Unbalanced symbol equation: $HCl + CaCO_3 \rightarrow CaCl_2 + H_2O + CO_2$. Step 3 — Balanced: $2HCl + CaCO_3 \rightarrow CaCl_2 + H_2O + CO_2$. The same three-step process is applied to Experiment 2d: hydrochloric acid + sodium hydrogen carbonate → sodium chloride + water + carbon dioxide. Symbol: $HCl + NaHCO_3 \rightarrow NaCl +$	Begin with: 'Let's turn your observations into chemistry language — equations.' Write the three-step process heading on the board. For Step 1, ask: 'What are our reactants — shout them out.' For Step 2, ask: 'Who remembers the formula for calcium carbonate?' WAIT 12 seconds. Cold-call 2 students. Write the unbalanced equation, then ask: 'Is this balanced? Count atoms on each side — tell your partner.' WAIT 15 seconds. Cold-call one pair to explain their counting. Correct and balance live on the board, narrating each step. For the four-reaction summary table, say: 'You have now gathered evidence for four types of acid reaction across six lessons. In your groups, fill in the summary table — you	Three-step equation scaffolding (word → unbalanced symbol → balanced symbol): this progressive approach prevents cognitive overload and makes the logic of balancing transparent. The four-reaction summary table is a consolidation strategy — it forces students to organise and connect across six lessons of evidence, building a coherent schema for 'how acids behave'. The supporting phenomena (Kericho, Mombasa) use the transfer strategy: applying classroom chemistry to novel real-world contexts tests depth of understanding rather than recall.	Collect the four-reaction summary tables (or photograph them). Check for: (a) correct and balanced equations for all four reaction types; (b) correct identification of products — particularly that both carbonate and hydrogen carbonate reactions produce $CO_2 + H_2O$ + salt; (c) quality of Kericho and Mombasa written responses — do students correctly apply the equation to a novel context? Students who cannot yet write balanced equations are identified for targeted support in the next lesson's warm-up.

	H_2O + CO_2 (already balanced). Students then open their notebooks to Lessons 4 and 5 equations and, working in groups, produce a FOUR-REACTION SUMMARY TABLE with columns: Reaction Type | General Word Equation | Example with HCl | Products. The teacher then introduces the two supporting phenomena: (i) 'In Kericho, farmers add powdered limestone ($CaCO_3$) to acidic tea-farm soils — using exactly this reaction to neutralise excess acid and release CO_2. Which equation explains this?' (ii) 'In Mombasa, car batteries contain H_2SO_4. If battery acid spills on a concrete wall (which contains $CaCO_3$), what will happen? Write the word equation.' Students write responses individually, then share with the group. The teacher returns to the canteen cups: 'The Stoney Tangawizi cup — carbonated water contains carbonic acid (H_2CO_3). If we added baking soda to that cup, predict what would happen using today's equations.'	have 7 minutes. Use your notebooks from Lessons 4 and 5.' Circulate and prompt: 'What are the products of acid + metal oxide? How is that different from acid + metal carbonate?' For the Kericho phenomenon, ask: 'Farmers in Kericho buy bags of lime ($CaCO_3$) and spread it on soil. Today you have done that reaction on your bench. What products form in the soil?' WAIT 15 seconds. Cold-call one student. For the canteen cup connection, hold up the Stoney Tangawizi cup from Lesson 1: 'If I added a pinch of baking soda to this, what would happen and why? Connect it to today's equations.' WAIT 15 seconds. Cold-call 2 students.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Simple probability: yellow marble] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Properties of Acids (Flexbook)] Source: CK-12	The class gathers at the Driving Question Board. The teacher reveals a pre-prepared sticky-note cluster labelled 'ACID REACTIONS — EVIDENCE SUMMARY (Lessons 4, 5, 6)'. Students contribute new sticky notes for: (a) 'Acids + carbonates → salt + water + CO_2 (limewater test confirms CO_2)'; (b) 'Acids + hydrogen carbonates → salt + water + CO_2'; (c) 'CO_2 is in Stoney Tangawizi — carbonation!'; (d) 'Liming of Kericho tea-farm soils uses acid–carbonate reaction'; (e) 'Four acid reaction types now identified'. The teacher asks the	Stand at the DQB and say: 'Six lessons in — look at how much evidence we have collected. Let's add today's discoveries.' Hand out sticky notes (two colours: one for 'We now know', one for 'We still wonder'). Ask: 'Which of the eight canteen cups from Lesson 1 do today's reactions help us explain? Talk to your partner — 30 seconds.' Cold-call 3 pairs. For each response, physically move to the canteen cup diagram on the board and draw a connecting arrow to today's evidence cluster. Ask: 'What question does today's lesson raise that we have NOT yet	DQB as an evolving public knowledge artefact: by physically posting and connecting evidence across lessons, students visualise the cumulative growth of their understanding. The 'Still Wondering' notes legitimise ongoing uncertainty and frame upcoming lessons as purposeful. Connecting evidence explicitly to the canteen cups keeps the anchoring phenomenon alive and relevant after six lessons.	Teacher reads all new DQB sticky notes. Indicators of strong understanding: (a) students correctly connect CO_2 production to carbonated drinks; (b) students identify baking soda ($NaHCO_3$) as the hydrogen carbonate that reacted today — same as the canteen cup substance; (c) 'Still wondering' notes reveal genuine curiosity about pH and strength (foreshadowing Lessons 7–8). Misconceptions to watch: students who conflate 'CO_2 produced' with 'the liquid turns colour' (conflating gas production with indicator colour change — to be addressed

🔍 Search ARES for similar readings	class to look at the driving question: 'What invisible properties of everyday Kenyan liquids cause them to turn such different colours — and why does it matter for our health, our farms, and our environment?' and asks: 'Which cups from the canteen do today's reactions help us understand better?' Students discuss in pairs and contribute 2–3 sticky notes connecting today's evidence to specific cups (e.g., Stoney Tangawizi as carbonic acid source, baking soda as hydrogen carbonate). The teacher also posts a 'STILL WONDERING' note: 'We can explain WHAT acids react with — but we still haven't explained WHY the cups turned different colours. What determines whether something is strongly acidic or just slightly acidic? Next lessons…'	answered?' WAIT 15 seconds. Accept 2–3 responses and post them as 'STILL WONDERING' sticky notes. Close by saying: 'Our DQB is growing. We now understand the chemical personality of acids — four reaction types. Coming up: we will find out what makes some acids stronger than others — and why that matters for the colour of that green cup in the middle.'		in Lesson 7).
Model Building Phase 📹 **VIDEO:** Simple probability: yellow marble Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students retrieve their running particle/molecular model drawings from previous lessons (begun in Lesson 2 and updated each lesson). Today they add a new panel labelled 'Acid + Carbonate / Hydrogen Carbonate'. In this panel, students draw: (a) H^+ ions (from HCl) approaching a $CaCO_3$ formula unit; (b) the CO_3^{2-} ion reacting with two H^+ ions to form H_2CO_3, which immediately decomposes into H_2O and CO_2 (shown as bubbles leaving the solution); (c) Ca^{2+} and Cl^- ions remaining in solution as the salt ($CaCl_2$). Students annotate their model with the balanced equation below the diagram. They then write a 'MODEL STATEMENT': 'When an acid reacts with a carbonate, the H^+ ions attack the carbonate ion (CO_3^{2-}), forming unstable carbonic acid which	Distribute previous model drawings (kept in a class folder). Say: 'Add a new panel today — Experiments 2c and 2d. Draw the particles, not just the formula. Show me the H^+ ion finding the CO_3^{2-} ion.' Circulate and prompt: 'Where does the CO_2 come from in your drawing? Show me the carbonate ion breaking apart.' WAIT 12 seconds per group. Cold-call one student to hold up their model and explain it to the class: 'Walk us through your drawing — what is the H^+ doing?' After the model statement is written, ask: 'Compare your model from Lesson 4 (acid + metal) to today's model (acid + carbonate). What is DIFFERENT about the products?' WAIT 15 seconds. Cold-call 2 students. Expect: 'In Lesson 4 we got hydrogen gas; today we got carbon dioxide and water.' Affirm	Cumulative particle model revision: by updating the same running model across all lessons, students build a coherent, growing visual representation of acid behaviour at the particle level. This combats fragmentation — each lesson's chemistry is not isolated but woven into a single explanatory framework. The Lesson 1 canteen-cup annotation strategy (returning to the original phenomenon drawing) reinforces that every lesson is a step toward explaining the original mystery, maintaining narrative coherence across the unit.	Teacher reviews model drawings (collected or photographed). Strong indicators: (a) H^+ ions correctly shown reacting with CO_3^{2-}; (b) CO_2 shown as bubbles leaving the solution (not staying dissolved); (c) salt ions shown remaining in solution; (d) model statement uses particle-level language ('H^+ ions', 'carbonate ion', 'unstable carbonic acid'). Weak indicators: students who draw only the equation without particle-level detail, or who show CO_2 staying in solution — these students need a follow-up question in the next lesson's warm-up.

	breaks down into water and carbon dioxide gas. The metal ion and the negative ion from the acid stay in solution as a dissolved salt.' Finally, students annotate their earlier Lesson 1 canteen-cup drawing, adding a speech bubble to the Stoney Tangawizi cup: 'Contains H_2CO_3 (carbonic acid) — if a base (like baking soda) is added, CO_2 will fizz out.' This connects the model back to the phenomenon.	and consolidate: 'Exactly — the type of gas produced depends on what the acid reacts with. This pattern will help us identify unknown substances — a chemist's superpower.' Close by asking students to annotate the Stoney Tangawizi cup on their Lesson 1 phenomenon drawing.		

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) EQUATION WRITING — Were students able to independently write and balance the two new equations, or did they need heavy scaffolding? If most students could not balance without step-by-step teacher guidance, plan a 10-minute equation-balancing warm-up at the start of Lesson 7. (2) LIMEWATER TEST — Did all groups obtain a clear positive result (milky limewater)? If some groups had unclear results, check whether delivery tubes were properly submerged, whether marble chips were fresh (old chips with surface coatings may react slowly), and whether limewater was freshly prepared. Consider a class demonstration as backup evidence for groups with failed experiments. (3) REAL-WORLD CONNECTION DEPTH — When asked about the Kericho soil liming and Mombasa battery acid scenarios, did students apply the equation correctly to a novel context, or did they only recall the classroom example? If transfer was weak, the next lesson's DQB review should explicitly ask students to generate a third real-world application of their own. (4) DQB QUALITY — Were students' 'Still Wondering' notes focused on the next logical question (acid/base strength, pH scale) or were they off-topic? If the DQB reveals confusion between gas production and indicator colour change, address this explicitly in Lesson 7 before introducing the pH scale. (5) MODEL COHERENCE — Review 5–6 student model drawings. Can students now show all four acid reaction types across their model panels in a coherent, particle-level narrative? Students who are falling behind on the running model may need a peer-support pairing in Lesson 7.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed that: (1) when dilute HCl was added to marble chips ($CaCO_3$), vigorous fizzing/effervescence occurred at the marble surface, the marble chips gradually dissolved and became smaller, and the gas produced turned limewater from clear to milky white; (2) when dilute HCl was added to baking soda ($NaHCO_3$), very rapid effervescence occurred (faster than with marble chips), the powder dissolved quickly into solution, and the gas again turned limewater milky white; (3) in both cases the limewater test confirmed the gas was carbon dioxide (CO_2).
What did I learn?	Students learned that: (1) acids react with metal carbonates to produce a salt, water, and carbon dioxide gas — e.g., $2HCl + CaCO_3 \rightarrow CaCl_2 + H_2O + CO_2$; (2) acids react with metal hydrogen carbonates to produce a salt, water, and carbon dioxide gas — e.g., $HCl + NaHCO_3 \rightarrow NaCl + H_2O + CO_2$; (3) the standard test for CO_2 is to bubble the gas through limewater — a milky precipitate of calcium carbonate ($CaCO_3$) confirms CO_2; (4) these reactions explain why Kericho farmers add lime to acidic soils (neutralises excess acid, releases CO_2) and why battery acid (H_2SO_4) can corrode carbonate-containing materials at Mombasa port; (5) combining evidence from Lessons 4, 5, and 6, acids react with four types of substances — metals, metal oxides, metal carbonates, and metal hydrogen carbonates — each producing a salt, and each producing specific additional products (H_2 gas, water, or water + CO_2).

How does this explain the phenomenon?	Students can now explain that: (1) H^+ ions from the acid attack the carbonate ion (CO_3^{2-}), forming unstable carbonic acid (H_2CO_3) which immediately decomposes into water and carbon dioxide gas — this is why fizzing is observed; (2) the more reactive or finely divided the carbonate, the faster the reaction (explaining why powdered $NaHCO_3$ reacts faster than solid marble chips); (3) the CO_2 gas produced during the Stoney Tangawizi carbonation process is related to the same chemistry — carbonic acid (H_2CO_3) dissolved under pressure; (4) returning to the canteen phenomenon: the acidic cups (lemon juice, Stoney Tangawizi, black tea, and the soft drink) would all react with carbonates and hydrogen carbonates — explaining both their indicator colours and their real-world chemical behaviour in contexts like digestion, agriculture, and industry.

LESSON 7 (80 minutes): Consolidation of Chemical Properties of Acids and Bases — Building the Reaction Summary Model	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To consolidate and connect all four chemical properties of acids (reactions with metals, carbonates, metal oxides, and bases) and the key properties of bases, linking every reaction back to the Arrhenius definition of acids and bases so that students can explain the colour-changing phenomenon using H^+ and OH^- as the active chemical agents.
Knowledge	Students will know: (1) the four characteristic chemical reactions of acids — with reactive metals, metal carbonates, metal oxides, and alkalis (neutralisation) — with generalised word and symbol equations and observable products; (2) the chemical properties of bases — reaction with acids (neutralisation), effect on indicators, and reaction with ammonium salts; (3) that H^+ ions are the active agent in all acid reactions and OH^- ions are the active agent in base reactions, consistent with the Arrhenius definition; (4) the identity and use of gas tests for hydrogen and carbon dioxide as evidence of acid reactions.
Skills	Students will be able to: (1) write and balance generalised chemical equations for all four types of acid reactions and for neutralisation; (2) construct a structured graphic/summary model organising reaction types, equations, products, and observable evidence; (3) apply the Arrhenius definition to explain why H^+ ions cause colour changes in indicators and drive the observed reactions; (4) compare and contrast the chemical properties of acids and bases using evidence gathered in previous lessons; (5) conduct and interpret the burning splint and limewater gas tests.
Attitudes	Students will appreciate: (1) the value of systematic evidence consolidation — that science builds understanding incrementally and that each experiment contributes a piece of a larger picture; (2) the relevance of acid and base chemistry to Kenyan daily life — from digestion of ugali and sukuma wiki, to soil pH in Rift Valley farms, to safe use of Jik bleach and liquid soap; (3) the importance of accuracy and precision in recording chemical equations and observations.
Key Inquiry Question	What are the chemical properties of acids and bases, and how does the Arrhenius definition — H^+ ions for acids, OH^- ions for bases — explain every colour change and every reaction we have observed so far in this unit?
Purpose in Storyline	By Lesson 7, students have gathered experimental evidence across six prior lessons: they have observed indicator colour changes, measured pH, and conducted individual acid reactions. This lesson is the mid-unit consolidation point. Students now synthesise all gathered evidence into a single graphic summary model, draw explicit connections between the Arrhenius definition and every observed reaction, and update the Driving Question Board to reflect their growing explanatory power. By the end of this lesson, students should be able to account for WHY each of the eight canteen cups turned its specific colour — not just describe it — using H^+ ion concentration, the pH scale, and the chemical properties that follow from Arrhenius theory.
Safety Notes	No new hazardous chemicals are introduced in this lesson. If gas tests are demonstrated: (1) Dilute HCl and zinc/marble chips — ensure adequate ventilation; keep acid volumes below 10 mL; teacher demonstrates or supervises closely. (2) The burning splint test for hydrogen — keep all other flammable materials away; use a long splint; teacher performs demonstration only. (3) Limewater for CO_2 test — non-hazardous; students may handle under supervision. (4) Remind students that Jik (sodium hypochlorite) referenced in the phenomenon is corrosive — never taste or handle without gloves. All MSDS protocols for dilute acids and bases remain in force from previous lessons.

B. LESSON OVERVIEW

This is Lesson 7 of 10 in the Acids and Bases unit and serves as the mid-unit EXPLAIN + MODEL BUILDING consolidation lesson. Students do not conduct new experiments; instead, they synthesise and make sense of all evidence gathered in Lessons 1–6. The lesson opens by returning to the anchoring phenomenon — the eight colour-changing canteen cups — and challenging students to now explain, not just describe, each colour using the Arrhenius definition. Through a structured Gallery Walk of student-generated evidence from previous lessons, a teacher-facilitated whole-class discussion, collaborative graphic model construction (the Acid Reactions Summary Model), and a Driving Question Board update, students build an integrated understanding of the chemical properties of acids and bases. Gas tests for H_2 and CO_2 are revisited via a brief teacher demonstration to cement the link between observable evidence and chemical equations. The lesson closes with students writing an evidence-based explanatory paragraph connecting the phenomenon to the Arrhenius definition — a key formative writing task.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Arrhenius Acids and Bases - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students are shown a projected image of the eight colour-changing canteen cups from the anchoring phenomenon (lemon juice = red, chai = orange, milk = yellow-green, Stoney Tangawizi = orange-red, tap water = green, baking soda = blue, liquid soap = blue-violet, Jik = purple). Each student silently and individually completes a 'Then vs. Now' two-column card: in the LEFT column (labelled 'Lesson 1 — I thought…') they write their original prediction for why the cups turned different colours. In the RIGHT column (labelled 'Lesson 7 — Now I think…') they write their current best explanation, using as much Arrhenius and pH vocabulary as they can. Students then share their 'Now I think' explanation with a shoulder partner for 2 minutes before three students are cold-called to share with the class.	Display the phenomenon image on the projector and say: 'Look carefully at these eight cups. You first saw these in Lesson 1 and you made predictions. Today is Lesson 7 — you have done experiments, measured pH, observed reactions. I want to know: has your thinking changed? Take your Then vs. Now card and in SILENCE — no talking — fill in both columns. You have 4 minutes.' [Set a visible 4-minute timer. Circulate and read cards silently — note 2–3 cards that show strong Arrhenius reasoning and 1–2 that still use pre-scientific language, for targeted cold-calling.] After 4 minutes: 'Now share your RIGHT column — your Lesson 7 thinking — with your shoulder partner. You have 2 minutes. GO.' [Wait 2 minutes. Circulate and listen for key vocabulary: H^+, OH^-, pH, concentration, Arrhenius.] Cold-call 3 students: 'Amina, tell the class your Lesson 7 thinking for the red cup — lemon juice.' [WAIT 12 seconds after each cold call.] After each response, ask the class: 'Does anyone want to add to what Amina said? Raise your hand.' Targeted probing question: 'WHY does a higher concentration of H^+ ions produce red and not	THEN vs. NOW Metacognitive Card — This strategy makes student thinking visible and forces explicit comparison between prior conceptions and current evidence-based understanding. It activates all prior learning from Lessons 1–6 as a cognitive anchor for the consolidation work ahead. By sharing with a partner before whole-class cold-call, all students rehearse their reasoning, increasing confidence and the quality of whole-class contributions.	Teacher reads cards during circulation to identify: (A) students who can correctly link H^+ ion concentration to low pH and red/orange indicator colours — these students are ready for model construction; (B) students who still describe only the colour without a causal explanation — these students need targeted questioning during Phase 3; (C) students who confuse strength with concentration — flag for explicit address in Phase 3 discussion. Observable indicator: at least 60% of cold-called responses should include 'H^+ ions' or 'Arrhenius' language by the end of Phase 1.

		green? Connect it to the pH scale.' Do NOT correct or confirm yet — hold the tension: 'We are going to spend today's lesson building the explanation that can answer that question fully.'		
Observe Phase 📹 **VIDEO:** Arrhenius Acids and Bases - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Six 'Evidence Stations' are set up around the classroom — one for each of the previous lessons. Each station displays: a printed summary of the key observation/data from that lesson (e.g., Station 3: 'Reaction of HCl with zinc metal — observations, word equation, symbol equation, gas test result'), a large blank sticky note pad, and the prompt: 'What did we OBSERVE? What did we CONCLUDE? What question does this raise?' Students work in groups of 4 and rotate through all six stations (5 minutes per station, 2 stations visited per group in parallel to save time — each group visits at least 3 stations directly and reviews the other 3 via a whole-class share-out). At each station, groups add one sticky note summarising the key evidence in their own words. Stations cover: (1) Indicator colours and pH values of the 8 canteen liquids; (2) Arrhenius definition — H^+ and OH^-; (3) Acid + metal reactions (HCl + Zn, HCl + Mg); (4) Acid + carbonate reactions (HCl + marble chips/sodium carbonate); (5) Acid + metal oxide and acid + alkali (neutralisation); (6) Properties of bases — indicators, taste/touch, reaction with acids.	Before the Gallery Walk, explain clearly: 'Each station holds evidence from one of your previous lessons. Your job is NOT to re-learn it — your job is to HARVEST it. Read, discuss with your group, and post one sticky note answering the three prompts. Be specific — write equations, name gases, state observations. You have 5 minutes per station. I will call time.' [Ring a bell or clap to signal rotation.] Circulate actively. At Station 3 (acid + metal), prompt: 'What is the name of the gas produced? How do you test for it? Write the generalised equation on your sticky note.' At Station 4 (acid + carbonate), prompt: 'What two gases could you test for here? How do you distinguish them? Bongani, tell me the limewater test.' [WAIT 10 seconds.] At Station 6 (base properties), push: 'You wrote that bases are slippery. WHY are they slippery — what ion causes this? Connect it to Arrhenius.' After the rotation, lead a 5-minute whole-class share-out: one group reports per station; others add or challenge. Record key equations and observations on the board as students speak — do not pre-write them.	EVIDENCE GALLERY WALK with Sticky Note Harvest — This strategy externalises the distributed knowledge students have built across six lessons and makes it collectively visible in one physical space. By requiring groups to write equations and observations in their own words, it checks for understanding rather than recall. The rotation structure ensures every student engages with every category of evidence before the model-building phase, preventing gaps in the summary model.	Teacher checks sticky notes at each station during rotation for: (A) correct generalised equations with correct products (e.g., Acid + Metal → Salt + Hydrogen; Acid + Carbonate → Salt + Water + Carbon dioxide); (B) correct gas test descriptions (burning splint squeaky pop for H_2; limewater turns milky for CO_2); (C) correct Arrhenius language at Station 2 and Station 6. Groups with incomplete or incorrect sticky notes are flagged — teacher pulls them aside during Phase 3 for targeted support. Observable indicator: at least 4 of 6 stations should have sticky notes with correct equations and at least one Arrhenius reference.
Explain Phase 📹 **VIDEO:** Arrhenius Acids and Bases -	The class reconvenes. The teacher facilitates a structured whole-class discussion to build a consolidated understanding of all	Open the discussion: 'From your Gallery Walk, you have harvested evidence from six lessons. Now we build the complete picture	SOCRATIC SUMMARY TABLE CONSTRUCTION — Rather than transmitting information via lecture, the teacher uses targeted Socratic	Teacher checks tables during construction by circulating during writing moments. Key indicators: (A) All four acid reaction types are

[Overview](#) Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Properties of Acids (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	four acid reaction types and base properties. Students have their exercise books open and construct a structured 'Acid Properties Summary Table' as the discussion proceeds — they do NOT copy from the board; instead, they complete the table by contributing answers during the discussion. The table has columns: Reaction Type | Reactants | Products | Generalised Word Equation | Generalised Symbol Equation | Observable Evidence | Gas Test (if applicable). Midway through the discussion, the teacher performs a brief demonstration: HCl + marble chips (CO_2 test with limewater) and HCl + zinc (H_2 test with burning splint). Students observe and record in their tables. The discussion then pivots to base properties and closes with an explicit Arrhenius connection: 'In EVERY acid reaction, what is the particle actually doing the reacting? In EVERY base reaction, what particle is responsible?' Students answer individually in writing before the cold-call.	together. I am NOT going to lecture — YOU are going to build this table by answering my questions. Every answer goes into your table. Let's begin.' [Work through each row of the table via Socratic questioning. For each reaction type, ask:] (1) 'Who can give me the generalised WORD equation for acid plus metal? Wanjiru?' [WAIT 12 seconds. If incorrect, say: 'Close — who can refine that? Kamau?'] (2) 'What is the symbol equation? Include state symbols.' [Write on board only AFTER a student has given it verbally.] (3) 'What is the observable evidence? What would you SEE and HEAR?' (4) 'What is the gas test? Describe it step by step.' Mid-lesson demonstration: 'Watch carefully — I am going to test for carbon dioxide using limewater. Tell me what you expect to see BEFORE I do it.' [WAIT 10 seconds for written prediction.] Perform demonstration. 'What did you observe? Does it match the equation? Ochieng, explain WHY the limewater turns milky.' [WAIT 12 seconds.] H_2 demonstration: 'I will now collect the gas from HCl and zinc and apply a burning splint. Tell me the expected result.' [Perform safely. Discuss.] Arrhenius pivot: 'Now look at all four rows of your table. In every single reaction, what particle from the acid is doing the work? Write your answer — one sentence — before I ask anyone.' [Give 30 seconds writing time.] Cold-call 4 students. Consolidate: 'H^+ ions are the active agent in ALL acid reactions. This is the power of the Arrhenius definition — it unifies everything in your table under one	questioning to draw the summary table content from students' own prior knowledge and evidence. This approach (1) reinforces that students already hold the knowledge from previous lessons, (2) requires active retrieval rather than passive copying, and (3) creates a student-generated reference artefact. The live demonstration re-anchors abstract equations to observable physical evidence, preventing the equations from becoming disconnected symbols. The explicit Arrhenius pivot at the end provides the unifying conceptual thread that connects all four reaction types and both gas tests into a single explanatory framework.	represented with correct word AND symbol equations; (B) Both gas tests are correctly described; (C) The sentence 'H^+ ions are the active agent' (or equivalent) appears in students' books; (D) Students can connect the gas test results back to specific equations without being told. Exit check: teacher asks 3 random students to point to a row in their table and read the full equation aloud — all three should be correct. If fewer than 2 are correct, the teacher pauses and re-teaches that equation before moving to Phase 4.

		idea.' Then do the same for bases: 'What particle is the active agent in base reactions? Write it. OH^- ions. This is why bases neutralise acids — OH^- removes H^+.' Close: 'Now look back at the eight canteen cups. The red ones — lemon juice, Stoney Tangawizi — are red because they have HIGH H^+ ion concentration. The purple one — Jik — is purple because it has HIGH OH^- concentration. Every colour is caused by an ion. You can now explain the phenomenon.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Arrhenius Acids and Bases - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Properties of Acids (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Students return to the class Driving Question Board, which has been displayed since Lesson 1. The DQB is organised into three columns matching the three KICD Key Inquiry Questions: (1) 'What are acids and bases?', (2) 'What are the chemical properties of acids and bases?', (3) 'How is the strength of acids and bases determined?' Students individually write two types of contributions on different coloured sticky notes: BLUE notes = 'We can now answer this' (with a brief answer); YELLOW notes = 'This new question has emerged'. Each student writes at least one blue note and one yellow note. Groups then place their notes on the DQB collaboratively, clustering similar answers and new questions. The teacher facilitates a 5-minute whole-class review: 'What can we now fully explain? What still puzzles us?' The teacher explicitly annotates which questions Column 2 of the DQB can now be considered 'answered' based on today's lesson, and flags unresolved questions in Column 3 as targets for Lessons 8–10.	Transition: 'Take out two sticky notes — blue and yellow. We are going to update our Driving Question Board. In Lesson 1, we had many questions and no answers. Today is Lesson 7. What CAN we answer now? On your BLUE note, write one thing you can now fully explain about the phenomenon — be specific, use equations or ion names. On your YELLOW note, write one question that today's lesson raised for you — something you still do not fully understand.' [Give 3 minutes for writing. Circulate and read notes.] 'Now come to the board — column by column — and place your notes. If your note is similar to one already there, stack it on top.' [Allow 2 minutes for placement.] Facilitate review: 'Look at Column 2. We now have [count] blue notes saying we can explain the chemical properties. Let me read one: [read a strong student note]. Class — do you agree this question is answered? Thumbs up/thumbs down.' [WAIT 8 seconds.] 'Column 3 — strength of acids and bases. I see yellow notes with questions about strong vs. weak acids, and about pH and	DRIVING QUESTION BOARD UPDATE with Colour-Coded Sticky Notes — The DQB is the unit's living knowledge map. By distinguishing 'answered' questions (blue) from 'new questions' (yellow), this strategy (1) makes learning progress visible and motivating; (2) prevents students from treating each lesson as isolated; (3) ensures the driving question remains the north star of all learning; and (4) generates student-owned questions that the teacher can use to preview and motivate upcoming lessons. The public, collaborative nature of the DQB creates shared intellectual accountability.	Teacher reads all blue notes during placement for accuracy. Key indicators: (A) Blue notes correctly reference H^+ or OH^- ions, specific reaction types, or the Arrhenius definition — not just colour observations; (B) Yellow notes reflect genuine conceptual puzzlement (e.g., 'Why does strong acid have more H^+ than weak acid if both are acids?') rather than recall gaps; (C) Column 2 of the DQB should have a majority of blue notes, indicating the class as a whole judges the chemical properties questions as answered. If fewer than half the class places a blue note on Column 2, the teacher notes this as a gap requiring revisiting at the start of Lesson 8.

		concentration. These are exactly what Lessons 8 and 9 will address. You are asking exactly the right questions.' Annotate the DQB with a marker: draw a green tick next to Column 2 questions now resolved; draw a star next to Column 3 questions targeted for upcoming lessons.		
Model Building Phase 📹 **VIDEO:** [Arrhenius Acids and Bases - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Properties of Acids (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Students construct their mid-unit Graphic Summary Model — a structured visual that synthesises all chemical properties of acids and bases onto one A4 page in their exercise books. The model uses a central node labelled 'ACID (produces H^+ in water — Arrhenius)' with four radiating branches, one for each reaction type, each showing: the reaction type name, generalised word equation, generalised symbol equation, specific Kenyan-context example, observable evidence, and gas test (where applicable). A separate node for 'BASE (produces OH^- in water — Arrhenius)' shows properties branching out similarly. A connecting arrow between the two nodes is labelled 'NEUTRALISATION: $H^+ + OH^- \rightarrow H_2O$' and annotated with 'This is why antacids relieve stomach acid — OH^- from the base neutralises H^+ in gastric acid.' At the bottom of the model, students draw the eight canteen cups from the phenomenon and annotate each cup with: the dominant ion (H^+ or OH^-), the approximate pH, and the indicator colour. Students work individually for 12 minutes, then share models with a partner for peer feedback using a 'Two Stars and a Wish' protocol before a whole-class share of two volunteer	Distribute A4 plain paper (or direct students to a full page in their exercise book). 'You are going to build your Acid and Base Summary Model. This is YOUR model — it should be in your own words, your own layout, but it MUST include everything we have discussed today. Use the checklist on the board as your guide.' [Display checklist: ✓ Central Arrhenius node for acid; ✓ 4 branches — one per reaction type; ✓ Word AND symbol equation on each branch; ✓ Kenyan example on each branch; ✓ Observable evidence; ✓ Gas test where applicable; ✓ Base node with properties; ✓ Neutralisation arrow; ✓ 8 canteen cups annotated with ion, pH, colour.] 'You have 12 minutes. Work in silence. I am circulating.' [Circulate actively. Specific interventions: (1) If a student cannot write the symbol equation for acid + carbonate: 'Look at your summary table from Phase 3. What are the products? Write them one by one.' (2) If a student omits the Kenyan example: 'Lemon juice is a Kenyan example for acid + carbonate — do you remember what happens when you squeeze lemon on a piece of marble? Write that.' (3) If a student's canteen cup annotations are only colours: 'Add the ion. WHY is lemon juice red?	GRAPHIC SUMMARY MODEL with Peer Feedback (Two Stars and a Wish) — The graphic model is the core knowledge-integration artefact of the unit. Research on science learning shows that students who construct visual-structural models of their knowledge develop more durable and transferable understanding than those who only read or copy notes. By requiring both word equations AND symbol equations, AND Kenyan context examples, AND observable evidence on each branch, the model forces multi-representational understanding. The 'Two Stars and a Wish' peer feedback protocol introduces a collaborative quality-assurance step that improves model accuracy and completeness without relying solely on the teacher. The explicit instruction that 'this model will keep growing' creates a sense of cumulative intellectual purpose that motivates continued engagement.	Teacher evaluates models during circulation and via the two projected models using three criteria: (1) COMPLETENESS — all four acid reaction branches present with equations; base node present; neutralisation arrow present; all 8 cups annotated. (2) ACCURACY — equations are balanced; correct products named; correct gas tests; correct Arrhenius nodes; correct ion annotations on cups. (3) CONNECTIVITY — the model shows explicit links between Arrhenius definition and observable evidence (not just a list of facts). Observable indicator: by end of Phase 5, at least 80% of students should have a model with all four acid branches and the Arrhenius central node correctly labelled. Students who are missing more than two branches are identified for follow-up support at the start of Lesson 8. The explanatory paragraph written in the next 5 minutes (see Teacher Reflection) serves as the lesson's summative formative writing task.

	models via the visualiser.	Because it has high concentration of which ion? Write H^+ next to the lemon juice cup.'] After 12 minutes: 'Share your model with your shoulder partner. Give two stars — two things they did well — and one wish — one thing they should add or correct. You have 3 minutes.' [Circulate and listen to feedback exchanges.] 'Who would like to share their model with the class? I need two volunteers.' [Project two models via visualiser. For each, ask the class: 'What is strong about this model? What would you add?' Give 2 minutes per model.] Close: 'Keep this model. In Lesson 9, you will add the pH scale and acid strength to this model. It will keep growing. By Lesson 10, this model will explain EVERYTHING about those eight cups.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions before planning Lesson 8:

1. ARRHENIUS INTEGRATION: Could students articulate — in their own words, without prompting — that H^+ ions are the active agent in acid reactions and OH^- ions are the active agent in base reactions? Or did they consistently need the teacher's prompt to make this connection? If the latter, plan a brief 5-minute retrieval activity at the start of Lesson 8 where students write one sentence connecting each reaction type to the Arrhenius definition.

2. EQUATION ACCURACY: Which of the four acid reaction types caused the most errors in the summary table and the graphic model? (Common weak points: acid + metal oxide equation; balancing the acid + carbonate equation with three products.) Prepare one targeted worked example for Lesson 8 addressing the most frequently missed equation.

3. DQB QUALITY: Were the yellow 'new questions' on the DQB genuinely conceptual (relating to acid/base strength, pH, concentration) or were they superficial (repeating questions already answered)? High-quality yellow notes indicate students are ready for the strength and pH content of Lessons 8–9. Superficial notes suggest the class needs more consolidation before advancing.

4. KENYAN CONTEXT CONNECTIONS: Did students spontaneously use Kenyan examples (lemon juice, Stoney Tangawizi, Jik, antacid tablets for stomach problems) in their models and explanations, or did they default to generic 'acid' and 'base'? Strong contextual integration predicts better transfer to the final unit assessment.

5. PHENOMENON LINK: When students annotated the eight canteen cups at the bottom of their graphic model, were the annotations causal (H^+ concentration → low

pH → red colour) or merely descriptive (lemon juice is red)? The goal of this lesson is causal explanation. If more than 30% of students gave only descriptive annotations, dedicate the opening 8 minutes of Lesson 8 to a structured class discussion explicitly building the causal chain: [Arrhenius acid/base] → [H^+/OH^- concentration] → [pH value] → [indicator colour] → [cup colour in phenomenon].

6. GAS TESTS: Were students able to describe BOTH gas tests (squeaky pop for H_2; limewater turns milky for CO_2) unprompted? These tests appear in KCSE examinations and must be automatised. If not, add a 3-minute paired oral drill at the start of Lesson 8.

LESSON 8 PREVIEW NOTE: Lesson 8 will introduce the distinction between strong and weak acids/bases (degree of ionisation), connecting to pH values and to the specific colours observed in the phenomenon for dilute vs. concentrated acid solutions. Ensure the DQB's Column 3 yellow notes are available for reference at the start of Lesson 8.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	In the Gallery Walk (Phase 2), students revisited evidence from all six prior lessons: the rainbow of indicator colours produced by the eight canteen liquids (lemon juice = red, Stoney Tangawizi = orange-red, chai = orange, milk = yellow-green, tap water = green, baking soda = blue, liquid soap = blue-violet, Jik = purple); the results of reactions of HCl with zinc metal (effervescence, squeaky pop confirms H_2), with marble chips/sodium carbonate (effervescence, limewater turns milky confirms CO_2), with copper oxide (black solid dissolves in blue solution), and with NaOH solution (no visible change — neutralisation); and the indicator behaviour of bases. In Phase 3, the teacher demonstrated the CO_2 limewater test and the H_2 burning splint test live, adding fresh direct observation to consolidate prior learning.
What did I learn?	Students learned that all four types of acid reactions — acid + metal, acid + carbonate, acid + metal oxide, and acid + alkali (neutralisation) — can be unified under the Arrhenius definition: in every case, H^+ ions from the acid are the active chemical agent. They learned the generalised word and symbol equations and observable evidence for each reaction type, including the specific gas tests. They learned that bases are characterised by OH^- ions (Arrhenius), react with acids in neutralisation ($H^+ + OH^- \rightarrow H_2O$), and produce distinctive colour changes in indicators opposite to those of acids. Students consolidated knowledge that the colour of each canteen cup in the phenomenon is causally determined by the concentration of H^+ or OH^- ions — not by appearance, taste, or smell alone.
How does this explain the phenomenon?	Students constructed a Graphic Summary Model synthesising all chemical properties of acids and bases onto one structured visual. They explicitly explained that the eight canteen cups turn different colours because they contain different concentrations of H^+ or OH^- ions: liquids with high H^+ concentration (lemon juice, Stoney Tangawizi) produce red/orange colours because they have low pH; liquids with high OH^- concentration (Jik, liquid soap) produce blue/purple colours because they have high pH; tap water, with equal H^+ and OH^-, produces green (neutral, pH 7). Students explained the neutralisation reaction between stomach acid (HCl, producing H^+) and antacid medication (a base, producing OH^-) as a real-world application. They updated the Driving Question Board to mark Column 2 (chemical properties) as substantially answered, and generated new questions about acid/base STRENGTH targeting Lessons 8–9.

LESSON 8 (40 minutes): Introduction to pH: Measuring the Strength of Acids and Bases in Kenyan Everyday Liquids	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students will use universal indicator and pH paper to measure the pH of a range of solutions and construct a pH number line populated with Kenyan real-world examples, advancing their explanation of the canteen cup colour differences.
Knowledge	Students will be able to: (1) define pH as a numerical scale from 0 to 14 that measures the relative concentration of H+ ions in a solution; (2) state that pH below 7 indicates an acidic solution, pH 7 indicates a neutral solution, and pH above 7 indicates a basic solution; (3) recall that universal indicator shows a continuous colour spectrum across the pH scale, not just two colours; (4) place at least eight Kenyan real-world liquids on a pH number line with correct relative positions.
Skills	Students will be able to: (1) safely use pH indicator paper and a universal indicator colour chart to determine the approximate pH of given solutions; (2) record and tabulate pH data accurately; (3) construct an annotated pH number line using experimental and reference data; (4) use pH data as evidence to rank solutions in order of increasing acidity or basicity.
Attitudes	Students will appreciate that a single number can carry significant scientific meaning about the invisible particle-level composition of a liquid, and will demonstrate curiosity about how pH measurement connects to health, farming, and environmental decisions in Kenya.
Key Inquiry Question	What does the pH number actually tell us about the strength of an acid or a base, and how is pH measured in the laboratory?
Purpose in Storyline	This lesson is the observational evidence-gathering lesson for pH. Students have spent Lessons 3 to 7 learning what acids and bases ARE and HOW they react. Now they confront the question of HOW MUCH — how acidic or how basic is a given liquid? By measuring the pH of the same canteen liquids from Lesson 1 and connecting the numbers to the colours they originally observed, students gain the quantitative dimension missing from all previous lessons. This sets up Lesson 9, where the degree of ionisation (strong vs. weak) will explain WHY some acids have a much lower pH than others even at similar concentrations.
Safety Notes	(1) Jik bleach solution must be pre-diluted by the teacher to approximately 1 percent and handled only with gloves; students must not allow it to contact skin or eyes. (2) Lemon juice and vinegar are safe but students must not taste any solution in the laboratory. (3) Remind students that the liquid soap solution can be slippery if spilled — wipe surfaces immediately. (4) Dispose of all test solutions into the designated waste cup, not the sink, until the teacher has checked concentrations. (5) Wash hands at the end of the practical session.

B. LESSON OVERVIEW

Lesson 8 opens with a short revisit of the Lesson 1 canteen cups phenomenon: students recall the colour spectrum they observed and are reminded that they can now explain WHAT caused those colours (H+ and OH- ions) but not yet WHY some cups were more intensely coloured than others or why the colours were arranged in such a precise order. The teacher plays a short introductory video on pH (approximately 5 minutes) that introduces the pH scale as a tool for quantifying hydrogen ion concentration. Students then conduct Experiment 3: they use universal indicator solution and pH indicator paper to test eight solutions (lemon juice, black tea, cow milk, a carbonated soft drink, tap water, baking soda solution, liquid soap solution, and dilute Jik bleach — the same set as Lesson 1). They record observed colours, match them to a pH colour chart, and assign a pH number to each solution. Using these data plus teacher-provided reference values, students construct a class pH number line from 0 to 14 on a strip of paper pinned to the classroom wall, populating it with the Kenyan real-world examples: stomach acid, freshly squeezed lime juice, black tea, cow milk, blood, Lake Magadi water, and liquid soap. The lesson closes with students updating the Driving Question Board with a refined question: what does the pH NUMBER actually tell us about the strength of the acid or base — and this becomes the focus of Lesson 9.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Experimental probability Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Hydrogen and Alkali Metals (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students open their notebooks to the Lesson 1 prediction table and review the original colours they recorded for each canteen cup. The teacher projects or draws on the board a blank pH number line from 0 to 14 with a simplified colour bar beneath it (red at 0, moving through orange, yellow, green, blue, to violet at 14). Students are given 2 minutes to individually write responses to two predict prompts in their notebooks: (1) Without yet knowing the numbers, arrange the eight canteen liquids in order from what you think is the most acidic to the most basic, based on everything you have learned so far. (2) Predict: where on a 0 to 14 scale do you think lemon juice sits? Where does baking soda sit? Where does Jik bleach sit? Give a reason for each guess. After writing, pairs share their rankings briefly. The teacher takes a quick show-of-hands survey: How many pairs put lemon juice below 5? How many put bleach above 10? This surfaces the range of prior conceptions and creates a genuine need to measure.	Teacher begins by holding up the Lesson 1 phenomenon photograph or re-displaying the eight coloured cups image and says: 'You told me in Lesson 1 that you were surprised by those colours. In Lessons 3 to 7 you built an explanation — H+ ions cause acid colours, OH- ions cause basic colours. But look at the red cup and the orange cup side by side. Both are acidic. Both have more H+ than OH-. So why are they DIFFERENT colours?' WAIT TIME: 20 seconds. Teacher then says: 'Today we are going to get numbers. We are going to measure exactly how much the balance between H+ and OH- has shifted in each liquid. Before we measure, let us predict.' Teacher distributes or projects the predict prompts and circulates during the 2-minute writing time, noting interesting predictions to use during debrief without naming students. Teacher conducts the show-of-hands survey and records approximate class consensus on the board next to the blank number line: 'Let us see if our data today confirms or surprises us.'	Elicitation of prior knowledge through ranked ordering tasks. By asking students to rank before measuring, the teacher creates cognitive anticipation — a gap between what students think and what data will show. The two specific predict prompts (lemon juice and bleach) target the extreme ends of the scale where misconceptions about 'strong' versus 'concentrated' are most common.	Teacher scans notebooks during the 2-minute writing period and notes whether students use the language of H+ and OH- in their reasoning or revert to taste-based or appearance-based reasoning. The show-of-hands survey gives a rapid class-level read of pre-knowledge. Any student who places bleach below 7 or lemon juice above 7 is flagged for targeted questioning during the experiment phase.
Observe Phase 📹 **VIDEO:** Experimental probability Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:**	Students watch a short teacher-selected or teacher-recorded introductory video on pH (approximately 4 to 5 minutes). The video covers: the origin of the term pH (power of Hydrogen), the 0 to 14 scale, the colour transitions of universal indicator, and one Kenyan real-world example (stomach acid at pH 1 to 2, used in digesting nyama choma). After the	During video viewing, teacher pauses at the colour bar segment and asks: 'Pause — look at that colour at pH 7. What colour is it? Green. Which cup in our canteen phenomenon was green?' (Tap water.) 'What does that tell you?' WAIT TIME: 15 seconds. Teacher then resumes the video. During the practical, teacher circulates actively. Key teacher moves: (1) At	Triangulation of two measurement methods (pH paper and universal indicator) builds measurement literacy and reinforces that the pH number is a property of the solution, not of the indicator. The Teacher Reference Card strategy extends the data set beyond what is testable in a 40-minute lesson while connecting to Kenyan geographic and health contexts	Teacher uses the freeze-and-hands move to check the Jik bleach result in real time. Teacher checks three specific data points in each group notebook during circulation: (1) Does the recorded pH of tap water sit at or very close to 7? (2) Is lemon juice recorded below 4? (3) Is baking soda solution recorded above 7? Groups with discrepancies are

Hydrogen and Alkali Metals (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	video, groups of three receive a tray containing eight small cups pre-labelled A through H (lemon juice, black tea, cow milk, carbonated soft drink, tap water, baking soda solution, liquid soap solution, dilute Jik bleach), a strip of pH indicator paper, a dropper bottle of universal indicator solution, a printed pH colour reference chart, a dropper, and a waste cup. Procedure: Step 1 — using a fresh strip of pH paper dipped once into each solution, students observe the colour change and match it to the colour chart to assign a pH range. Step 2 — for confirmation, students add 3 drops of universal indicator to a small volume of each solution in a clean cup and compare the resulting colour to the printed chart. Step 3 — students record all results in a results table with columns: Solution, pH paper colour, Approximate pH from paper, Universal indicator colour, Approximate pH from UI, Best estimate pH, Classification (acid / neutral / base). Groups take a full 12 to 14 minutes for the practical work. While working, students also receive a Teacher Reference Card with pH values for six additional Kenyan real-world liquids they cannot test in class: stomach acid (pH 1.5), fresh lime juice as used in Mombasa pilau preparation (pH 2.0), Lake Magadi water which is highly alkaline due to sodium carbonate (pH 10 to 11), human blood (pH 7.4), urine of a well-hydrated person (pH 6.0), and the water from a Kericho tea farm soil runoff (pH 4.5). Students add these to their results tables as reference rows.	any group struggling to match pH paper colour to the chart, teacher models holding the paper against a white background and asks: 'Is this closer to orange or yellow? Orange means more acidic — what pH range does orange correspond to on your chart?' (2) At groups finishing early, teacher prompts: 'You have assigned a number. Now look at your Lesson 4 results table. Does the litmus result for lemon juice agree with your pH result today? It should — explain why in one sentence.' (3) Teacher calls a 30-second freeze halfway through the practical: 'Everyone stop. Raise your hand if your Jik bleach result is showing pH above 12.' Teacher uses the distribution of hands to confirm or correct understanding of the basic end of the scale in real time. Teacher uses the phrase: 'The number is telling you something about invisible particles — more H+ ions shifts the number DOWN toward 0; more OH- ions shifts the number UP toward 14. Keep that particle picture in your mind as you record.'	(Lake Magadi, Kericho, Mombasa pilau). The mid-practical freeze is a high-leverage formative move that surfaces measurement errors before they become embedded misconceptions.	prompted to re-test. Teacher also checks that students are recording BOTH the pH paper result AND the universal indicator result, not just one.

Explain Phase 📹 **VIDEO:** Experimental probability Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Hydrogen and Alkali Metals (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	After the practical, the class gathers as a whole group. The teacher projects or draws a large pH number line on the board (0 to 14) with the colour bar. Groups share their best-estimate pH values for each of the eight tested solutions. The teacher facilitates a class negotiation of agreed values by asking: 'Group 1 said lemon juice is pH 2. Group 3 said pH 3. What do both groups agree on?' (Both agree it is acidic, below 7.) 'What might explain the small difference?' (Concentration of juice, freshness, how much water was added.) The teacher then positions each solution on the board number line, writes the name and pH, and adds the six reference values from the Teacher Reference Card. Students copy this annotated number line into their notebooks. The teacher then leads a whole-class explanation discussion using three focusing questions displayed on the board: (1) 'Why does the universal indicator show so many different colours rather than just red for acid and blue for base?' (2) 'What does it mean to say that stomach acid at pH 1.5 has MORE H+ ions than lemon juice at pH 2 — are we talking about a small difference or a large difference?' (Teacher introduces the idea — without requiring logarithm mathematics — that each step on the pH scale represents a tenfold change, using a simple analogy: walking 1 step vs. walking 10 steps vs. walking 100 steps.) (3) 'Lake Magadi water is pH 10 to 11. What does that tell us about OH- ions there?' Students write a three-sentence explanation in their notebooks answering question 1 using the	Teacher opens the explain discussion by returning to the canteen cups photograph and says: 'We now have numbers. Can we go back to the original question — why did the cups turn DIFFERENT colours? Before today, your answer was H+ ions. Now you have a more powerful answer. What is it?' WAIT TIME: 25 seconds. Teacher uses cold-calling (random selection from a name jar) for the first response, then opens to the class for elaboration. For question 2 about the tenfold change, teacher uses the following analogy script: 'Imagine the school corridor. pH 7 is one person standing there. pH 6 is ten people. pH 5 is one hundred people. pH 4 is one thousand people. We are talking about H+ ion concentration multiplying by ten for every single step down the scale. So the difference between lemon juice and stomach acid is not one unit — it is a tenfold difference in how many H+ ions are packed into every litre of liquid.' Teacher checks comprehension by asking: 'So if pH 2 has ten times more H+ than pH 3, how many times more H+ does pH 1 have than pH 3?' WAIT TIME: 20 seconds. (Answer: one hundred times.) Teacher does not require students to calculate logarithms — this is conceptual preparation for Lesson 9. For question 3 about Lake Magadi, teacher connects explicitly to Kenyan geography: 'Lake Magadi in the Rift Valley is one of the most alkaline water bodies on Earth. The water is rich in sodium carbonate — trona — which dissolves to release OH- ions. Very few organisms can survive	The board-based class negotiation of pH values uses social sensemaking — groups compare data, identify sources of variation, and arrive at a shared understanding. The tenfold analogy builds intuition for logarithmic scaling without requiring formal mathematics. The Lake Magadi context anchors the abstract number in a concrete, memorable Kenyan geographic reality that students can visualise. The three-sentence writing task is a micro-writing-to-learn strategy that consolidates the explain phase before the DQB update.	The teacher listens for whether students use the phrase 'concentration of H+ ions' or 'H+ ions' when explaining colour differences, rather than just saying 'it is more acidic.' The cold-call response to the opening question reveals whether students can spontaneously integrate pH numbers into their existing Arrhenius explanation. The mental calculation check (how many times more H+ does pH 1 have than pH 3?) gives a quick conceptual comprehension read without requiring written computation.

	language of H+ ion concentration and colour change.	there. But some specialised algae and flamingos that feed on the algae thrive. The pH tells us exactly why it is challenging for most life.' Teacher closes the explain phase by saying: 'Every colour in our canteen cups from Lesson 1 now has a number attached to it. But one question remains unanswered on our DQB. Let us go there now.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Experimental probability Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Hydrogen and Alkali Metals (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students return to the class Driving Question Board displayed at the front of the room. The teacher reads aloud the DQB entry left from Lesson 7: 'We can now explain WHAT makes acids and bases react and WHAT H+ and OH- ions do — but we do not yet know HOW MUCH. How do we measure the strength of an acid or base?' Students are told: 'That question has been partially answered today. Let us update the board and generate the next question.' Each student takes a sticky note and writes one of the following: (a) a statement of evidence gathered today that partially answers the Lesson 7 DQB question, or (b) a new question generated by today results — something today data made them wonder about. Students post their notes on the DQB under a new heading: Lesson 8 Evidence and New Questions. The teacher reads five to seven notes aloud (selecting for diversity of content) and facilitates a brief 3-minute whole-class discussion to identify the most important unanswered question. The class agrees that the central new question is: 'What does the pH NUMBER actually tell us about the strength of the acid or base —	Teacher stands at the DQB and models reading the Lesson 7 entry with deliberate pacing, then says: 'Before you write your sticky note, look at your completed results table and your annotated pH number line. What do you now KNOW that you did not know at the start of this lesson? Write that as evidence. Then think: what does today data make you wonder about next?' WAIT TIME: 90 seconds of silent individual thinking before writing begins. Teacher circulates and reads notes over shoulders, using light redirection prompts for notes that are too vague: 'You wrote it is more accurate. Can you be more specific — more accurate than what? What were you measuring?' After posting, teacher selects notes that reference the tenfold difference idea, the comparison between HCl and lemon juice (both acidic but different pH), and the Lake Magadi OH- example. Teacher frames the consensus question for Lesson 9 by saying: 'You have noticed something important. Two substances can both be acidic — both below pH 7 — but their pH numbers are very different. That difference is not random. It is explained by something we have not yet	The two-option sticky note structure (evidence statement OR new question) ensures that all students engage with metacognitive reflection, not just question generation. Reading selected notes aloud and facilitating consensus around the Lesson 9 question models scientific discourse: the class is collectively constructing the next step in its own inquiry rather than receiving it from the teacher. The explicit teacher framing of the Lesson 9 target question provides a narrative bridge that maintains storyline coherence.	The sticky notes provide a written artefact of each student individual sensemaking. Teacher collects any notes that reveal persistent misconceptions (for example, a student who writes 'pH tells us how dangerous the liquid is' rather than connecting pH to H+ concentration) and plans a brief clarification at the start of Lesson 9. The consensus question process reveals whether the class as a whole has grasped the key tension — same category (acid), different pH — that motivates Lesson 9 content.

	and why does HCl at pH 1 behave differently from citric acid at pH 3 even though both are acidic?' This question is written in large letters on the DQB as the Lesson 9 target question.	studied: the degree to which an acid ionises. That is exactly where we are going in Lesson 9.'		
Model Building Phase 📹 **VIDEO:** Experimental probability Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Hydrogen and Alkali Metals (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	In the final 5 to 6 minutes, students open to the running Cumulative Model page in their notebooks — the concept model they have been building since Lesson 3. In Lesson 7 they added the four acid reaction summary box. Today they add a new section titled: The pH Scale. They draw a horizontal number line from 0 to 14, colour-code it using their universal indicator colour chart, label the three zones (acidic 0 to below 7, neutral at 7, basic above 7 to 14), and place at least five of the Kenyan real-world examples in correct positions. They also write one connecting sentence beneath the number line: 'The pH number represents the relative concentration of H+ ions in a solution — a lower pH means a higher concentration of H+ ions, linking directly to the Arrhenius definition of an acid as a substance that produces H+ ions in water.' Students who finish early add an arrow annotation showing the direction of increasing H+ concentration (toward 0) and increasing OH- concentration (toward 14).	Teacher says: 'Your cumulative model is growing. In Lesson 3 you added the Arrhenius definition. In Lessons 4 to 6 you added the four reaction types. In Lesson 7 you unified them. Today you add the measurement tool that will let you QUANTIFY how acidic or basic any solution is. Add the pH section now — five minutes.' Teacher circulates and checks that the colour gradient is approximately correct and that students are placing examples in sensible relative positions. Teacher prompts early finishers: 'Add an annotation — what happens to the number of H+ ions as you move from pH 7 toward pH 0? Draw an arrow and label it.' Teacher closes the lesson with a direct callback to the phenomenon: 'Look at your model and then imagine you are back at the canteen table in Lesson 1. The lemon juice cup turned red — your model now tells you that red means pH around 2, which means a very high concentration of H+ ions. The tap water cup turned green — pH 7 — balanced H+ and OH-. Next lesson you will be able to explain not just the colour and the number, but WHY different acids produce such different pH values even when they are poured from similarly sized bottles. See you in Lesson 9.'	The cumulative model is the primary knowledge-integration artefact for the unit. Adding the pH section alongside the Arrhenius definition and the reaction types box forces students to see pH as part of the same explanatory framework, not as a separate topic. The connecting sentence requirement ensures that students do not treat the pH scale as a standalone memorisation task but link it explicitly to the particle-level model. The lesson-closing callback to the phenomenon maintains the narrative thread of the entire unit storyline.	Teacher scans completed pH sections for three specific items: (1) Is the colour gradient in the correct direction (red at 0, green at 7, violet or blue at 14)? (2) Are Kenyan real-world examples placed in correct relative positions — is stomach acid clearly to the left of lemon juice, and is bleach clearly to the right of baking soda? (3) Does the connecting sentence include both pH and H+ ion concentration language? Students whose models lack the connecting sentence are asked to add it before the next lesson.

D. TEACHER REFLECTION

After the lesson, the teacher should consider: (1) Did students successfully use both the pH paper and the universal indicator, and did they notice that the two methods gave consistent results? If groups struggled with one method, note which — this informs Lesson 9 practical decisions. (2) Did the tenfold change analogy land? If most students could not answer the mental calculation check (how many times more H+ does pH 1 have than pH 3?), plan a brief re-entry in Lesson 9 before introducing strong vs. weak ionisation. (3) Were the DQB sticky notes evidence-based and specific, or were they vague? Vague notes suggest students are not yet connecting the pH number to the particle model — build in a brief explicit bridge at the start of Lesson 9. (4) Did the Lake Magadi and Kericho references generate genuine student interest or curiosity comments? If so, these contexts should be foregrounded even more strongly in Lesson 9 when agricultural pH is the central application. (5) Was the 40-minute timing manageable? The video, practical, explain discussion, DQB update, and model building are ambitious for 40 minutes. If time ran short, the model building section can be set as a 5-minute starter activity at the beginning of Lesson 9.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students measured the pH of eight Kenyan everyday liquids using pH indicator paper and universal indicator solution, and recorded the colour changes and approximate pH values for each solution. They also received reference pH data for six additional Kenyan real-world liquids including stomach acid, lime juice used in Mombasa cooking, Lake Magadi water, and Kericho tea farm soil runoff.
What did I learn?	Students learned that pH is a numerical scale from 0 to 14 that measures the relative concentration of H+ ions in a solution. A pH below 7 indicates an acidic solution, pH 7 indicates a neutral solution, and pH above 7 indicates a basic solution. Each step on the pH scale represents approximately a tenfold change in H+ ion concentration. Universal indicator shows a continuous colour spectrum that allows finer discrimination between solutions than litmus alone.
How does this explain the phenomenon?	Students used their pH data to explain why the eight canteen cups from the Lesson 1 phenomenon turned different colours: each colour corresponds to a specific pH value that reflects the relative concentration of H+ ions in that liquid. They connected pH measurement back to the Arrhenius definition by explaining that substances with more H+ ions in solution have lower pH values and produce redder colours with universal indicator, while substances with more OH- ions have higher pH values and produce bluer or violet colours. They updated the Driving Question Board with a new focused question about what the pH number reveals about the strength — not just the presence — of an acid or base.

LESSON 9 (40 minutes): Strong vs. Weak Acids and Bases: Why the Same pH Step Can Make or Break a Tea Farm	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students will use pH data collected in Lesson 8 to construct a particle-level explanation of why some acids and bases produce lower or higher pH values than others, connecting degree of ionisation to the Arrhenius definition and to real Kenyan agricultural and health contexts.
Knowledge	Students will be able to: (1) distinguish between strong and weak acids and strong and weak bases using the concept of degree of ionisation; (2) explain that strong acids and bases ionise completely in aqueous solution while weak acids and bases ionise partially; (3) relate degree of ionisation to relative H+ or OH- ion concentration and therefore to pH; (4) give named Kenyan-relevant examples of strong and weak acids and bases; (5) explain why soil pH 4.2 is damaging to tea plants while pH 5.5 to 6.0 is optimal.
Skills	Interpreting particle diagrams to compare ion concentrations; analysing and extending pH data from Lesson 8; constructing and annotating particle diagrams for strong vs. weak acid solutions at equal concentration; drawing evidence-based conclusions; updating a cumulative explanatory model.
Attitudes	Students will appreciate that precise scientific distinctions — complete vs. partial ionisation — have direct consequences for farming livelihoods and human health in Kenya; they will value the power of a model that can explain both a classroom demonstration and a real agricultural problem.
Key Inquiry Question	How is the strength of acids and bases determined?
Purpose in Storyline	Lesson 9 is the final Explain phase before the synthesis lesson. Students now have pH measurements (Lesson 8), indicator colour data (Lesson 4), and a full repertoire of acid and base reactions (Lessons 5 and 6). This lesson provides the missing conceptual bridge: why do two acids both turn universal indicator red but give different pH readings? The answer — degree of ionisation — completes the particle-level Arrhenius model and gives students the language to write a full scientific explanation of the eight cups from Lesson 1 in Lesson 10.
Safety Notes	This is a non-practical, sensemaking lesson. No chemicals are handled. Teacher should remind students never to taste or smell concentrated acids or bases directly. If particle diagram worksheets are distributed, ensure no chemical bottles are brought to desks.

B. LESSON OVERVIEW

Using their own pH data strips and readings from Lesson 8 as the launching evidence, students are guided through a structured teacher-facilitated sensemaking sequence. They first predict why two acids measured at different pH values must differ in something beyond their chemical formula. The teacher then introduces degree of ionisation using a dual particle diagram displayed on the board, contrasting hydrochloric acid (strong, complete ionisation) with ethanoic acid from Kenyan white vinegar (weak, partial ionisation), and sodium hydroxide (strong base) with ammonia solution used in cleaning products (weak base). Students annotate their own diagrams, write ionisation equations with reversible arrows for weak acids and one-directional arrows for strong acids, and interpret a pH data table linking degree of ionisation to H+ concentration to pH number to indicator colour — thereby reconnecting to the anchoring phenomenon. The agricultural sensemaking anchor — why a Kericho tea farm with soil pH 4.2 suffers crop failure while pH 5.5 to 6.0 supports optimal growth — gives concrete meaning to the abstract concept. The lesson closes with a DQB update and a model-building entry that prepares students for the final synthesis in Lesson 10.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Arrhenius Acids and Bases - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Strong and Weak Acids and Acid Ionization Constant (Ka) (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students open their Lesson 8 notebooks and locate their pH data table. The teacher projects or writes two data rows on the board: Row 1 — hydrochloric acid solution, pH 1, universal indicator colour: red; Row 2 — lemon juice (citric acid), pH 2 to 3, universal indicator colour: orange-red. The teacher prompts: Both of these are acids. Both turn universal indicator into red or orange tones. Both contain H+ ions according to the Arrhenius definition. Yet their pH values are different. In your notebook, write one or two sentences to answer this question: If both solutions contain H+ ions, what could explain the difference in their pH readings? Students write independently for 90 seconds, then share briefly with a shoulder partner. Three volunteers share with the class. Predicted answers are likely to include: one has more acid dissolved, one is more concentrated, or one releases more H+ ions. The teacher accepts all predictions without evaluating them and writes key student phrases on the board under the heading OUR PREDICTIONS. This activates prior knowledge of the pH scale (Lesson 8) and the Arrhenius definition (Lesson 3) and creates cognitive readiness for the ionisation explanation to follow.	Teacher opens by pointing to the eight cups from the anchoring phenomenon photograph or drawing on the board: Remember our eight cups from Lesson 1. We now know that the colour each cup showed is linked to the pH of the liquid. But today I want to push us one level deeper. Two of those cups — the lemon juice and a strong acid — both turned reddish colours. Yet if you measured their pH carefully, as we did in Lesson 8, they would give different numbers. Why? Write your best thinking now. [WAIT TIME: 90 seconds of silent individual writing — teacher circulates, reads over shoulders, selects two contrasting responses to highlight.] After sharing: I am hearing two big ideas — concentration and something about how much H+ is actually released. Both of those ideas are going to matter today. Hold onto them.	Prediction-before-instruction activates the Arrhenius model from Lesson 3 and the pH scale from Lesson 8. Anchoring the prediction in the students own data (their Lesson 8 pH table) makes the cognitive need genuine rather than artificial. Writing before sharing reduces anchoring bias toward the first student who speaks.	Teacher reads student notebook entries during the 90-second silent period. Checks whether students are connecting pH difference to H+ ion concentration difference (desired) or attributing it solely to concentration of solute (a common misconception that will be addressed in the Explain phase). This informs pacing of the explanation.
Observe Phase 📹 **VIDEO:** Arrhenius Acids and Bases - Overview Source: CK-12	The teacher distributes the Lesson 9 Particle Diagram Worksheet (or draws the diagrams on the board for students to copy). The worksheet shows four beakers drawn side by side, each containing the same number of	Teacher projects or draws the four beakers. Points to the pH data on the board: Here is your evidence. Same starting concentration — same number of molecules going in. HCl gives pH 1. Ethanoic acid gives pH 3. That is not a small	The observe phase here is data-driven observation of a quantitative anomaly — the 100-times difference in H+ concentration inferred from pH data students collected themselves. This grounds the abstract concept of	Teacher listens to paired discussions and cold-calls two pairs: one that reached the complete vs. partial ionisation idea and one that is still uncertain. This reveals whether the class is ready for the formal explanation or needs

🔍 Search ARES for similar videos 📖 **HTML:** Strong and Weak Acids and Acid Ionization Constant (Ka) (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	formula units before ionisation: Beaker A — 6 HCl molecules dissolving; Beaker B — 6 CH3COOH (ethanoic acid) molecules dissolving; Beaker C — 6 NaOH formula units dissolving; Beaker D — 6 NH3 molecules dissolving in water. Below each beaker is an empty product diagram box. The teacher says: Before I explain anything, I want you to look at the evidence we already have. From your Lesson 8 pH data, a 0.1 mol per litre hydrochloric acid solution gives pH 1, while a 0.1 mol per litre ethanoic acid solution gives approximately pH 3. That means same concentration — same number of formula units going in — yet the HCl solution has about 100 times more H+ ions in it. Similarly, 0.1 mol per litre sodium hydroxide solution gives pH 13, while 0.1 mol per litre ammonia solution gives approximately pH 11. Students are asked: In Beaker A (HCl), if 6 formula units produce many more H+ ions than 6 formula units of ethanoic acid in Beaker B — what must be happening differently inside those two beakers? Students discuss in pairs for 60 seconds. The teacher collects responses. Expected output: HCl must be breaking apart completely; ethanoic acid must only partly break apart. Students have now reasoned toward the concept before being told it — this is the key observable inference of the lesson.	difference. On the pH scale, each step is a tenfold change in H+ concentration. pH 1 versus pH 3 means the HCl solution has 100 times more H+ ions than the ethanoic acid solution. [WAIT TIME: pause 5 seconds, let that number settle.] So I am asking you: if the same number of molecules went in, but one produced 100 times more H+ ions — what must be happening inside the beakers differently? Talk to your partner. [WAIT TIME: 60 seconds paired discussion.] After collecting responses: Someone just said one acid breaks all the way apart and the other only breaks part way. That is exactly the insight we need. Let us now give that idea a precise scientific name and draw it.	ionisation in their own empirical evidence. The particle diagram task is scaffolded observation of a model, not free construction, ensuring all students access the key visual before the formal explanation.	another data prompt. Teacher notes any student who uses the phrase strong and weak correctly before being taught — these students can be invited to elaborate during the Explain phase.
Explain Phase 📹 **VIDEO:** Arrhenius Acids and Bases - Overview	The teacher formally introduces the terms strong acid, weak acid, strong base, and weak base, anchoring each in degree of ionisation. Using the particle	Teacher completes the first particle diagram (HCl) live on the board, thinking aloud: I am going to draw 6 HCl formula units going into water. Now I will draw what	The particle diagram is the central representational tool — it makes the abstract concept of degree of ionisation visible and countable. Live construction of the diagram by	After the table is completed, teacher asks students to individually write one sentence: Explain in particle terms why lemon juice has a higher pH than

Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Strong and Weak Acids and Acid Ionization Constant (Ka) (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	diagrams on the board or worksheet, the teacher completes each product beaker: Beaker A (HCl, strong acid) — all 6 formula units are shown fully separated into H+ and Cl- ions; Beaker B (ethanoic acid, weak acid) — only 1 or 2 of the 6 molecules are shown ionised, with 4 or 5 remaining as intact CH3COOH molecules, and a reversible double-headed arrow shown; Beaker C (NaOH, strong base) — all 6 formula units fully separated into Na+ and OH- ions; Beaker D (ammonia/NH3, weak base) — only 1 or 2 NH3 molecules have reacted with water to form NH4+ and OH-, with the reversible arrow shown. Students annotate their diagrams with the labels: complete ionisation, partial ionisation, strong, weak, and reversible equilibrium symbol. The teacher then guides students to write four ionisation equations: HCl(aq) arrow H+(aq) + Cl-(aq) [one-directional arrow, labelled strong]; CH3COOH(aq) double arrow CH3COO-(aq) + H+(aq) [reversible arrow, labelled weak]; NaOH(aq) arrow Na+(aq) + OH-(aq) [one-directional, strong]; NH3(aq) + H2O(l) double arrow NH4+(aq) + OH-(aq) [reversible, weak]. Teacher emphasises: The reversible arrow for weak acids and bases tells us the reaction does not go to completion — at any moment, most molecules are still intact, so fewer H+ or OH- ions are present. This is why the pH is closer to 7 than a strong acid or base of the same concentration. The teacher then projects or writes the following summary linking table on the board: columns are — Substance, Kenyan source, Strong	comes out. HCl is a strong acid — it ionises completely. So every single HCl unit breaks apart. I draw 6 H+ ions and 6 Cl- ions. No HCl molecules remain. One-directional arrow — no going back. Now ethanoic acid. Same number going in. But watch — only about 1 in 100 molecules actually ionises at this concentration. I will exaggerate slightly to make it visible: maybe 1 or 2 out of 6. Most molecules stay intact. I draw a reversible double arrow — meaning the forward and reverse reactions are both happening. Some molecules ionise, some ions recombine. The result? Far fewer H+ ions in solution. Lower H+ concentration means higher pH — further from 0, closer to 7. [WAIT TIME: 10 seconds of silent student observation and annotation.] Now you annotate your beakers. Write complete ionisation or partial ionisation inside each product box. [WAIT TIME: 60 seconds.] For the agricultural connection: I want to tell you about a real problem in Kericho. [Narrates scenario.] Now, here is a thinking question — do not answer yet. Write it down first: Why would a farmer adding too much lime to fix the acidity actually create a new problem? Think about what a strong base does to pH. [WAIT TIME: 45 seconds silent writing, then 30 seconds paired discussion, then cold-call two students.]	the teacher models scientific thinking explicitly. The linking table consolidates multiple representations (particle, equation, number, colour) into one reference. The Kericho agricultural scenario provides a systems-level application that requires students to reason in both directions — too little and too much — demonstrating deep rather than surface understanding. Connecting back to the neutralisation reaction from Lesson 5 and the pH scale from Lesson 8 explicitly builds the cumulative model.	hydrochloric acid even though both are acids. Teacher scans three to five notebooks to assess whether students are using degree of ionisation and H+ ion concentration correctly. Students who write only that lemon juice is less concentrated are flagged for targeted questioning — the teacher re-asks: What if we made them the same concentration? What would still be different? This targets the critical misconception that strength and concentration are the same thing.

	or Weak, Degree of ionisation, Approximate pH at 0.1 mol/L. Rows are: HCl (laboratory acid, not commonly consumed, strong, complete, pH 1); ethanoic acid (white vinegar sold in Kenyan supermarkets, weak, partial, pH 3); citric acid (fresh lemon juice from Kenyan farms, weak, partial, pH 2 to 3); NaOH (caustic soda / oven cleaner, strong, complete, pH 13); ammonia solution (multipurpose cleaner used in Kenyan homes and schools, weak, partial, pH 11). Students copy and annotate the table. Agricultural connection is introduced: The teacher narrates the Kericho tea scenario. Kericho County in the Rift Valley is one of Kenyas most important tea-growing regions. Tea plants grow best in slightly acidic soils with pH 5.5 to 6.0. In some fields, poor drainage and decomposing organic material can drive soil pH down to 4.2. At pH 4.2, the H+ ion concentration is about 6 times higher than at pH 5.0 and almost 100 times higher than at pH 6.0. What is causing this soil acidity? Mostly organic acids and dissolved carbon dioxide forming carbonic acid — both weak acids. But because they are produced continuously by decomposing matter, the cumulative effect drives pH low enough to damage plant roots. Farmers apply agricultural lime (calcium hydroxide, a strong base) to neutralise this excess acidity, raising pH to the optimal range. Students connect: the strong base fully ionises to produce OH- ions that neutralise H+ ions — the neutralisation reaction from Lesson 5. Students are then asked to answer in their notebooks: Why would applying			

	too much lime be equally damaging? Expected answer: too much strong base would raise pH above 7, making soil alkaline, which also harms tea plant nutrient uptake. This two-sided thinking reinforces that both extremes of the pH scale, driven by either strong acids or strong bases, cause agricultural harm.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Arrhenius Acids and Bases - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Strong and Weak Acids and Acid Ionization Constant (Ka) (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The teacher draws attention to the DQB displayed in the room. The teacher reads aloud the question that was added at the end of Lesson 8: What does the pH number actually tell us about the strength of the acid or base? The teacher asks the class: Can we now answer this question? What would we write? Students compose a two-sentence answer in their notebooks first (45 seconds), then one student is invited to read their answer aloud. The class refines it collaboratively until they agree on a consensus statement such as: The pH number tells us the relative concentration of H+ ions in a solution, which depends on both the concentration of the acid or base and its degree of ionisation. A strong acid at a given concentration ionises completely and produces more H+ ions, giving a lower pH, while a weak acid at the same concentration ionises only partially and gives a higher pH. The teacher writes this on the DQB card, replacing the original question, and pins it prominently. The teacher then asks: After everything we have learned today, does anyone want to add a NEW question to the DQB for our last lesson? Students are given 30 seconds to think. Expected new	Teacher stands at the DQB and points to the Lesson 8 question card: This was our cliffhanger from last lesson. We said we did not yet know what the pH number really tells us about strength. Do we know now? Write your answer — two sentences — before we talk. [WAIT TIME: 45 seconds silent writing.] Selects a student who wrote a strong response: Read yours to us. To class: Does everyone agree? What would you add or change? [WAIT TIME: 20 seconds for hands.] Facilitates a brief refinement. Then: What new questions does today raise for you? We have one lesson left. What do you still want to know? [WAIT TIME: 30 seconds.] Accepts two or three new question cards. Points to the original eight-cup question and says: In Lesson 10 we will finally be able to explain every single cup — every colour — using everything we have built. You are nearly there.	The DQB update makes student knowledge growth visible and metacognitive. Moving a question from open to answered is a concrete indicator of learning progress. Asking what new questions have arisen prevents premature closure and models authentic scientific inquiry — good answers generate new questions. Previewing Lesson 10 as the moment of full explanation sustains motivation and narrative coherence.	The two-sentence DQB answer is a quick written formative assessment of conceptual integration. Teacher checks for presence of the three key ideas: H+ ion concentration, degree of ionisation, and the link to pH number. Absence of any of these three in a students written answer signals incomplete understanding that teacher will address in the Model Building phase or through individual feedback.

	questions might include: What are the actual uses of strong and weak acids and bases in Kenya? Can a weak acid ever be dangerous? How do farmers actually test their soil pH? These are previewed as questions Lesson 10 will address. The teacher also points to the original anchoring phenomenon question on the DQB — the eight cups — and asks: Are we almost able to explain every cup? What do we still need from Lesson 10? This plants anticipation for the final lesson.			
Model Building Phase 📹 **VIDEO:** Arrhenius Acids and Bases - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Strong and Weak Acids and Acid Ionization Constant (Ka) (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	In the final 7 minutes of the lesson, students open their cumulative model — the graphic organiser or concept diagram they have been building since Lesson 3 and expanding each lesson. Today they add a new section labelled LESSON 9: STRONG AND WEAK ACIDS AND BASES. In this section they: (1) draw or paste two small particle diagrams side by side — one showing a strong acid (fully ionised) and one showing a weak acid (partially ionised) at equal concentration, labelling H+ ions, intact molecules, and the type of arrow; (2) write a rule box: Strong acid or base = complete ionisation = more H+/OH- ions = lower/higher pH. Weak acid or base = partial ionisation = fewer H+/OH- ions = pH closer to 7; (3) add a Kenyan examples row: strong acids — HCl (lab use), H2SO4 (car battery acid, Lesson 5 supporting phenomenon); weak acids — citric acid (lemon juice, lemon juice cup from Lesson 1), ethanoic acid (white vinegar, Kenyan supermarkets), carbonic acid (Stoney Tangawizi, the soft drink from Lesson 1), organic	Teacher says: Last 7 minutes. Open your cumulative model. We are adding one more layer today — the most powerful explanatory layer we have built so far. Label a new section: Lesson 9, Strong and Weak. I want two particle diagrams, a rule box, and a Kenyan examples row. If you have already done those, draw the arrow back to the anchoring phenomenon and label it. [WAIT TIME: 5 minutes of independent work, teacher circulates.] After 5 minutes: One more thing before we close. Look at your model. Can you now point to where the Jik bleach from Lesson 1 lives? Where does the lemon juice live? Where does baking soda live? Point to each one and say to your partner: strong or weak, and why. [WAIT TIME: 45 seconds paired activity.] Closes: In Lesson 10 we bring everything together. You will watch a video on the uses of acids and bases across Kenya, complete your final model, and then — finally — write a complete scientific explanation of every one of those eight cups. You are ready.	The cumulative model is the primary tool for knowledge integration across the unit. Adding to it each lesson makes learning incremental and visible. The Kenyan examples row grounds abstract categories in the familiar substances of the anchoring phenomenon. The backward arrow to the phenomenon reinforces that every concept learned has explanatory purpose. The extension challenge — classifying all eight cups — is a low-stakes formative rehearsal for the Lesson 10 final task, reducing cognitive load when the full explanation is required.	Teacher uses the final circulation to check three things in student models: (1) that the arrow directions in ionisation equations are correct (one-directional for strong, reversible for weak); (2) that students have correctly classified lemon juice as a weak acid and Jik bleach as a strong base; (3) that students can explain in words why a weak acid has a higher pH than a strong acid at the same concentration. Any student who has incorrectly classified a substance or drawn the wrong arrow is given immediate verbal feedback and asked to self-correct before closing their notebook.

	acids in Kericho tea-farm soil; strong bases — NaOH (caustic soda / Jik bleach, the bleach cup from Lesson 1), $Ca(OH)_2$ (agricultural lime, Kericho scenario); weak bases — ammonia solution (household cleaners), soap solution (Lesson 1 cup); (4) draw one arrow from the new section back to the ANCHORING PHENOMENON section of their model, labelling it: degree of ionisation explains different pH values in the eight cups. Students who finish early are challenged to annotate which of the eight original cups were strong acids, weak acids, strong bases, or weak bases and to estimate whether the farmer in Kericho is dealing with a strong or weak acid problem in the soil and why. Teacher circulates, asks targeted questions: Show me on your model where the bleach (Jik) fits. Is it strong or weak? How do you know from its pH? This directly reconnects to the anchoring phenomenon and previews Lesson 10.			

D. TEACHER REFLECTION

After the lesson, consider: (1) Did the 100-times H^+ concentration difference framing (pH 1 vs pH 3) create genuine cognitive need for the ionisation explanation, or did some students remain confused about the distinction between concentration and strength? If the misconception persists, prepare a short remediation prompt for the opening of Lesson 10: same concentration, different pH — what must differ? (2) Were students able to correctly apply the strong vs. weak distinction to the original eight cups in the model-building phase? If the bleach (Jik, NaOH-based) and baking soda ($NaHCO_3$, weak base) classification was uncertain, this signals a gap to address in the Lesson 10 DQB closure. (3) Did the Kericho agricultural scenario create meaningful engagement and systems-level thinking (both over-acidification and over-liming as problems), or did students treat it as a detail to copy rather than a reasoning task? If the latter, consider posing it as a structured problem in the Lesson 10 review: a farmer has soil pH 4.2 — how much lime is needed, and what is the risk of overdoing it? (4) Are cumulative models sufficiently detailed and coherent to support the Lesson 10 final explanation task? Scan five models at random after the lesson and identify one strength and one gap in each.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	From our Lesson 8 pH data, two acids at the same concentration gave different pH readings: hydrochloric acid gave pH 1 while ethanoic acid (vinegar) gave approximately pH 3, and two bases at the same concentration gave different pH readings: sodium hydroxide gave pH 13 while ammonia solution gave approximately pH 11. These differences could not be explained by concentration alone.
What did I learn?	Strong acids (such as HCl and H_2SO_4) and strong bases (such as NaOH and $Ca(OH)_2$) ionise completely in aqueous solution, producing a high concentration of H^+ or OH^- ions and therefore giving pH values far from 7. Weak acids (such as ethanoic acid in vinegar, citric acid in lemon juice, and carbonic acid in Stoney Tangawizi) and weak bases (such as ammonia solution and soap) ionise only partially, producing fewer H^+ or OH^- ions and giving pH values closer to 7. The degree of ionisation is shown by a one-directional arrow for strong acids and bases and a reversible double arrow for weak acids and bases.
How does this explain the phenomenon?	The different colours produced in the eight cups of the anchoring phenomenon reflect not only whether a substance is an acid or base, but also how strongly it ionises. The lemon juice turned orange-red rather than deep red because citric acid is a weak acid that ionises partially, producing fewer H^+ ions and a higher pH than a strong acid at similar concentration. The Jik bleach turned deep purple-blue because NaOH is a strong base that ionises completely, producing a very high OH^- concentration and a pH well above 7. In Kericho, soil acidity caused by weak organic acids is still damaging at pH 4.2 because even partial ionisation of many acid molecules accumulates enough H^+ ions to harm tea plant roots, and farmers apply calcium hydroxide (a strong base) to neutralise this acidity through the neutralisation reaction studied in Lesson 5.

LESSON 10 (80 minutes): Uses of Acids and Bases + Final Model Build & Explanation

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To consolidate understanding of the uses of acids and bases in industrial, agricultural, domestic, and medical contexts, and to produce a final scientific explanation of the anchoring phenomenon.
Knowledge	Students know that acids and bases have wide practical uses in everyday Kenyan life — including hydrochloric acid in digestion, carbonic acid in carbonated drinks such as Stoney Tangawizi, sulphuric acid in car batteries, calcium hydroxide (lime) in soil treatment on Rift Valley farms, sodium hydroxide in soap manufacture, and bases in antacid medicines and toothpaste. Students know that the pH scale (0–14) quantifies acid/base strength, that universal indicator produces a spectrum of colours correlated to pH, and that Arrhenius acids donate H^+ ions while Arrhenius bases produce OH^- ions in aqueous solution.
Skills	Students can: (1) match real Kenyan substances to their uses based on acidic or basic character; (2) construct and annotate a final cumulative particle/concept model; (3) write a full evidence-based scientific explanation of the colour-changing cups phenomenon; (4) evaluate the completeness of their Driving Question Board answers against lesson evidence.
Attitudes	Students appreciate the relevance of chemistry to health, agriculture, and environmental stewardship in Kenya. They demonstrate intellectual honesty by comparing their Lesson 1 predictions with their final explanations and acknowledging growth in understanding.
Key Inquiry Question	What are acids and bases? What are the chemical properties of acids and bases? How is the strength of acids and bases determined?
Purpose in Storyline	This is the culminating lesson of the sub-strand. All prior lessons have stripped away layers of mystery around the eight colour-changing cups. In this lesson, students integrate everything — Arrhenius definitions, indicators, pH scale, and chemical properties — to explain every cup scientifically and to connect that chemistry to uses they encounter daily across Kenya. The Driving Question Board is formally closed, and students produce their most complete model yet, demonstrating full sub-strand mastery.
Safety Notes	No new hazardous substances are introduced in this lesson. If the teacher demonstrates the eight-cup phenomenon again as a callback, standard precautions apply: avoid ingesting any indicator-treated liquids, handle Jik (sodium hypochlorite) solution with gloves, and ensure adequate ventilation. Students working with written/digital materials only — no open chemicals during the main lesson activities.

B. LESSON OVERVIEW

Lesson 10 is the final, integrative lesson of Sub-Strand 3.1. Students open by briefly revisiting their very first Lesson 1 predictions about the eight colour-changing cups. They then watch a short video and complete a structured reading on the uses of acids and bases across Kenyan industrial, agricultural, domestic, and medical contexts — grounding abstract chemistry in real-life relevance. A collaborative DQB Closure activity has every pair match each original Driving Question Board question to an evidence-based answer drawn from across the unit. Students then revise and finalise their cumulative particle/concept models (built across lessons), annotating the additions made in each lesson and comparing the final model with their Lesson 1 naive sketch. The lesson culminates in the Final Explanation writing task: students independently write a full scientific explanation of every colour observed in the eight cups from Lesson 1, explicitly using the Arrhenius definition, pH values, indicator behaviour, and chemical properties of acids and bases. Teacher facilitates whole-class sharing of final explanations and conducts a structured celebration of intellectual growth.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Arrhenius Acids and Bases - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students are shown a projected image of the eight cups from Lesson 1 (lemon juice, black chai, cow's milk, Stoney Tangawizi, tap water, baking soda solution, liquid soap, and Jik bleach solution — all with universal indicator added, producing the full colour rainbow). Each student individually retrieves their original Lesson 1 prediction card/notebook entry. Without looking at notes from other lessons, they write a brief 'Before & After' statement: what they originally thought caused the colour differences, and what they now believe. Pairs then share with each other. The teacher cold-calls 4 students to share their 'Before' statements aloud, capturing the naïve ideas on the board. This activates prior knowledge across the full unit and creates a contrast frame for the final explanation.	Project the eight-cup image prominently. Say: 'Take out your notebook and find your very first prediction from Lesson 1. Read it silently. Then write two sentences — what did you think THEN, and what do you think NOW?' Allow 3 minutes silent individual writing. Then say: 'Turn to your partner. Share your Before and your Now. You have 90 seconds.' After pairs share, cold-call exactly 4 students for their 'Before' statements. Write them verbatim on the board under the heading 'What We Thought in Lesson 1'. Do NOT correct or praise yet — preserve the contrast. Say: 'Hold on to your NOW explanations. By the end of this lesson, you will have written the most complete answer you have ever given in Chemistry.' Allow 10 seconds of WAIT TIME after posing each cold-call prompt.	Strategy: Prediction Contrast / Before-and-After Elicitation. This strategy activates the full conceptual journey of the unit by making prior misconceptions explicit and visible. Placing Lesson 1 ideas on the board creates a public 'before' baseline that will be directly compared to the final explanation, making intellectual growth concrete and motivating.	Observable indicator: Students can articulate at least one specific conceptual shift from Lesson 1 (e.g., 'I thought it was just about taste, but now I know it is about H^+ and OH^- ions and the pH number'). Teacher listens for use of unit vocabulary (ions, pH, Arrhenius, indicator) in 'Now' statements. Students who cannot name any conceptual shift are flagged for additional support during the final explanation writing phase.
Observe Phase 📹 **VIDEO:** Arrhenius Acids and Bases - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students watch a 6–8 minute curated video (hosted on the ARES platform) featuring Kenyan contexts: a Kericho tea farmer applying agricultural lime (calcium hydroxide) to acidic soil; a Nairobi garage mechanic explaining sulphuric acid in a car battery; a pharmacist in Mombasa describing antacid tablets neutralising excess stomach hydrochloric acid; a soft-drinks factory showing carbon dioxide dissolved under pressure to form carbonic acid in sodas like Stoney Tangawizi; and a soap-making cooperative in Kisumu using sodium hydroxide. After the video, students complete a two-page structured reading (on ARES) on the same contexts, which includes a table of common	Before the video, say: 'As you watch, your job is to spot every acid or base being used in a real Kenyan setting. On your paper, make a quick tally — left column: acids I saw; right column: bases I saw.' Play the video without pausing unless a technical issue arises. After the video, allow 10 seconds WAIT TIME then cold-call 3 students: 'Name one acid you spotted and how it was being used.' Then distribute or direct students to the structured reading. Say: 'Read carefully. Underline every use of an acid in RED. Underline every use of a base in BLUE. Then fill in your Uses Sort table. You have 10 minutes.' Circulate actively — check that students are correctly classifying	Strategy: Annotated Reading with Colour-Coding + Evidence Sort Table. Colour-coding forces students to make classification decisions at every sentence, not passively skim. The Sort Table structures information gathering so that students are simultaneously building the evidence base needed for the final explanation and the DQB closure. Connecting each industrial/medical use back to acid/base character reinforces that the chemistry of the eight cups is the same chemistry operating in society.	Observable indicator: Completed Uses Sort table with at least 6 correct entries. Teacher spot-checks 5 tables during circulation — look specifically for correct pH assignment (e.g., stomach HCl ~pH 1–2; antacids ~pH 8–9; agricultural lime ~pH 12) and correct Arrhenius justification. Students who cannot justify a classification are re-directed to their particle model notes from earlier lessons.

	acids and bases, their pH ranges, and their uses. Students annotate the reading: they underline uses of acids in one colour and uses of bases in another. They also complete a quick 'Uses Sort' table with columns: Substance | Acid or Base | Approximate pH | Use in Kenya.	substances. Prompt students who misclassify: 'Remember — what does an Arrhenius acid release in water? Does this substance do that?' Use 12-second WAIT TIME before offering hints.		
Explain Phase 📹 **VIDEO:** Arrhenius Acids and Bases - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The full Driving Question Board (DQB) — accumulated across all 10 lessons — is displayed (projected or physical board). Each question card is visible. Working in their established pairs, students receive a set of 'Answer Evidence Cards' — small slips pre-printed by the teacher — each containing one key piece of evidence or explanation from across the unit (e.g., 'Arrhenius acids release H^+ ions in water', 'pH below 7 = acidic; pH above 7 = basic', 'Universal indicator turns red-orange at low pH', 'Bases turn litmus blue', 'HCl in the stomach helps digest food', 'Agricultural lime neutralises acidic Rift Valley soil'). Pairs must match each Answer Evidence Card to the question it best answers on the DQB. When a question has been fully answered, they mark it with a green tick. Questions that remain partially answered are marked with an amber dot. Pairs then share matches with the class; the teacher leads a whole-class confirmation of each answer, correcting misconceptions in real time. Finally, students write a 3-sentence 'Unit Summary Statement' in their notebooks capturing the three most important ideas of the sub-strand.	Display the DQB prominently. Say: 'Every question on this board was asked by YOU across this unit. Today we are going to close the board — that means answering every single question with real evidence from our lessons.' Distribute Answer Evidence Cards. Say: 'Work with your partner. Match each card to the question it answers. You have 8 minutes. Some questions may need more than one card.' Set a visible timer. After 8 minutes, cold-call 5 pairs in sequence to share one match each. For each match, ask the class: 'Does everyone agree? Thumbs up, thumbs sideways, or thumbs down.' Address any thumbs-sideways or thumbs-down immediately. Say: 'Before we move on — who can explain WHY this evidence answers this question?' Allow 12-second WAIT TIME. Once all questions are resolved, say: 'In your notebook, write your 3-sentence Unit Summary Statement. Start with: The most important idea in this sub-strand is…' Allow 4 minutes. Cold-call 3 students to read their summaries.	Strategy: Driving Question Board Closure with Evidence Matching. This strategy transforms the DQB from an ongoing question-tracking tool into a summative sensemaking event. By physically matching evidence to questions, students perform the cognitive work of synthesis — not just recall. The green-tick / amber-dot system makes gaps in understanding visible without shame, and the whole-class confirmation consolidates a shared, accurate model of the sub-strand.	Observable indicator: At least 90% of DQB questions receive a green tick (fully answered) by end of activity. Teacher notes which questions remain amber — these are addressed explicitly during the Final Explanation phase. The 3-sentence Unit Summary Statement is collected as a written formative artefact; teacher checks for presence of: (1) a reference to ions/Arrhenius, (2) a reference to pH, and (3) a reference to real-world use or health/environmental relevance.
Driving Question	Students retrieve their Lesson 1 naive model sketch (a simple	Say: 'Place your Lesson 1 model on the LEFT side of your desk and	Strategy: Model Comparison and Annotated Revision (Model	Observable indicator: Final model contains all four required elements

Board (DQB) Creation 📹 **VIDEO:** [Arrhenius Acids and Bases - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Properties of Acids (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	drawing/diagram they made at the start of the unit to explain the colour-changing cups). They place this next to their most recent cumulative model (built and revised across lessons). Using a structured 'Model Evolution' annotation sheet (on ARES), students annotate their final model by labelling: (A) concepts added in early lessons (e.g., the definition of acid/base, litmus colour changes); (B) concepts added in middle lessons (e.g., Arrhenius ion definitions, pH scale, indicator colours); and (C) concepts added in final lessons (e.g., strength vs. concentration, neutralisation, real-world uses). Students draw arrows on their final model showing how the presence of H^+ ions in lemon juice leads to a low pH, which leads to universal indicator turning red, which explains the colour of Cup 1. They repeat this chain for at least three more cups. The final model must include: particle-level representation of H^+ and OH^- ions, pH values for each of the eight cups (approximate), indicator colour predictions, and one real-world use for each substance.	your final model on the RIGHT. Look at both. In your head, count how many new ideas appear in your final model that are completely absent from your Lesson 1 model.' Allow 15 seconds silent observation. Cold-call 3 students: 'Name ONE thing in your final model that was not in your Lesson 1 model.' Affirm and add to a running 'Growth List' on the board. Then say: 'Now annotate your final model using the labels A, B, and C on your annotation sheet. Draw the cause-and-effect arrow chains for at least four of the eight cups. You have 12 minutes.' Circulate and prompt: 'Your arrow chain should start with: what particles are in this liquid? Then what does that mean for pH? Then what does the indicator do? Then what colour do we see?' Provide 10-second WAIT TIME before helping any student who is stuck.	Evolution Protocol). This strategy makes the process of conceptual change visible to the student. By annotating which ideas arrived at which stage of the unit, students develop metacognitive awareness of how scientific understanding is built incrementally. Drawing explicit cause-and-effect chains from ions → pH → colour directly prepares students for the Final Explanation writing task in Phase 5.	(particle-level ions, approximate pH values, indicator colour predictions, one real-world use per substance) for at least 6 of the 8 cups. Teacher reviews 6 models during circulation, checking specifically that arrow chains are directional and causal (not just associative lists). Students whose chains lack the particle-level step (H^+/OH^- → pH) are prompted to revise before moving to Phase 5.
Model Building Phase 📹 **VIDEO:** [Arrhenius Acids and Bases - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Properties of Acids (Flexbook)] Source: CK-12 🔍 Search ARES	This is the culminating performance task of the sub-strand. Students work individually in silence. They are given the 'Final Explanation' prompt sheet (on ARES): 'Return to the eight cups from Lesson 1. For each cup, write a full scientific explanation of the colour you observed when universal indicator was added. Your explanation must include: (1) the name and nature of the dissolved substance (acid or base); (2) the Arrhenius definition — which ions does it release in	Distribute the Final Explanation prompt sheet. Say: 'This is your chance to show everything you have learned in this sub-strand. Work in silence. Use your final model, your annotation sheet, and your Uses Sort table — but write the explanation in YOUR OWN WORDS. You have 20 minutes.' Set a visible timer. Circulate silently for the first 5 minutes — do not answer questions; allow productive struggle. After 5 minutes, you may offer one-on-one prompts to students who are stuck:	Strategy: Final Explanation Writing + Structured Peer Feedback (One Star, One Step). The Final Explanation is the highest-order sensemaking activity of the unit — it requires students to synthesise six distinct conceptual threads (substance identity, Arrhenius definition, pH, indicator behaviour, real-world use, societal relevance) into a single coherent written argument for each of eight substances. Peer feedback using One Star, One Step trains students to evaluate scientific	Observable indicator: Each student's Final Explanation for at least 6 of 8 cups contains all six required elements. Teacher collects all Final Explanation sheets as a summative formative artefact. During the peer-feedback segment, teacher listens for: (a) correct use of 'Arrhenius acid/base' terminology; (b) accurate pH values (within ±1 unit of accepted values); (c) correct indicator colour for that pH range; (d) a Kenya-relevant real-world use. Students who cannot produce

for similar readings	water?; (3) the approximate pH of the liquid; (4) the colour of the universal indicator at that pH and why; (5) one real-world use of this substance relevant to Kenya; and (6) what the colour tells us about health, farming, or environment.' Students write their explanations for all eight cups (lemon juice, chai, milk, Stoney Tangawizi, tap water, baking soda solution, liquid soap, Jik bleach). After 20 minutes of individual writing, two volunteers share their explanation for one cup each (different cups). The class provides structured peer feedback using 'One Star, One Step' (one thing done well, one thing to improve). The lesson closes with the teacher reading aloud the unit driving question one final time — 'What invisible properties of everyday Kenyan liquids cause them to turn such different colours — and why does it matter for our health, our farms, and our environment?' — and cold-calling 3 students to answer it in one sentence each, without notes.	'Start with: this liquid contains… and it releases… in water.' After 20 minutes, say: 'Pens down. Who would like to share their explanation for one cup?' Select 2 volunteers for different cups. After each share, direct the class: 'Give One Star — something scientifically correct and specific. Give One Step — one way to make the explanation more complete or more precise.' Allow 8-second WAIT TIME for peer feedback. Finally, project the driving question. Say: 'This is the question we have been answering for ten lessons. In ONE sentence — without your notes — answer it. Hands up only after I count to ten silently.' Cold-call 3 students. Affirm the scientific accuracy of each response specifically: 'Excellent — you used the word pH correctly' or 'Good — can you add which ion makes it acidic?'	explanations using specific criteria, not vague impressions. The closing cold-call on the driving question — done without notes — assesses genuine internalisation of the sub-strand's core ideas.	the Arrhenius ion definition for any cup receive targeted written feedback on their sheet for remediation. The final three driving-question responses (cold-called, no notes) serve as an oral exit ticket — teacher records whether each response correctly names ions, pH, and societal relevance.

D. TEACHER REFLECTION

After delivering this final lesson, reflect on the following questions:

1. FINAL EXPLANATION QUALITY: Did students' written explanations for the eight cups demonstrate genuine conceptual integration — or were students retrieving isolated facts? Review 5–6 Final Explanation sheets tonight. Look for students who correctly link ions → pH → indicator colour → societal relevance as a causal chain. Flag students whose explanations are merely lists of memorised facts without causal connective language.

2. DQB CLOSURE: Were there any questions on the DQB that remained amber (partially answered) even after the closure activity? If so, which concepts were still unclear? This reveals gaps in the unit's instructional sequence that should be addressed in future delivery or in a short remediation session before any sub-strand assessment.

3. MODEL EVOLUTION: Did students' final models visibly and meaningfully surpass their Lesson 1 naive sketches? If many students' final models looked only marginally more complex than their starting point, consider whether the model-building phases across the unit were sufficiently scaffolded. Future delivery might benefit

from more frequent, structured model revision moments at the end of each lesson.

4. KENYAN CONTEXT CONNECTIONS: Did students draw on Kenyan examples (Rift Valley soil, Stoney Tangawizi, Jik, antacids, soap) naturally in their explanations — or did they default to generic/Western examples? Strong localisation of the final explanation indicates that Kenya-relevant contexts were successfully embedded throughout the unit.

5. DRIVING QUESTION RESPONSE: In the final cold-call, could students answer the driving question — 'What invisible properties of everyday Kenyan liquids cause them to turn such different colours — and why does it matter?' — in one sentence without notes? If not, which component (invisible properties / colour change / societal relevance) was most often missing? Use this to focus the opening of any review or assessment lesson.

6. AFFECTIVE DIMENSION: Did students visibly recognise and express their own growth when comparing Lesson 1 and Lesson 10 models? Building this metacognitive moment is as important as the content — it develops students' identity as learners of science. Note which students seemed most surprised by their own growth; these are students who may need regular encouragement about their capability in Chemistry.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students watched a video showing Kenyan contexts — a Kericho tea farmer applying agricultural lime to acidic soil, a Nairobi garage mechanic using sulphuric acid in a car battery, a Mombasa pharmacist describing antacid tablets, a soft-drinks factory producing carbonic acid in Stoney Tangawizi, and a Kisumu soap-making cooperative using sodium hydroxide. Students also observed the eight-cup universal indicator demonstration (via projected image callback from Lesson 1), noting the full colour spectrum from red (lemon juice, pH ~2) through green (tap water, pH ~7) to purple (Jik bleach, pH ~12–13).
What did I learn?	Students learned that acids and bases have wide practical uses across Kenyan life: hydrochloric acid (pH ~1–2) aids digestion in the stomach; carbonic acid gives carbonated drinks like Stoney Tangawizi their fizz (pH ~3–4); sulphuric acid powers car batteries; calcium hydroxide (agricultural lime) neutralises acidic soils in the Rift Valley to improve crop yields; sodium hydroxide is used in soap manufacture; and bases such as sodium bicarbonate and magnesium hydroxide are active ingredients in antacid medications that neutralise excess stomach acid. Students consolidated the Arrhenius definitions (acids release H^+; bases release OH^-), the pH scale (0–14), and the relationship between pH and universal indicator colour, and applied these to explain every colour in the eight cups from Lesson 1.
How does this explain the phenomenon?	Students explained that the colour differences among the eight cups arise because each liquid contains dissolved substances that release different amounts of H^+ or OH^- ions in water. Liquids with high H^+ ion concentration (low pH) — such as lemon juice, Stoney Tangawizi, and chai — cause universal indicator to turn red, orange, or yellow. Tap water, which is neutral (pH ~7), produces a stable green colour. Liquids with high OH^- ion concentration (high pH) — such as baking soda solution, liquid soap, and Jik bleach — cause universal indicator to turn blue or purple. This matters for health (stomach acid, antacids), for farming (soil pH and lime application), and for the environment (industrial effluent pH and water quality in rivers like the Nairobi River).

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.